MASTER DOCUMENT: Microalgae Biosorption of NdFeB Magnet Leachate Contaminants for Wastewater Regulatory Compliance

Summary

This master document consolidates the previously generated deliverables into one continuous, compliance-framed package focused exclusively on the wastewater-treatment step (biosorption/bioabsorption and associated process/measurement design), and not on downstream metal recovery/refinement. The compiled material includes: (i) a study evidence table, (ii) a narrative synthesis emphasizing multi-metal effects (especially Fe and B), (iii) a discharge-limit compilation from EU BREF snippets and US EPA metal-finishing limits, and (iv) an end-to-end experimental, analytical, and pilot-validation plan including DOE/RSM designs, ICP-MS SOP elements, cost drivers, procurement lists, and a data pipeline for compliance dashboards. Core constraints highlighted across the artifacts are that Fe precipitation/competition is strongly pH-dependent in NdFeB-derived solutions and that boron removal collapses in multi-metal mixtures (e.g., down to ~0.2% in multimetal conditions for some microalgae), making boron a likely rate-limiting compliance parameter unless a dedicated boron-polishing step is integrated.

Table of contents

This document is organized as a single set of chapters to match the originally requested master deliverable structure.

- Chapter 1: Literature Review Evidence Table
- Chapter 2: Narrative Literature Synthesis
- Chapter 3: Chinese Literature, Regulatory Compliance Gap Matrix, and Experimental Protocol
- Chapter 4: Reduced Factorial Screening Design, Go/No-Go Decision Tree, and Cost-Effectiveness Analysis
- Chapter 5: JMP/Minitab-Importable DSD Run Order Table, Budget Sensitivity Analysis, and Technology Selection Decision Matrix
- Chapter 6: Response Surface Methodology Design, ICP-MS Analytical SOP, and 24-Month Timeline
- Chapter 7: Reagent and Consumables Procurement List
- Chapter 8: Data Analysis Pipeline
- Chapter 9: Grant Proposal Abstract and Specific Aims
- Chapter 10: Research Strategy
- Chapter 11: Pilot-Scale Equipment Procurement List
- Chapter 12: Synthetic Leachate Preparation SOP

Chapter 1

This chapter reproduces the extracted evidence table as a single markdown table with the requested columns.

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Chlorella vulgaris (SAG 211-12)	Real NdFeB magnet leachate (aqueous solution derived from permanent Nd–Fe–B magnets; selective leaching used to remove Fe/B/Dy/Co interference)	Nd (50–250 mg/L; stock standardized to 1000 mg/L Nd); Fe, B, Dy, Co analyzed but reported below detection limit in prepared solutions	Nd: % removal not tabulated; >90% of Nd biosorption occurred before equilibrium (equilibrium reached within ~30 min across temperatures)	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	pH control and (in general) Fe competition/precipitation: Fe can precipitate as pH rises, reducing competition; however Fe was removed by selective leaching so competitive effects were not present in tested leachate	Oven-dried (non-living) biomass (24 h at 90℃)	Batch (Erlenmeyer flasks)	Up to 90 min (samples taken to 90 min); equilibrium within ~30 min	0.5–3 g/L (dry biomass)	Pseudo-second-order kinetics (R^2>0.99); Langmuir isotherm (qmax 188.68 mg/g; KL 0.029 L/mg)

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Chlorella vulgaris (dried biomass)	Mixed leachate solution derived from neodymium magnets (batch and continuous sorption systems)	Nd (initial 250 mg/L)	Not mentioned in source (reported as uptake capacity, e.g., q = 157.21 mg/g at pH 5).	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	Slow Nd sorption kinetics reduced capacity in continuous vs batch operation	Dried biomass	Batch and continuous sorption systems	Not reported (described qualitatively as slow kinetics)	0.5 g/L	Not reported (q reported; kinetics described as slow)
Galdieria sulphuraria (SAG 21.92)	Synthetic multi-metal solution (5 ppm each of Cu, La, Nd, Dy); tested at different pH	Cu 5 ppm; La 5 ppm; Nd 5 ppm; Dy 5 ppm	Overall removal efficiency: <9% at pH 2.5; up to 86% at pH 5.0 (solution-wide metric; per-metal % not specified)	Not mentioned in source (uptake reported as mg/g in some conditions).	Not assessed against discharge limits (not mentioned in source).	Orthophosphate limits REE solubility/availability and thus biosorption under growth-compatible conditions (phosphate + high acidity)	Living biomass (grown, centrifuged, washed, resuspended to OD 10)	Batch (conical flasks / capped bottles; 30 mL reaction volume)	Samples at 3 h and 20 h	Not mentioned in source.	Not mentioned in source.

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Galdieria sulphuraria (SAG 21.92)	Synthetic La-spiked nutrient-replete growth medium (phosphate present)	La added to reach 20 µM and 100 µM (depending on condition/pH)	No measurable La biosorption observed at initial pH 1.5 (under all growth conditions)	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	Orthophosphate limits REE solubility and availability for biosorption	Living cultures (nutrient replete medium)	Batch culture flasks (100 mL cultures in conical flasks)	At least 6 days (duration stated; time-resolved removal not reported)	Not mentioned in source.	Not mentioned in source.

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Anabaena sp. (SAG 12.82)	Synthetic REE solutions: REE nitrate salts dissolved in modified Bold's Basal Medium (BBM)	Eu^{3+} (0.111 mM and 0.191 mM), Sm^{3+} (0.139 mM), Nd^{3+} (0.103 mM)	REE^{3+} biosorption reported as almost complete (>99%) within the first hour (with additional detail for Eu/Sm/Nd time courses)	Eu residual reported as 76 ppb after >1 day; Sm residual 1.52×10^-3 mM in early samples (Nd residual not numerically specified)	Not assessed against discharge limits (not mentioned in source).	Nutrient/starving state affects performance (biosorption more effective in culture media than starving culture); mechanism not fully resolved	Living biomass (biomass stayed alive during experiments; separated by centrifugation)	Batch flask cultures (250 mL flasks with 150 mL medium)	Minutes to ≤1 h for >99% uptake depending on REE (e.g., Nd within 30 min; Eu within ~1 h)	Not mentioned in source.	Not reported (tracked by ICP-MS/LIBS; no model fit parameters given)

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Anabaena cylindrica (SAG 1403.2)	Synthetic REE solutions: REE nitrate salts dissolved in modified BBM	Eu^{3+} (0.111 mM and 0.191 mM), Sm^{3+} (0.139 mM), Nd^{3+} (0.103 mM)	REE^{3+} biosorption reported as almost complete (>99%) within the first hour (with additional detail for Eu/Sm/Nd time courses)	Eu residual reported as 76 ppb after >1 day; Sm residual 1.52×10^-3 mM in early samples (Nd residual not numerically specified)	Not assessed against discharge limits (not mentioned in source).	Nutrient/starving state affects performance; mechanism not fully resolved	Living biomass (biomass stayed alive during experiments; separated by centrifugation)	Batch flask cultures (250 mL flasks with 150 mL medium)	Minutes to ≤1 h for >99% uptake depending on REE (e.g., Nd within 30 min; Eu within ~1 h)	Not mentioned in source.	Not reported (tracked by ICP-MS/LIBS; no model fit parameters given)

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Nostoc sp. 20.02	Synthetic REE solutions (single-metal screening at 10 mM; plus equimolar mixed-metal competition tests)	REEs: Ce, Nd, Tb, La; competition tests: Ce in presence of Al, Pb, Ni, Zn (mixed-metal solutions)	Not reported as % removal; reported as adsorption capacity (e.g., 84.2–91.5 mg·g−1, species-dependent)	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	Mixed-metal competition reduces REE uptake; at higher concentrations other metals can replace Ce on binding sites	Lyophilized (dried) biomass	Batch tube incubation (2 mL solutions, shaking)	Standard incubation 3 h; kinetics sampled 2–60 min	10–20 mg in 2 mL (~5–10 g/L)	Isotherms fitted (Langmuir/Freundlich); Langmuir reported as best correlation in this study

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Synechococcus elongatus (UTEX 2973)	Synthetic REE solutions (single-metal screening at 10 mM; plus equimolar mixed-metal competition tests)	REEs: Ce, Nd, Tb, La; competition tests: Ce in presence of Al, Pb, Ni, Zn (mixed-metal solutions)	Not reported as % removal; reported as adsorption capacity (mg·g−1) across tested conditions	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	Mixed-metal competition reduces REE uptake; other metals can replace Ce at higher total concentrations	Lyophilized (dried) biomass	Batch tube incubation (2 mL solutions, shaking)	Standard incubation 3 h; kinetics sampled 2–60 min	10–20 mg in 2 mL (~5–10 g/L)	Isotherms fitted (Langmuir/Freundlich); Langmuir reported as best correlation
Calothrix brevissima (SAG 34.79)	Synthetic single-metal solutions (Nd^{3+}/Eu^{3+} and other metals as individual solutions; stock solutions 10 mM)	Single-metal solutions of Nd^{3+}, Eu^{3+} and others: Au^{3+}, Pb^{2+}, Pt^{4+}, Fe^{2+}, Cu^{2+}, Ni^{2+} (10 mM stocks)	Not reported as % removal; Nd sorption capacity reported (Qmax Nd ≈ 0.47±0.01 mmol·g−1)	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	Nickel identified as least effectively bound ion (potential limiting/competing contaminant in mixed wastewaters)	Lyophilized, ground biomass (stored at −20℃)	Batch sorption assays; biomass grown in shake flasks/bubble columns/1-L airlift reactor	3 h (Nd experiments) and 24 h (some Qmax experiments)	5–20 g/L tested; commonly 5 g/L	Isotherm fitting used (Langmuir/Freundlich); Langmuir-like behavior reported for some species (kinetic models not reported)

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Chlorella kessleri (SAG 211-11g)	Synthetic single-metal solutions (Nd^{3+}/Eu^{3+} and other metals as individual solutions; stock solutions 10 mM)	Single-metal solutions of Nd^{3+}, Eu^{3+} and others: Au^{3+}, Pb^{2+}, Pt^{4+}, Fe^{2+}, Cu^{2+}, Ni^{2+} (10 mM stocks)	Not reported as % removal; Nd sorption capacity reported (Qmax Nd ≈ 0.37±0.04 mmol·g−1)	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	Nickel identified as least effectively bound ion; common wastewater metals (Pb/Fe/Cu/Ni) noted as potentially problematic competitors requiring pre-removal in application contexts	Lyophilized, ground biomass (stored at −20℃)	Batch sorption assays; biomass grown in shake flasks/bubble columns/15-L airlift reactor	3 h (Nd experiments) and 24 h (some Qmax experiments) depending on test	5–20 g/L tested; commonly 5 g/L	Nd isotherm for Chlorella kessleri better fitted with Freundlich isotherm (kinetic models not reported)

Species	Waste Type	Metals Tested	Removal Efficiency (%)	Final Concentration (mg/L)	Compliance Status	Limiting Factor	Biomass Form	Reactor Type	Contact Time	Biomass Dosage (g/L)	Kinetic/Isotherm Fit
Spirulina platensis	Synthetic single-metal dysprosium solutions (Dy(NO3)2·6H2O; batch lab experiments)	Dy (10–100 mg/L)	Max Dy removal 50% at pH 3.0 during 1 h; fast initial removal 38% within 5 min	Not mentioned in source.	Not assessed against discharge limits (not mentioned in source).	pH strongly affects removal capacity; increased pH beyond optimum reduced removal capacity	Dried, homogenized biomass (105℃, 24 h; non-living)	Batch experiments (stirred, 50 mL solution)	5–60 min emphasized (max removal reported at 60 min; rapid uptake within 5 min)	0.5 g biosorbent in 50 mL (10 g/L)	Not supported in available quote set beyond qualitative kinetics statement; model details not retained here

Chapter 2

1. State of evidence

The directly relevant evidence base for magnet-derived leachate treatment using microalgae biosorption is thin in the retrieved excerpts, with only a small subset using aqueous solutions explicitly derived from Nd–Fe–B magnets and tested with microalgal biomass. In those magnet-derived studies, the biosorbent was dried *Chlorella vulgaris* biomass and experiments were performed in batch flasks and also in continuous sorption configurations.

Most other included studies use synthetic single-metal or synthetic multi-metal solutions (typically REE nitrate/sulfate salts in defined media) and are therefore not faithful analogs of full-composition NdFeB manufacturing leachates that include Fe-dominant matrices and process chemicals. A smaller subset uses real industrial wastewater (non-magnet) matrices, showing that microalgal biosorption has been tested in complex real waters, but not specifically NdFeB manufacturing wastewaters in the provided excerpts.

The table below is an evidence map (not a performance ranking) that highlights which retrieved studies most closely resemble NdFeB-leachate compliance treatment.

Evidence category	Example species	Matrix type in excerpt	Why it matters for NdFeB wastewater compliance
Magnet-derived leachate biosorption	*Chlorella vulgaris*	Aqueous solution derived from real Nd–Fe–B magnets; selective leaching used such that Fe/B/Dy/Co were below detection in prepared solutions	Direct link to magnet source material, but not representative of Fe-rich leachates because Fe and other co-contaminants were removed in solution preparation
Synthetic multi-metal REE + base metals	*Galdieria sulphuraria*	5 ppm each of Cu, La, Nd, Dy at different pH	Demonstrates pH sensitivity and multi-metal effects for REE + Cu, but lacks Fe/B and industrial organics
Synthetic REE salts in growth media	*Anabaena* spp.	REE(NO3)3·6H2O salts in modified BBM (Nd, Sm, Eu concentrations reported)	Shows rapid REE uptake in culture media, but does not represent industrial leachate chemistry (e.g., Fe-dominant matrices and chelators/coolants are absent in the excerpted conditions)

2. REE biosorption performance

For neodymium (Nd), dried *Chlorella vulgaris* showed high uptake capacities when tested on magnet-derived aqueous solutions, with Nd kinetics described as rapid (equilibrium within ~30 minutes) and with "more than 90%" of biosorption occurring before equilibrium. In the same Nd–*C. vulgaris* biosorption study, the Nd sorption kinetics fit a pseudo-second-order model (R^2>0.99) and the isotherm fit a Langmuir model with qmax 188.68 mg/g (KL 0.029 L/mg).

A separate magnet-derived "mixed leachate solution" study also used dried *Chlorella vulgaris* and reported a maximum Nd uptake of q = 157.21 mg/g at pH 5 with 0.5 g/L biosorbent dosage (initial Nd 250 mg/L). That work explicitly noted that slow Nd sorption kinetics limited performance in continuous assays relative to batch, underscoring that hydraulic/contact-time constraints are likely to matter for compliance operations.

Other taxa (including cyanobacteria) show very fast REE removal in defined media. In modified BBM with REE nitrate salts, *Anabaena* sp. (SAG 12.82) achieved almost complete (>99%) biosorption of trivalent REEs within the first hour, and specifically reported >99% Nd removal within 30 minutes after metal-salt addition (with initial Nd^{3+} 0.103 mM). For europium, the same study reported a residual concentration of 76 ppb after more than one day, indicating persistence of a small dissolved fraction after extended contact in that system.

Screening-style adsorption tests with lyophilized cyanobacterial biomass reported high REE capacities under acidic-to-near-neutral conditions. For example, *Nostoc* sp. 20.02 was reported to have the highest adsorption capacity (84.2–91.5 mg g−1) in a screening study covering Ce, Nd, Tb, and La (single-element tests) and also included mixed-metal competition tests.

However, multi-ion competition can significantly reduce REE binding compared with single-element tests. In mixed-metal solutions, cerium adsorption capacity dropped significantly relative to single-element solutions, and at a metal concentration of 4 mM other metals (Al, Pb, Ni, Zn) "seemed to replace cerium" on binding sites as Ce binding capacity dropped across tested cyanobacterial biomasses.

3. Iron interference

The most directly NdFeB-linked biosorption study in the excerpts highlights a central difficulty for realistic magnet leachates: iron chemistry changes with pH, and Fe can precipitate as pH rises, altering competitive interactions and potentially confounding adsorption/uptake measurements. That study described that as pH increased from 2 to 5, iron ions started to precipitate, which "leads to a decrease in the competition," but it also stated that a selective leaching procedure was applied to prepare an aqueous solution without iron from Nd–Fe–B magnets. Consistent with that, it reported a magnet-derived solution (pH 2.4) with "no iron detected," and analysis indicated Fe (and also B, Dy, Co) were below detection limits in the prepared solutions due to selective leaching.

4. Boron

Boron behavior in multi-contaminant waters appears consistently challenging in the retrieved excerpts. In a synthetic multi-contaminant mixture, boron removal efficiencies decreased sharply when moving from monometallic to multimetallic solutions, including a drop from 25.6% to 0.2% for *Chlamydomonas reinhardtii* and from 21.4% to 6.6% for *Chlorella vulgaris* under multimetal conditions, with a similar drop for *Scenedesmus almeriensis* (22.9% to 3.6%). A related extract emphasizes that B removal decreased "remarkably" in multimetallic solutions, down to 0.2% in *C. reinhardtii* (with exceptions for a Chlorophyceae isolate).

5. Co-contaminant effects

Direct evidence on NdFeB-specific co-contaminants (metalworking fluids, emulsified oils, surfactants, and chelating agents such as EDTA/NTA) is not present in the retrieved excerpts used for the evidence table; instead, several studies use clean synthetic metal-salt systems with mineral acids/bases for pH control (e.g., Dy(NO3)2·6H2O with HNO3/NaOH).

6. Regulatory compliance assessment

The retrieved excerpts rarely frame results in terms of discharge permitting, and instead emphasize uptake capacity, percent removal, and kinetic/isotherm fits (e.g., pseudo-second-order kinetics and Langmuir isotherms). Numeric discharge limits compiled elsewhere in this package include US EPA 40 CFR Part 433 limits for Cu/Ni/Cr/Zn/TSS/O&G (daily max and monthly average values shown in extracted rows) and EU STM BREF ranges for multiple parameters (B, Cu, Ni, Al, total Cr, Zn, COD, hydrocarbons, suspended solids).

7. Process configuration and scale

Across studies, biosorption was primarily tested in batch systems, with contact times from minutes to hours and with dried biomass often used to emphasize surface binding. In the magnet-derived Nd biosorption study, experiments were conducted in 500 mL Erlenmeyer flasks using oven-dried biomass, with dry biosorbent dosages ranging from 0.5 to 3 g/L and contact up to 90 minutes (with equilibrium within ~30 minutes). Continuous operation is directly relevant to wastewater compliance, and one magnet-derived leachate study explicitly reported Nd removal in both batch and continuous sorption systems using dried *Chlorella vulgaris* biomass, while also warning that slow kinetics reduced capacity under continuous conditions compared to batch assays.

8. Critical evidence gaps

The excerpts show proof-of-principle that microalgae can bind/accumulate REEs rapidly under controlled conditions, but they also show that the only NdFeB-origin dataset in this set used selective leaching that removed Fe/B/Dy/Co to below detection, meaning the Fe-dominant competitive binding and Fe(OH)3 co-precipitation regime expected in realistic magnet wastewaters was not tested in that "magnet-derived" aqueous matrix. Multiple additional sources state, more generally, that no peer-reviewed study has tested microalgae against the full chemical matrix of NdFeB manufacturing leachate, reinforcing this gap at the application level.

9. Comparison to conventional treatment

Conventional precipitation is widely described as mature and effective for heavy metal removal, but with sludge-generation drawbacks. A review extract reports precipitation removal efficiencies on the order of ~95–99% for most heavy metals in wastewater, while multiple sources emphasize large slurry/sludge production and difficult dewatering for hydroxide sludges.

Chapter 3

This chapter reproduces the Chinese/regulatory and protocol artifact in consolidated form.

Summary

The extracted evidence base contains strong laboratory proof that microalgae and cyanobacteria can sorb/accumulate REEs in controlled conditions, but the dataset remains thin for the central compliance question of meeting numeric discharge limits in real NdFeB leachates that retain the full Fe–B–Co–Ni–Cu–Al matrix and process organics; this limitation is also reflected in broader statements that no peer-reviewed study has tested microalgae against the full chemical matrix of NdFeB manufacturing leachate. Across extracted multi-contaminant studies that do report boron behavior, boron is consistently weak in mixed-metal systems (e.g., down to ~0.2%), implying it is likely rate-limiting unless a separate boron-focused polishing step is added.

Part 1

Chinese-language and China-affiliated sources retrieved in the workflow did not yield a full, NdFeB shop-floor wastewater composition table (Fe, B, REEs, Co/Ni/Cu/Al, plus COD/oils/chelators) within the provided extracts, and China-specific biological treatment material in the extracted snippets is dominated by non-NdFeB contexts or by patents focused on chemical recovery rather than biological polishing for discharge compliance.

Part 2

The discharge-limit snippets available in the extracted materials include US EPA 40 CFR Part 433 rows for Cu, Ni, Zn, total Cr, TSS, and oil & grease (daily max and monthly average) and an EU STM BREF table snippet listing ranges for B, Cu, Ni, Al, total Cr, Zn, COD, total hydrocarbons, and suspended solids, specified as daily composites unfiltered prior to analysis taken after treatment and before any dilution.

Part 3

A compliance-oriented program must begin with stream-resolved characterization (metals + COD/TSS/oils and relevant additives) and then use screening/optimization designs that treat residual mg/L versus discharge limits as the core response variables, because regulatory targets are concentration-based.

Chapter 4

This chapter reproduces the reduced-design and go/no-go logic, together with a comparative cost-effectiveness scaffold.

Part 1

Definitive screening designs are described as three-level designs where run count relates to the number of variables (k), e.g., N = 2k + 1 or 2k + 3, and JMP guidance states that for odd k the minimum number of runs in the screening design is 2k + 3. DSDs are further described as being able to estimate main effects, interactions, and quadratic effects with a low number of experiments, and as having good properties for finding significant linear effects even with active second-order effects.

Part 2

The go/no-go logic is structured around staged gates where early viability and single-metal feasibility precede multi-metal compliance feasibility, recognizing that high percent removal can still leave residual concentrations above strict BAT-AEL/NPDES-style limits (e.g., when limits are sub-mg/L). Regulatory numeric limits used for such gates in this package include Part 433 values and STM BREF values as extracted.

Part 3

Chemical precipitation is described as broadly used and mature for heavy-metal removal; precipitation is associated with sludge-generation burdens (large slurry/sludge volumes and difficult dewatering). Boron is described as difficult to remove; boron removal can require specialized technologies and boron-selective resins/fibers have reported boron removal rates in the range of ~93–98%.

Chapter 5

This chapter reproduces the DSD run-order table (270 runs) and associated import and MCDA decision-matrix notes.

Design basis

Blocking is described as appropriate when an identifiable nuisance source of variation will vary across runs, and blocking reduces correlation between nuisance variation and factor effects; JMP also documents "Randomize within Blocks" behavior for run order in fixed-block designs, and Minitab documents defining custom screening designs from spreadsheets.

Complete 270-run table

The 270-run DSD run-order table is reproduced below as provided in the prior artifact (15 DSD runs × 6 species blocks × 3 replicates). The factor ranges and levels reflect a three-level design structure consistent with DSD practice (low/center/high), and the run-order approach is described as randomized within blocks consistent with JMP guidance.

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg/L	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
1	C. vulgaris	1	9	3	100	Yes	5	0	5	1	
2	C. vulgaris	1	2	3	100	No	0	1	5	24	
3	C. vulgaris	1	13	5	55	No	2.5	0.5	2.75	12.5	
4	C. vulgaris	1	5	7	10	Yes	0	1	0.5	24	
5	C. vulgaris	1	11	5	55	No	2.5	0.5	2.75	12.5	
6	C. vulgaris	1	4	7	100	Yes	0	0	0.5	1	
7	C. vulgaris	1	7	7	100	No	5	1	0.5	1	
8	C. vulgaris	1	1	7	10	No	5	0	5	24	
9	C. vulgaris	1	15	3	10	No	0	0	0.5	1	
10	C. vulgaris	1	10	7	10	No	0	1	5	1	
11	C. vulgaris	1	6	3	100	No	5	0	0.5	24	
12	C. vulgaris	1	3	3	10	Yes	5	1	0.5	1	
13	C. vulgaris	1	12	5	55	No	2.5	0.5	2.75	12.5	
14	C. vulgaris	1	8	3	10	Yes	0	0	5	1	
15	C. vulgaris	1	14	7	100	Yes	5	1	5	24	
16	C. vulgaris	2	3	3	10	Yes	5	1	0.5	1	
17	C. vulgaris	2	11	5	55	No	2.5	0.5	2.75	12.5	
18	C. vulgaris	2	1	7	10	No	5	0	5	24	
19	C. vulgaris	2	15	3	10	No	0	0	0.5	1	
20	C. vulgaris	2	6	3	100	No	5	0	0.5	24	
21	C. vulgaris	2	8	3	10	Yes	0	0	5	1	
22	C. vulgaris	2	13	5	55	No	2.5	0.5	2.75	12.5	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg/L	E_EDTA_mM	F_Dosage_g/L	G_Time_h
23	C. vulgaris	2	5	7	10	Yes	0	1	0.5	24
24	C. vulgaris	2	9	3	100	Yes	5	0	5	1
25	C. vulgaris	2	12	5	55	No	2.5	0.5	2.75	12.5
26	C. vulgaris	2	14	7	100	Yes	5	1	5	24
27	C. vulgaris	2	2	3	100	No	0	1	5	24
28	C. vulgaris	2	7	7	100	No	5	1	0.5	1
29	C. vulgaris	2	4	7	100	Yes	0	0	0.5	1
30	C. vulgaris	2	10	7	10	No	0	1	5	1
31	C. vulgaris	3	12	5	55	No	2.5	0.5	2.75	12.5
32	C. vulgaris	3	6	3	100	No	5	0	0.5	24
33	C. vulgaris	3	1	7	10	No	5	0	5	24
34	C. vulgaris	3	9	3	100	Yes	5	0	5	1
35	C. vulgaris	3	15	3	10	No	0	0	0.5	1
36	C. vulgaris	3	4	7	100	Yes	0	0	0.5	1
37	C. vulgaris	3	11	5	55	No	2.5	0.5	2.75	12.5
38	C. vulgaris	3	2	3	100	No	0	1	5	24
39	C. vulgaris	3	8	3	10	Yes	0	0	5	1
40	C. vulgaris	3	13	5	55	No	2.5	0.5	2.75	12.5
41	C. vulgaris	3	10	7	10	No	0	1	5	1
42	C. vulgaris	3	7	7	100	No	5	1	0.5	1
43	C. vulgaris	3	3	3	10	Yes	5	1	0.5	1
44	C. vulgaris	3	14	7	100	Yes	5	1	5	24

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pre_mM	D_Coagulant_pct	E_EDTA_mM	F_Dosage_g_L	G_Time_h
45	C. vulgaris	3	5	7	10	Yes	0	1	0.5	24
46	G. sulphuraria	1	14	7	100	Yes	5	1	5	24
47	G. sulphuraria	1	1	7	10	No	5	0	5	24
48	G. sulphuraria	1	7	7	100	No	5	1	0.5	1
49	G. sulphuraria	1	3	3	10	Yes	5	1	0.5	1
50	G. sulphuraria	1	11	5	55	No	2.5	0.5	2.75	12.5
51	G. sulphuraria	1	9	3	100	Yes	5	0	5	1
52	G. sulphuraria	1	2	3	100	No	0	1	5	24
53	G. sulphuraria	1	13	5	55	No	2.5	0.5	2.75	12.5
54	G. sulphuraria	1	5	7	10	Yes	0	1	0.5	24
55	G. sulphuraria	1	10	7	10	No	0	1	5	1
56	G. sulphuraria	1	4	7	100	Yes	0	0	0.5	1
57	G. sulphuraria	1	15	3	10	No	0	0	0.5	1
58	G. sulphuraria	1	6	3	100	No	5	0	0.5	24
59	G. sulphuraria	1	12	5	55	No	2.5	0.5	2.75	12.5

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_pct	E_EDTA_mM	F_Dosage_g_L	G_Time_h
60	G. sulphuraria	1	8	3	10	Yes	0	0	5	1
61	G. sulphuraria	2	12	5	55	No	2.5	0.5	2.75	12.5
62	G. sulphuraria	2	1	7	10	No	5	0	5	24
63	G. sulphuraria	2	4	7	100	Yes	0	0	0.5	1
64	G. sulphuraria	2	9	3	100	Yes	5	0	5	1
65	G. sulphuraria	2	15	3	10	No	0	0	0.5	1
66	G. sulphuraria	2	6	3	100	No	5	0	0.5	24
67	G. sulphuraria	2	11	5	55	No	2.5	0.5	2.75	12.5
68	G. sulphuraria	2	8	3	10	Yes	0	0	5	1
69	G. sulphuraria	2	3	3	10	Yes	5	1	0.5	1
70	G. sulphuraria	2	14	7	100	Yes	5	1	5	24
71	G. sulphuraria	2	5	7	10	Yes	0	1	0.5	24
72	G. sulphuraria	2	10	7	10	No	0	1	5	1
73	G. sulphuraria	2	2	3	100	No	0	1	5	24

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
74	G. sulphuraria	2	7	7	100	No	5	1	0.5	1	
75	G. sulphuraria	2	13	5	55	No	2.5	0.5	2.75	12.5	
76	G. sulphuraria	3	8	3	10	Yes	0	0	5	1	
77	G. sulphuraria	3	1	7	10	No	5	0	5	24	
78	G. sulphuraria	3	14	7	100	Yes	5	1	5	24	
79	G. sulphuraria	3	5	7	10	Yes	0	1	0.5	24	
80	G. sulphuraria	3	11	5	55	No	2.5	0.5	2.75	12.5	
81	G. sulphuraria	3	3	3	10	Yes	5	1	0.5	1	
82	G. sulphuraria	3	13	5	55	No	2.5	0.5	2.75	12.5	
83	G. sulphuraria	3	6	3	100	No	5	0	0.5	24	
84	G. sulphuraria	3	10	7	10	No	0	1	5	1	
85	G. sulphuraria	3	2	3	100	No	0	1	5	24	
86	G. sulphuraria	3	7	7	100	No	5	1	0.5	1	
87	G. sulphuraria	3	12	5	55	No	2.5	0.5	2.75	12.5	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
88	G. sulphuraria	3	15	3	10	No	0	0	0.5	1	
89	G. sulphuraria	3	9	3	100	Yes	5	0	5	1	
90	G. sulphuraria	3	4	7	100	Yes	0	0	0.5	1	
91	Desmodesmus sp.	1	7	7	100	No	5	1	0.5	1	
92	Desmodesmus sp.	1	1	7	10	No	5	0	5	24	
93	Desmodesmus sp.	1	12	5	55	No	2.5	0.5	2.75	12.5	
94	Desmodesmus sp.	1	14	7	100	Yes	5	1	5	24	
95	Desmodesmus sp.	1	3	3	10	Yes	5	1	0.5	1	
96	Desmodesmus sp.	1	10	7	10	No	0	1	5	1	
97	Desmodesmus sp.	1	5	7	10	Yes	0	1	0.5	24	
98	Desmodesmus sp.	1	13	5	55	No	2.5	0.5	2.75	12.5	
99	Desmodesmus sp.	1	8	3	10	Yes	0	0	5	1	
100	Desmodesmus sp.	1	11	5	55	No	2.5	0.5	2.75	12.5	
101	Desmodesmus sp.	1	4	7	100	Yes	0	0	0.5	1	
102	Desmodesmus sp.	1	9	3	100	Yes	5	0	5	1	
103	Desmodesmus sp.	1	6	3	100	No	5	0	0.5	24	
104	Desmodesmus sp.	1	2	3	100	No	0	1	5	24	
105	Desmodesmus sp.	1	15	3	10	No	0	0	0.5	1	
106	Desmodesmus sp.	2	1	7	10	No	5	0	5	24	
107	Desmodesmus sp.	2	14	7	100	Yes	5	1	5	24	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
108	Desmodesmus sp.	2	8	3	10	Yes	0	0	5	1	
109	Desmodesmus sp.	2	4	7	100	Yes	0	0	0.5	1	
110	Desmodesmus sp.	2	11	5	55	No	2.5	0.5	2.75	12.5	
111	Desmodesmus sp.	2	7	7	100	No	5	1	0.5	1	
112	Desmodesmus sp.	2	13	5	55	No	2.5	0.5	2.75	12.5	
113	Desmodesmus sp.	2	2	3	100	No	0	1	5	24	
114	Desmodesmus sp.	2	10	7	10	No	0	1	5	1	
115	Desmodesmus sp.	2	5	7	10	Yes	0	1	0.5	24	
116	Desmodesmus sp.	2	15	3	10	No	0	0	0.5	1	
117	Desmodesmus sp.	2	6	3	100	No	5	0	0.5	24	
118	Desmodesmus sp.	2	12	5	55	No	2.5	0.5	2.75	12.5	
119	Desmodesmus sp.	2	3	3	10	Yes	5	1	0.5	1	
120	Desmodesmus sp.	2	9	3	100	Yes	5	0	5	1	
121	Desmodesmus sp.	3	11	5	55	No	2.5	0.5	2.75	12.5	
122	Desmodesmus sp.	3	2	3	100	No	0	1	5	24	
123	Desmodesmus sp.	3	8	3	10	Yes	0	0	5	1	
124	Desmodesmus sp.	3	15	3	10	No	0	0	0.5	1	
125	Desmodesmus sp.	3	4	7	100	Yes	0	0	0.5	1	
126	Desmodesmus sp.	3	10	7	10	No	0	1	5	1	
127	Desmodesmus sp.	3	13	5	55	No	2.5	0.5	2.75	12.5	
128	Desmodesmus sp.	3	1	7	10	No	5	0	5	24	
129	Desmodesmus sp.	3	7	7	100	No	5	1	0.5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_EDTA_mM	F_Dosage_g/L	G_Time_h
130	Desmodesmus sp.	3	12	5	55	No	2.5	0.5	2.75	12.5
131	Desmodesmus sp.	3	5	7	10	Yes	0	1	0.5	24
132	Desmodesmus sp.	3	14	7	100	Yes	5	1	5	24
133	Desmodesmus sp.	3	6	3	100	No	5	0	0.5	24
134	Desmodesmus sp.	3	9	3	100	Yes	5	0	5	1
135	Desmodesmus sp.	3	3	3	10	Yes	5	1	0.5	1
136	A. platensis	1	11	5	55	No	2.5	0.5	2.75	12.5
137	A. platensis	1	1	7	10	No	5	0	5	24
138	A. platensis	1	6	3	100	No	5	0	0.5	24
139	A. platensis	1	14	7	100	Yes	5	1	5	24
140	A. platensis	1	3	3	10	Yes	5	1	0.5	1
141	A. platensis	1	9	3	100	Yes	5	0	5	1
142	A. platensis	1	12	5	55	No	2.5	0.5	2.75	12.5
143	A. platensis	1	5	7	10	Yes	0	1	0.5	24
144	A. platensis	1	2	3	100	No	0	1	5	24
145	A. platensis	1	15	3	10	No	0	0	0.5	1

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mM	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
146	A. platensis	1	7	7	100	No	5	1	0.5	1	
147	A. platensis	1	13	5	55	No	2.5	0.5	2.75	12.5	
148	A. platensis	1	10	7	10	No	0	1	5	1	
149	A. platensis	1	4	7	100	Yes	0	0	0.5	1	
150	A. platensis	1	8	3	10	Yes	0	0	5	1	
151	A. platensis	2	13	5	55	No	2.5	0.5	2.75	12.5	
152	A. platensis	2	7	7	100	No	5	1	0.5	1	
153	A. platensis	2	1	7	10	No	5	0	5	24	
154	A. platensis	2	10	7	10	No	0	1	5	1	
155	A. platensis	2	4	7	100	Yes	0	0	0.5	1	
156	A. platensis	2	12	5	55	No	2.5	0.5	2.75	12.5	
157	A. platensis	2	15	3	10	No	0	0	0.5	1	
158	A. platensis	2	3	3	10	Yes	5	1	0.5	1	
159	A. platensis	2	8	3	10	Yes	0	0	5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg_L	E_EDTA_mM	F_Dosage_g_L	G_Time_h
160	A. platensis	2	5	7	10	Yes	0	1	0.5	24
161	A. platensis	2	14	7	100	Yes	5	1	5	24
162	A. platensis	2	2	3	100	No	0	1	5	24
163	A. platensis	2	9	3	100	Yes	5	0	5	1
164	A. platensis	2	6	3	100	No	5	0	0.5	24
165	A. platensis	2	11	5	55	No	2.5	0.5	2.75	12.5
166	A. platensis	3	14	7	100	Yes	5	1	5	24
167	A. platensis	3	7	7	100	No	5	1	0.5	1
168	A. platensis	3	1	7	10	No	5	0	5	24
169	A. platensis	3	10	7	10	No	0	1	5	1
170	A. platensis	3	4	7	100	Yes	0	0	0.5	1
171	A. platensis	3	12	5	55	No	2.5	0.5	2.75	12.5
172	A. platensis	3	15	3	10	No	0	0	0.5	1
173	A. platensis	3	3	3	10	Yes	5	1	0.5	1

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_pct	E_EDTA_mM	F_Dosage_g_L	G_Time_h
174	A. platensis	3	8	3	10	Yes	0	0	5	1
175	A. platensis	3	5	7	10	Yes	0	1	0.5	24
176	A. platensis	3	11	5	55	No	2.5	0.5	2.75	12.5
177	A. platensis	3	2	3	100	No	0	1	5	24
178	A. platensis	3	9	3	100	Yes	5	0	5	1
179	A. platensis	3	6	3	100	No	5	0	0.5	24
180	A. platensis	3	13	5	55	No	2.5	0.5	2.75	12.5
181	Nostoc sp.	1	14	7	100	Yes	5	1	5	24
182	Nostoc sp.	1	7	7	100	No	5	1	0.5	1
183	Nostoc sp.	1	2	3	100	No	0	1	5	24
184	Nostoc sp.	1	9	3	100	Yes	5	0	5	1
185	Nostoc sp.	1	4	7	100	Yes	0	0	0.5	1
186	Nostoc sp.	1	11	5	55	No	2.5	0.5	2.75	12.5
187	Nostoc sp.	1	6	3	100	No	5	0	0.5	24
188	Nostoc sp.	1	12	5	55	No	2.5	0.5	2.75	12.5
189	Nostoc sp.	1	1	7	10	No	5	0	5	24
190	Nostoc sp.	1	15	3	10	No	0	0	0.5	1
191	Nostoc sp.	1	10	7	10	No	0	1	5	1

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg/L	E_EDTA_mM	F_Dosage_g/L	G_Time_h
192	Nostoc sp.	1	5	7	10	Yes	0	1	0.5	24
193	Nostoc sp.	1	13	5	55	No	2.5	0.5	2.75	12.5
194	Nostoc sp.	1	3	3	10	Yes	5	1	0.5	1
195	Nostoc sp.	1	8	3	10	Yes	0	0	5	1
196	Nostoc sp.	2	10	7	10	No	0	1	5	1
197	Nostoc sp.	2	4	7	100	Yes	0	0	0.5	1
198	Nostoc sp.	2	13	5	55	No	2.5	0.5	2.75	12.5
199	Nostoc sp.	2	1	7	10	No	5	0	5	24
200	Nostoc sp.	2	8	3	10	Yes	0	0	5	1
201	Nostoc sp.	2	15	3	10	No	0	0	0.5	1
202	Nostoc sp.	2	6	3	100	No	5	0	0.5	24
203	Nostoc sp.	2	12	5	55	No	2.5	0.5	2.75	12.5
204	Nostoc sp.	2	3	3	10	Yes	5	1	0.5	1
205	Nostoc sp.	2	9	3	100	Yes	5	0	5	1
206	Nostoc sp.	2	5	7	10	Yes	0	1	0.5	24
207	Nostoc sp.	2	2	3	100	No	0	1	5	24
208	Nostoc sp.	2	14	7	100	Yes	5	1	5	24
209	Nostoc sp.	2	11	5	55	No	2.5	0.5	2.75	12.5
210	Nostoc sp.	2	7	7	100	No	5	1	0.5	1
211	Nostoc sp.	3	1	7	10	No	5	0	5	24
212	Nostoc sp.	3	11	5	55	No	2.5	0.5	2.75	12.5
213	Nostoc sp.	3	5	7	10	Yes	0	1	0.5	24

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_EDTA_mM	F_Dosage_g_L	G_Time_h
214	Nostoc sp.	3	13	5	55	No	2.5	0.5	2.75	12.5
215	Nostoc sp.	3	2	3	100	No	0	1	5	24
216	Nostoc sp.	3	9	3	100	Yes	5	0	5	1
217	Nostoc sp.	3	15	3	10	No	0	0	0.5	1
218	Nostoc sp.	3	6	3	100	No	5	0	0.5	24
219	Nostoc sp.	3	12	5	55	No	2.5	0.5	2.75	12.5
220	Nostoc sp.	3	4	7	100	Yes	0	0	0.5	1
221	Nostoc sp.	3	10	7	10	No	0	1	5	1
222	Nostoc sp.	3	14	7	100	Yes	5	1	5	24
223	Nostoc sp.	3	3	3	10	Yes	5	1	0.5	1
224	Nostoc sp.	3	8	3	10	Yes	0	0	5	1
225	Nostoc sp.	3	7	7	100	No	5	1	0.5	1
226	Coccomyxa sp.	3	15	3	10	No	0	0	0.5	1
227	Coccomyxa sp.	3	8	3	10	Yes	0	0	5	1
228	Coccomyxa sp.	3	3	3	10	Yes	5	1	0.5	1
229	Coccomyxa sp.	3	11	5	55	No	2.5	0.5	2.75	12.5
230	Coccomyxa sp.	3	6	3	100	No	5	0	0.5	24
231	Coccomyxa sp.	3	1	7	10	No	5	0	5	24
232	Coccomyxa sp.	3	13	5	55	No	2.5	0.5	2.75	12.5
233	Coccomyxa sp.	3	4	7	100	Yes	0	0	0.5	1
234	Coccomyxa sp.	3	9	3	100	Yes	5	0	5	1
235	Coccomyxa sp.	3	12	5	55	No	2.5	0.5	2.75	12.5

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g_L	H_Time_h
236	Coccomyxa sp.	1	2	3	100	No	0	1	5	24	
237	Coccomyxa sp.	1	14	7	100	Yes	5	1	5	24	
238	Coccomyxa sp.	1	7	7	100	No	5	1	0.5	1	
239	Coccomyxa sp.	1	5	7	10	Yes	0	1	0.5	24	
240	Coccomyxa sp.	1	10	7	10	No	0	1	5	1	
241	Coccomyxa sp.	2	2	3	100	No	0	1	5	24	
242	Coccomyxa sp.	2	13	5	55	No	2.5	0.5	2.75	12.5	
243	Coccomyxa sp.	2	7	7	100	No	5	1	0.5	1	
244	Coccomyxa sp.	2	4	7	100	Yes	0	0	0.5	1	
245	Coccomyxa sp.	2	15	3	10	No	0	0	0.5	1	
246	Coccomyxa sp.	2	9	3	100	Yes	5	0	5	1	
247	Coccomyxa sp.	2	11	5	55	No	2.5	0.5	2.75	12.5	
248	Coccomyxa sp.	2	1	7	10	No	5	0	5	24	
249	Coccomyxa sp.	2	6	3	100	No	5	0	0.5	24	
250	Coccomyxa sp.	2	14	7	100	Yes	5	1	5	24	
251	Coccomyxa sp.	2	8	3	10	Yes	0	0	5	1	
252	Coccomyxa sp.	2	3	3	10	Yes	5	1	0.5	1	
253	Coccomyxa sp.	2	10	7	10	No	0	1	5	1	
254	Coccomyxa sp.	2	5	7	10	Yes	0	1	0.5	24	
255	Coccomyxa sp.	2	12	5	55	No	2.5	0.5	2.75	12.5	
256	Coccomyxa sp.	3	7	7	100	No	5	1	0.5	1	
257	Coccomyxa sp.	3	13	5	55	No	2.5	0.5	2.75	12.5	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg/L	E_EDTA_mM	F_Dosage_g/L	G_Time_h
258	Coccomyxa sp.	3	2	3	100	No	0	1	5	24
259	Coccomyxa sp.	3	10	7	10	No	0	1	5	1
260	Coccomyxa sp.	3	5	7	10	Yes	0	1	0.5	24
261	Coccomyxa sp.	3	12	5	55	No	2.5	0.5	2.75	12.5
262	Coccomyxa sp.	3	3	3	10	Yes	5	1	0.5	1
263	Coccomyxa sp.	3	15	3	10	No	0	0	0.5	1
264	Coccomyxa sp.	3	8	3	10	Yes	0	0	5	1
265	Coccomyxa sp.	3	1	7	10	No	5	0	5	24
266	Coccomyxa sp.	3	6	3	100	No	5	0	0.5	24
267	Coccomyxa sp.	3	14	7	100	Yes	5	1	5	24
268	Coccomyxa sp.	3	9	3	100	Yes	5	0	5	1
269	Coccomyxa sp.	3	4	7	100	Yes	0	0	0.5	1
270	Coccomyxa sp.	3	11	5	55	No	2.5	0.5	2.75	12.5

MCDA notes

A weighted-summation approach is described as one of the simplest and most widely applied techniques of MCA/MCDA, and a typical MCDA process includes problem formulation, evaluation, weighting, calculation (e.g., in Excel), and sensitivity analysis before recommendations.

Chapter 6

This chapter compiles the Stage-2 optimization (RSM) framework, ICP-MS methods/QC, and the program timeline.

RSM design

Response surface methodology is described as a combination of mathematical and statistical techniques used for developing, improving, and optimizing processes. A Box–Behnken design can reduce the number of runs; BBD versus CCD run counts for 3 and 4 factors are stated in the provided design note (3 factors: BBD 15, CCD 20; 4 factors: BBD 27, CCD 31).

ICP-MS methods and QC

EPA Method 200.8 is identified as "Determination of Trace Elements in Waters and Wastes by ICP-MS," and complementary use of EPA 6020B (ICP-MS for solid waste extracts) and EPA 6010D (ICP-OES for higher-concentration matrices) is described in an applied protocol extract. Sample preservation guidance includes acidifying to pH < 2 (e.g., ~2 mL/L HNO3) and indicates that properly acid-preserved samples can be held up to six months; for boron, separate aliquots in polypropylene are recommended and borosilicate glassware should be avoided. Digestion requirements for total recoverable analytes are described, and boron memory effect mitigation includes nitric acid plus mannitol rinses and extended washout, with quartz/PTFE labware recommended for boron work.

Timeline

Phase anchors for a 24-month program include setup/procurement (M1–M2), leachate collection/characterization (M2–M4), DSD screening experiments (M4–M8), RSM optimization (M9–M13), and 30-day continuous pilot operation (M16–M20).

Chapter 7

This chapter reproduces the procurement-oriented content for analytical standards, acids, culture media, and sampling consumables.

Analytical standards and internal standards

Example multi-element REE standards are listed as containing 50 mg/L of REE suite elements in 2% nitric acid, and rare-earth calibration standards are available from vendors such as Inorganic Ventures (e.g., 10 ppm rare earth ICP standard, part CMS-1-125ML, $163). Example internal standard mix products include an Agilent internal standard mix (part 5188-6525, list price $282).

Trace-metal-grade acids

Examples of trace-metal-grade acids include Fisher TraceMetal nitric acid (catalog A509P500, $225) and hydrochloric acid (catalog A508P500, $197.70).

Culture media and filters

BG-11 medium kits are described as providing components to prepare 100 L of BG-11 medium and are priced at $1,500. PVDF syringe filters are available in 0.45 µm formats (e.g., Whatman GD/X, part 6900-2504, $276.95 per 50-pack).

Chapter 8

This chapter reproduces the data-pipeline content that links ICP-MS exports to validated datasets and compliance dashboards.

Agilent MassHunter export notes include that "CPS Data to CSV" generates raw count data per second and that "Tabulate Chart after Sample Acquisition" can be used to export .csv in CPS format; consistent export formats are highlighted as important for downstream import processes. EPA's TADA toolkit is described as providing tools to help visualize wastewater monitoring data and discharge limits, and includes a Shiny application pathway; Shiny is

described as a free/open-source tool for customizable interactive web apps using R and supports tabular inputs such as CSV/XLSX.

Chapter 9

This chapter reproduces the funding-opportunity and grant narrative snippets.

NSF Environmental Engineering is described as supporting research focused on reducing pollution impacts, including remediation and engineered processes; wastewater treatment via biogeochemical processes is highlighted, and biosorption proposals are guided to emphasize methods that transform pollutants and enhance resource recovery. EPA STAR is described as supporting innovative water research, including wastewater treatment technologies and recovery of critical minerals like rare earth elements. DOE expects up to 10 awards between $500k and $2.5M with a minimum 20% cost share under one critical-minerals funding notice, with a stated submission deadline (11/26/2024, 5 PM ET).

Chapter 10

This chapter reproduces the research-strategy oriented context used in the proposal narrative.

NdFeB magnets are described as critical components in generators and motors used in wind turbines and EVs, and the permanent magnet market is projected to grow from USD 35.0B (2022) to USD 78.9B (2032) with NdFeB projected at ~45% share by 2030 in one projection. NdFeB magnet chemical composition ranges include 53–62 wt% Fe, 24–29 wt% Nd, 0.85–0.96 wt% B, and 0.54–3.6 wt% Co; acid leachate compositions can be very high in Fe and Nd (e.g., Fe tens of g/L; Nd tens of g/L) with B and Co also present under reported leaching conditions. Multiple sources state no peer-reviewed study has tested microalgae against the full chemical matrix of NdFeB manufacturing leachate, consistent with the evidence-gap framing in earlier chapters.

Chapter 11

This chapter reproduces the pilot-equipment procurement evidence.

Pilot-scale bioreactor platforms are described as having working volumes of 10–50 L for process development, and photobioreactor systems list vessel capacities up to 100 L; a price commentary notes a 100-L bioreactor may cost $50,000–$100,000. Peristaltic pumps have reported flow ranges up to 3400 mL/min; Watson-Marlow 120 series pump systems have listed prices (e.g., $3,310–$3,530 and $4,630–$4,770) and tubing products have listed price ranges; online sensors/controllers have listed starting prices and DO probe listings include "from $1,218." Data loggers have listed prices (e.g., CR6 $1,900; CR1000 $1,495) and DAQ devices list sale prices (e.g., LabJack T7 $550).

Chapter 12

This chapter reproduces the synthetic leachate preparation SOP elements, emphasizing stability and contamination control.

Leachate compositions reported for NdFeB-related materials can be strongly Fe- and Nd-rich (e.g., Fe 31.6 g/L; Nd 15.1 g/L; B 0.535 g/L; Co 1.70 g/L under one condition) and can reach higher concentrations under other conditions; a separate major-element composition example reports iron 84.7 g/L, neodymium 31.7 g/L, dysprosium 11.5 g/L. The SOP emphasizes keeping preparation strongly acidic because precipitation phenomena occur as pH rises; Fe(III) precipitation is described as typically occurring around pH ~3.5–4.0 and high pH can lead to REE hydroxide formation, so dissolution and storage at low pH is needed to maintain solubility and avoid confounding precipitation

with biosorption. For boron contamination control and boron memory effects in analysis, multiple method notes emphasize avoiding borosilicate glassware (use polypropylene/quartz/PTFE) and using nitric acid + mannitol rinse sequences with extended washout times when analyzing boron-containing samples.

Microalgae Bioabsorption of NdFeB Magnet Leachate Contaminants for Wastewater Regulatory Compliance

Summary

The extracted evidence base contains strong laboratory proof that microalgae (and cyanobacteria) can sorb or bioaccumulate rare-earth ions (e.g., Nd, La, Sm, Eu) rapidly under controlled conditions, often with pseudo-second-order kinetics and Langmuir-type isotherms, but the dataset is thin for the central compliance question: achieving discharge compliance from **real** NdFeB manufacturing leachates that retain the full Fe–B–Co–Ni–Cu–Al matrix and process organics. For the only clearly NdFeB-derived microalgae biosorption study in the table (Chlorella vulgaris on magnet-derived leachate), the leachate was selectively prepared such that Fe, B, Dy, and Co were below detection limits; thus the study demonstrates high Nd uptake but does not resolve Fe/B interference or multi-contaminant compliance.

Across the extracted multi-contaminant microalgae studies that do report boron behavior, boron is consistently the weakest-performing contaminant in mixed-metal systems (e.g., B removal dropping to ~0.2% in multimetal conditions), implying boron will likely be rate-limiting for regulatory compliance unless a separate boron-focused polishing step is included.

Part 1 Chinese literature and patents

1.1 Landscape

The Chinese-language and China-affiliated sources retrieved in this workflow are dominated by (i) general microalgae wastewater treatment studies (nutrients/COD), (ii) magnetic-field or magnetic-nanoparticle enhancement and harvesting studies, and (iii) magnet-recycling patents that focus on chemical/electrochemical recovery pathways rather than biological polishing for discharge compliance.

Within the retrieved Chinese or China-relevant corpus, **direct Chinese publications on microalgae biosorption/bioabsorption of full-composition NdFeB manufacturing leachates** (Fe-dominant matrices plus REEs plus boron and process organics) were not present in the provided extracts; China-related sources instead discuss REE mining wastewater (e.g., La in high-ammonia matrices) or general heavy-metal remediation rather than NdFeB shop-floor leachates.

Institutions and locations explicitly identified in the retrieved China-affiliated sources include Jiangsu University (Tribonema sp. maintained there) and the Institute of Hydrobiology, Chinese Academy of Sciences (FACHB culture collection providing Chlorella sorokiniana).

1.2 Wastewater characterization data from Chinese sources

The retrieved Chinese sources do **not** provide a composition table for a real NdFeB manufacturing wastewater (mg/L of Fe, B, Nd, Dy, Co, Cu, Ni, Al, plus COD/O&G and chelators). Instead, the closest China-relevant quantitative wastewater characterization in the provided extracts is for rare-earth mining wastewater (La) and high-ammonia simulated matrices: a simulated REE mining wastewater is described with 1060 mg L−1 NH4+-N and La tested across 0−1000 mg L−1, while background values for southern China ion-adsorption ore leachates are described as "a few to several tens of mg L−1" La.

For NdFeB composition (solid feedstock, not wastewater), Chinese patents provide typical alloy compositions that indicate which elements are expected in resulting wastes/leachates. Examples include waste magnet steel composi-

tions listing Nd, Dy, Pr, Fe, Co, Cu, Al, and B in wt.% (e.g., Nd 22.5%, Dy 0.8%, Pr 7.1%, Fe 67.8%, Co 0.7%, Cu 0.1%, B 1%; and Nd 25.4%, Dy 5%, Fe 67.2%, Co 1.1%, Cu 0.2%, Al 0.1%, B 1%).

Because these are alloy compositions rather than aqueous measurements, they should be used only as a guide for expected contaminant suite, not for discharge/compliance calculations.

1.3 China regulation GB 26451

Within the retrieved regulatory extracts, GB 26451-2011 is explicitly identified as the "Emission Standards of Pollutants from Rare Earths Industry."

A secondary source citing GB 26451 provides one "Indirect discharge standards" row with numeric values presented as pH 6–9, NH4+-N 50 mg/L, COD 100 mg/L, As 0.1 mg/L, and Pb 0.2 mg/L.

The extracted GB 26451 snippets in this workflow do not provide numeric limits for boron, iron, cobalt, copper, nickel, aluminum, or rare earth elements (Nd, Dy, total REE), so GB 26451 cannot be used here to construct a complete NdFeB contaminant compliance matrix from the present extraction alone.

1.4 Patents and IP signals

The retrieved China patent set contains multiple NdFeB-related patents, but the patents identified here primarily concern recycling/recovery (chemical or electrochemical leaching and precipitation) rather than a biological microalgae-based polishing step for wastewater discharge compliance. For example, CN113957250B describes an electrochemical leaching loop with preferred pH 3.0–5.0 and electrolyte components including ammonium iron sulfate, citric acid, and boric acid, emphasizing leachate and rinse recycling to minimize wastewater.

Another patent (CN103789549A) targets recovery of Nd, Pr, Dy, Co, and Fe from NdFeB oil sludge using pH adjustment (e.g., "adjust the pH=8–9 under the OH–NH3 system" for combination precipitation), again reflecting chemical treatment rather than algae biosorption.

A separate Chinese patent (CN109896712A) is notable because it explicitly references dosing an "ABS microalgae/microbe agent" (ABS⊠⊠⊠⊠⊠) and combines it with magnetic separation, reporting rapid treatment (3–6 minutes) and high removals for SS, phosphorus/TP, and moderate COD/BOD removal in hospital wastewater; while not NdFeB-specific, it signals Chinese IP activity combining algae/biomass with magnetic separation hardware that could be adapted for industrial effluents.

Part 2 Regulatory compliance gap matrix

2.1 Discharge limits

The table below compiles numeric limits that were explicitly extractable from the provided regulatory snippets. Many NdFeB-relevant pollutants (notably Nd, Dy/total REE) are not covered in these excerpts; where limits are not present in the supplied text, cells are marked as "Not in extracted text."

Parameter	EU BAT-AEL or BREF values available in extracts	US EPA metal finishing values available in extracts	China GB 26451 values available in extracts
pH	Not in extracted text for NFM BAT-AEL; STM BREF excerpt provides ranges for treated effluent parameters but pH not shown in extracted lines	pH appears as a regulated parameter in Part 433 tables but numeric range is not included in the extracted row that only shows “pH	(1)
COD	Ferrous metals BAT conclusions: COD 30–190 mg/L (BAT-AEL range shown in extract); STM BREF excerpt: COD 100–500 mg/L	Not in extracted text for Part 433 COD (Part 433 extracts provide metals, TSS, O&G, pH line but not COD)	COD 100 mg/L (indirect discharge standards row)
TSS	Ferrous metals BAT conclusions: TSS 5–130 mg/L (BAT-AEL range shown in extract); STM BREF excerpt: suspended solids 5–30 mg/L	TSS 60 mg/L (daily max) and 31 mg/L (monthly avg)	Not in extracted text
Oil and grease	Ferrous metals BAT conclusions: hydrocarbon oil index 0.5–14 mg/L; STM BREF excerpt: total hydrocarbons 1–5 mg/L	Oil & Grease 52 mg/L (daily max) and 26 mg/L (monthly avg)	Not in extracted text
Fe	Ferrous metals BAT conclusions: Fe 1–5 mg/L; STM BREF excerpt: Fe 0.1–5 mg/L	Not in extracted text (Part 433 excerpts list Cu/Ni/Cr/TSS/O&G but not Fe numeric)	Not in extracted text
B	STM BREF excerpt: B 1–10 mg/L	Not in extracted text (Part 433 excerpt does not provide B numeric)	Not in extracted text (GB row shows NH4-N/COD/As/Pb only)
Cu	NFM BAT conclusions (Table 2 extract): Cu 0.05–0.5 mg/L (daily average BAT-AEL shown in extract); STM BREF excerpt: Cu 0.2–2.0 mg/L	Cu 3.38 mg/L (daily max) and 2.07 mg/L (monthly avg)	Not in extracted text

Parameter	EU BAT-AEL or BREF values available in extracts	US EPA metal finishing values available in extracts	China GB 26451 values available in extracts
Ni	NFM BAT conclusions (Table 2 extract): Ni 0.5 mg/L (daily average BAT-AEL shown in extract); Ferrous metals BAT conclusions: Ni 0.01–10 mg/L (range shown in extract); STM BREF excerpt: Ni 0.2–2.0 mg/L	Ni 3.98 mg/L (daily max) and 2.38 mg/L (monthly avg)	Not in extracted text
Co	NFM BAT conclusions (Table 2 extract): Co 0.1–0.5 mg/L (daily average BAT-AEL shown in extract)	Not in extracted text (Part 433 excerpt does not provide Co numeric)	Not in extracted text
Al	STM BREF excerpt: Al 1–10 mg/L	Not in extracted text	Not in extracted text
Cr	NFM BAT conclusions (Table 2 extract): total Cr 0.2 mg/L (daily average BAT-AEL shown in extract); Ferrous metals BAT conclusions: Cr 0.01–10 mg/L (range shown in extract)	total Cr 2.77 mg/L (daily max) and 1.71 mg/L (monthly avg)	Not in extracted text
Nd	Not in extracted text	Not in extracted text	Not in extracted text
Dy	Not in extracted text	Not in extracted text	Not in extracted text

2.2 Best microalgae performance versus limits

The table below uses only the "best available" values **explicitly present in the extracted evidence_table** (often % removal or final mg/L), and compares them with the available regulatory numbers above. Many parameters lack final concentrations or were never measured against discharge limits in the extracted studies; those gaps are explicitly retained as "Not reported."

Parameter	Best extracted microalgae result	Closest extracted limit value	Implication for compliance
Co	Chlorella vulgaris reduced Co2+ from 63.6 mg/L to 1.01 mg/L after 10 h in a photobioreactor (98.78% removal).	EU NFM BAT-AEL for Co 0.1–0.5 mg/L (daily average).	Even strong Co removal may still miss stricter EU BAT-AEL unless further polishing reduces Co below 0.5 mg/L (here residual 1.01 mg/L).
B	In multimetal systems, boron removal dropped drastically (e.g., down to ~0.2% reported for Chlamydomonas reinhardtii in multimetallic solutions; B identified as lowest yield ratio).	STM BREF excerpt B 1–10 mg/L.	Evidence suggests boron is difficult to remove biologically in mixed-metal matrices; compliance at 1–10 mg/L likely requires dedicated boron treatment beyond microalgae alone (evidence indicates very low removal in mixtures).
Cr	In real industrial water, Cr reduced from 135.56 to 45.67 mg/L by Chlorella vulgaris (dried biomass).	US EPA Part 433 total Cr 2.77 mg/L daily / 1.71 mg/L monthly.	Large compliance gap remains: residual 45.67 mg/L greatly exceeds typical metal-finishing limits, implying microalgae biosorption alone is insufficient at these Cr loads without additional treatment stages.
Nd	Chlorella vulgaris biosorbed Nd from magnet-derived leachate, reaching high uptake capacities (qexp ~157.4 mg/g at pH 5 and 250 mg/L Nd initial).	Nd numeric wastewater limits are not in extracted regulatory text.	Performance is promising for Nd uptake, but compliance cannot be assessed against numeric discharge limits from the provided regulatory extracts, and the leachate tested was Fe/B/Dy/Co below detection limits due to selective leaching.

Parameter	Best extracted microalgae result	Closest extracted limit value	Implication for compliance
Fe	Evidence includes partial Fe removal in a summarized study (Microcystis aeruginosa removal 54.14% from 0.35 mg/L over 4 days, reported in a review table).	Ferrous metals BAT conclusions Fe 1–5 mg/L; STM BREF excerpt Fe 0.1–5 mg/L.	The extracted evidence does not yet demonstrate Fe compliance performance at Fe levels relevant to NdFeB leachates (often Fe-dominant); moreover, key NdFeB biosorption experiments intentionally removed Fe from the test solution, leaving the core interference/compliance question open.
Cu, Ni, Al, O&G, COD, TSS	Not reported as final concentrations in the extracted microalgae NdFeB-relevant studies (Cu/Ni/Al primarily appear as co-metals in synthetic or non-NdFeB matrices, with results reported as % removal or capacities rather than final effluent concentrations).	Numeric limits exist for Cu/Ni and TSS/O&G under Part 433; ranges exist for Cu/Ni/Al/COD/SS under STM BREF excerpt.	Compliance feasibility cannot be quantified from the present extracted evidence because final concentrations in mg/L (post-treatment) are missing for these parameters in the NdFeB-context studies.

2.3 Feasibility rating

Because the extracted evidence is largely laboratory-scale and often not benchmarked against discharge limits, the feasibility rating below is conservative and explicitly tied to what is (and is not) supported in the provided extracts.

Contaminant class	Feasibility rating	Evidence basis in extracted corpus
REEs in clean or lightly contaminated matrices	YELLOW	High REE biosorption and rapid kinetics are demonstrated in synthetic media and some magnet-derived solutions, but compliance endpoints and multi-contaminant Fe/B/organics matrices are not demonstrated in the extracted studies.

Contaminant class	Feasibility rating	Evidence basis in extracted corpus
Co	YELLOW	Strong Co removal by Chlorella vulgaris is shown in synthetic Co solution, but residual 1.01 mg/L still exceeds the EU NFM BAT-AEL range of 0.1–0.5 mg/L and real wastewater matrices were explicitly identified as a needed next step.
Cr in industrial wastewater	RED	Real industrial wastewater Cr reduction to 45.67 mg/L still far above typical metal-finishing Cr limits (1.71 mg/L monthly average), implying algae biosorption alone is insufficient at high Cr loads in the extracted evidence.
Boron	RED	Mixed-metal studies show very low boron removal (down to ~0.2%) and boron identified as the lowest-performance contaminant under co-metal interference; this makes boron a likely compliance bottleneck under STM-like boron limits (1–10 mg/L) unless additional processes are used.
Fe-dominant matrices typical of NdFeB leachates	BLACK	The magnet-derived Nd biosorption study removed Fe from the prepared leachate (Fe below detection limit), so Fe competition and Fe removal in Fe-dominant NdFeB leachates is not demonstrated in the extracted evidence.
Cooling fluids, oils, chelators (EDTA/NTA), surfactants	BLACK	No extracted experiment in the evidence_table reports biosorption performance in the presence of EDTA/NTA, metalworking fluids, or emulsified oils; the review literature emphasizes that site-specific wastewater impurities require further testing, but does not provide these data in the excerpts here.

2.4 Rate-limiting contaminants

Ranking is limited to what can be justified from the extracted data. Boron emerges as the clearest rate-limiting contaminant under co-contaminant interference because boron removal collapses in multimetal systems and is explicitly

identified as the lowest-performing element in those experiments.

A second likely bottleneck is "matrix realism": the best NdFeB-specific biosorption study intentionally created a Fe-free magnet-derived Nd solution (Fe/B/Dy/Co below detection limit), which avoids the industrially dominant Fe interference that is expected in true NdFeB leachates.

Part 3 Experimental protocol

3.1 Characterization

A compliance-oriented program must start with stream-resolved characterization of real NdFeB manufacturing wastewater and leachates, because reviews emphasize the need for tests "using wastewater samples with varied impurities under conditions similar to definite wastewater sites."

Analytical scope should include (i) metals and metalloids expected from NdFeB alloys and plating (Fe, Nd, Pr, Dy, Co, Ni, Cu, Al, B), (ii) bulk parameters used in regulatory limits (COD, TSS, oil/hydrocarbon index), and (iii) chelators and surfactants where present, because multi-contaminant matrices and "the presence of other ions" are repeatedly highlighted as controlling algal uptake performance.

A practical detection-limit constraint is that studies measuring complex mixtures can reach ICP-MS limits of quantification around 0.1 μg/L with good replicate precision, indicating that sub-μg/L REE endpoints are technically measurable if methods are chosen appropriately.

3.2 Factorial screening

Screening should explicitly decouple (a) biosorption conditions from (b) growth conditions when phosphate is present, because nutrient media phosphate can limit REE solubility and cause phosphate precipitation that masks true biosorption; one study concludes biosorption should occur separately from biomass growth to avoid REE phosphate precipitation.

The screening matrix should include pH as a primary factor because Nd biosorption on Chlorella vulgaris shows an optimum around pH 5 with high uptake capacity, and because Fe behavior changes with pH via precipitation, which can change competition and apparent removal mechanisms.

A recommended initial screening design consistent with the user's requested structure is:

- Species panel: Chlorella vulgaris (high Nd uptake from magnet-derived solutions), Galdieria sulphuraria (acidophilic REE uptake but phosphate-limited under growth conditions), Desmodesmus/Scenedesmus spp. (commonly used in wastewater), and a cyanobacterium such as Nostoc sp. for high REE adsorption capacities in screening studies.
- Biomass forms: living (to evaluate bioaccumulation), dried/oven-dried (to evaluate "pure" biosorption), and immobilized beads (because immobilization is repeatedly used in REE-removal literature and is a practical route for column operation).
- Leachate fractions: 10%, 25%, 50%, 100% real leachate, because real-matrix toxicity is a recognized barrier and requires explicit tolerance testing rather than extrapolation from clean solutions.
- pH range: at minimum 2–7; this is motivated by (i) acidic leachates being common and (ii) multiple studies reporting strong pH dependence, including sharp performance loss at very low pH (e.g., no La biosorption at pH 1.5) and increased REE biosorption at pH 5 in multi-metal mixes.

- With and without Fe pre-removal: use a split-stream strategy because high-Fe matrices are expected in real NdFeB leachates, and the existing NdFeB biosorption study avoided Fe via selective leaching, leaving Fe interference unresolved; therefore Fe-precipitation pretreatment should be evaluated as a realistic prerequisite for biosorption polishing.

The response variables must be **residual concentrations** (mg/L) of each regulated parameter rather than % removal, because the regulatory endpoints in the extracted regulations are concentration-based (mg/L).

3.3 Optimization

Optimization should target rapid kinetics where possible because multiple biosorption studies show fast initial uptake (minutes to tens of minutes), and such kinetics support short-contact polishing stages if capacity and selectivity remain adequate in real matrices.

However, continuous-flow performance must be demonstrated explicitly because slow kinetics can reduce capacity in continuous systems relative to batch, as reported for Nd sorption onto Chlorella vulgaris in batch versus continuous sorption systems.

A compliance-driven optimization plan should therefore include:

- Dosage optimization (e.g., 0.5–3 g/L dried biomass was used in the NdFeB-derived Nd biosorption study), coupled with staged contactor design to manage breakthrough and allow operational flexibility.
- Multi-stage cascade tests (2–4 stages) designed to reach the strictest metal limit among the suite, recognizing that different ions can replace REEs on binding sites in mixed-metal solutions at higher concentrations.

3.4 Pilot validation

The extracted literature repeatedly notes the scarcity of pilot-scale algal bioremediation studies, so a 50–100 L continuous pilot with daily monitoring against numeric limits is required to support any compliance claim.

Pilot testing must include (i) stability testing under variable wastewater composition and (ii) a realistic harvesting/solids separation strategy, because multiple Chinese sources emphasize magnetic separation and magnetic flocculation as practical enablers for biomass separation and compact reactors.

3.5 Design constraints

Three design constraints are strongly implied by the extracted evidence. First, phosphate in growth media can suppress biosorption by limiting REE solubility, so nutrient dosing and process staging must be managed.

Second, boron is a persistent bottleneck in mixed-metal matrices (boron removal collapsing under co-metal interference), so the pilot must include a boron-specific polishing module or demonstrate boron compliance directly.

Third, toxicity thresholds matter: La at ≥250 mg/L triggered metabolic collapse in Chlorella sorokiniana under simulated REE wastewater, indicating that high-REE streams may require dilution, staged treatment, or the use of non-living biomass to avoid viability failure.

3.6 Timeline and milestones

A practical 24-month validation program can be structured around (1) characterization, (2) factorial screening, (3) optimization/continuous-flow proof, and (4) pilot validation, with an early "go/no-go" decision when no condition

achieves at least a substantial approach toward the strictest contaminant limit. The rationale is that the extracted evidence already shows some parameters can remain far above regulatory values after algae treatment (e.g., Cr), so early termination criteria prevent lengthy optimization when the approach is not technically viable for the specific wastewater.

Specific go no-go criterion

A defensible go/no-go milestone at month 9 is whether any tested configuration (species × biomass form × pH × pretreatment) can reduce each of the key regulated metals with known limits in the extracted frameworks (e.g., Cu, Ni, Cr, Co where a limit exists) to within a factor of two of the strictest applicable mg/L limit; the extracted data show that even strong removal can leave residual concentrations above strict BAT-AEL ranges (e.g., Co residual 1.01 mg/L vs EU 0.1–0.5 mg/L), which is precisely the kind of quantitative gap this milestone is intended to detect.

Research Report: Advancing Microalgae Treatment for NdFeB Wastewater Compliance

PART 1: Cost Estimate and Equipment Specifications for 24-Month Experimental Program

1.1 Analytical Costs

The analytical budget is a primary cost driver for the proposed 24-month program, dominated by Inductively Coupled Plasma (ICP) analysis. Costs per sample for multi-element analysis via ICP-Mass Spectrometry (ICP-MS) or ICP-Optical Emission Spectrometry (ICP-OES) vary by provider, turnaround time, and academic/commercial status, ranging from approximately $60 to over $550 per sample. Sample preparation, such as acid digestion, adds an additional $20 to $80 per sample.

A preliminary estimate of the number of aqueous samples required for the factorial screening in Phase 2 is substantial. The proposed design (6 species × 3 biomass forms × 4 dilutions × 6 pH levels × 2 Fe conditions × 3 coolant levels × 3 EDTA levels × 3 triplicates) generates a theoretical maximum of 19,440 samples. This is likely an overestimate, as a fractional factorial or staged design would be employed. A more realistic estimate for budgeting purposes across all phases is:

- **Phase 1 (Characterization):** ~50 samples (multiple waste streams, replicates).
- **Phase 2 (Screening):** ~2,500 samples (assuming a reduced fractional factorial design).
- **Phase 3 (Optimization):** ~1,000 samples (kinetic, dosage, and polishing tests).
- **Phase 4 (Pilot Monitoring):** ~500 samples (30-day continuous operation, daily/weekly sampling).

Total Estimated Samples: ~4,050.

Estimated Analytical Budget:

Item	Unit Cost (Range)	Quantity	Total Cost (Estimate)
ICP-MS/OES Analysis (External)	$90 - $180 / sample	4,050	$364,500 - $729,000
Sample Digestion/Prep	$25 - $50 / sample	4,050	$101,250 - $202,500
Other Analyses (COD, TSS, O&G)	$35 - $70 / sample	~1,000	$35,000 - $70,000
Subtotal			**$500,750 - $1,001,500**

This budget assumes external laboratory services. Acquiring an ICP-MS instrument carries a capital cost of approximately $150,000, plus ongoing operational and maintenance expenses. REE analysis requires ICP-MS with detection limits in the low parts-per-billion (ppb) range to verify compliance with stringent standards.

1.2 Culture Facility and Equipment

The program requires a dedicated microalgae culture facility with bench-scale and pilot-scale capabilities.

- **Bench-Scale Screening (Phase 2):** A cost-effective setup using multiple 1-10 L glass bubble-column or airlift photobioreactors (PBRs) is recommended. While commercial, fully-automated lab reactors can cost several thousand dollars each, simplified systems can be assembled for as little as $160 to $800 per unit. The budget should account for at least 12-18 bench-scale reactors to allow for parallel experiments.

- **Pilot Reactor (Phase 4)**: A 50-100 L continuous-flow pilot reactor is required. The cost for a reactor of this scale can range from $50,000 to $100,000, depending on materials (glass vs. stainless steel) and the level of instrumentation (pumps, sensors, control systems).
- **Harvesting Equipment**: A laboratory-scale centrifuge is essential for biomass harvesting and sample preparation, with associated costs for centrifuge tubes. Flocculation can be achieved with cost-effective reagents like calcium hydroxide.

1.3 Consumables and Reagents

This category represents a significant operational cost. Key consumables include:

- **Culture Media**: Standard media like Bold's Basal Medium (BBM) and BG-11.
- **Synthetic Leachate Salts**: Bulk quantities of $FeCl_3 \cdot 6H_2O$, $NdCl_3 \cdot 6H_2O$, H_3BO_3, $CoCl_2 \cdot 6H_2O$, and other metal salts are needed to prepare the recipes in Part 2.
- **Process Chemicals**: Acids (HNO_3, HCl), bases (NaOH), immobilization agents (sodium alginate), biomass pretreatment chemicals (NaOH), representative metalworking fluids, and chelating agents (EDTA). EDTA has been costed at approximately $1 per 100 g in related studies.
- **Gases**: CO_2 for culture pH control and growth, priced at approximately $45 per metric ton.

1.4 Waste Handling and Disposal

The metal-laden biomass and spent leachate will likely be classified as hazardous waste. Disposal costs are highly variable.

- **Waste Characterization**: TCLP (Toxicity Characteristic Leaching Procedure) testing is required for waste classification and costs approximately $60-$145 per sample.
- **Solid Waste Disposal**: Costs for heavy metal-bearing solids range from $164 per 55-gallon drum to over $190 per ton.
- **Liquid Waste Disposal**: Costs for heavy metal-bearing liquids are approximately $170 per 55-gallon drum or $0.83 to $1.70 per gallon.
- **Lab Packs**: Disposal of small-volume laboratory waste is often handled via lab packs, costing around $400-$600 per 55-gallon drum or on a per-kilogram basis (e.g., $15-$26/kg for corrosive/poisonous materials).

1.5 Personnel

The project is estimated to require two full-time equivalent (FTE) personnel. Based on 2023 hourly rates for environmental consulting and technical services, the roles would include:

- **1 FTE - Project Manager / Senior Scientist**: Responsible for experimental design, data analysis, and reporting. (Rate: ~$185/hr)
- **1 FTE - Laboratory Technician**: Responsible for algae cultivation, running experiments, and sample preparation. (Rate: ~$47 - $106/hr)

1.6 Total Budget Summary

The following table provides a rough order-of-magnitude budget for the 24-month program.

Category	Estimated 24-Month Cost (USD)	Notes
Analytical Services	$750,000	Based on ~4,000 samples at an average all-in cost of ~$187/sample.
Equipment (Capital)	$125,000	Includes one 100-L pilot reactor ($100k) and multiple bench-scale PBRs ($25k total).
Consumables & Reagents	$100,000	Estimate for media, salts, chemicals, gases.
Waste Handling & Disposal	$75,000	Includes characterization, transport, and disposal of solid/liquid hazardous waste.
Personnel (2 FTEs)	$920,000	Based on 2 FTEs x 2080 hrs/yr x 2 yrs at blended rate of $110/hr.
Subtotal	**$1,970,000**	
Contingency (15%)	**$295,500**	
Total Estimated Budget	**$2,265,500**	

PART 2: Synthetic NdFeB Leachate Recipe Table

2.1 Real NdFeB Leachate Composition Reference Data

Real NdFeB waste composition varies by source (grinding, cutting, end-of-life magnets) but is consistently dominated by iron, followed by neodymium and other rare earths. Bulk solid compositions reported in literature show Fe content from 56-74 wt%, Nd from 20-32 wt%, B from 0.7-1.2 wt%, and Co from <1 to 5 wt%. When leached, these solids produce highly acidic, metal-rich solutions. For example, leaching scrap magnet powder in strong acid can yield solutions with Fe concentrations as high as 60-85 g/L and Nd concentrations of 28-33 g/L. The following recipes are designed to simulate these challenging conditions for different waste streams.

2.2 Synthetic Recipe: Grinding Sludge Leachate

This recipe simulates the highly concentrated, acidic leachate from the direct dissolution of grinding sludge or magnet powder.

- **Objective**: Replicate a high-iron, high-REE, low-pH matrix.
- **Target pH**: 2.0 (Adjust with concentrated HNO_3)
- **Salts per 1 Liter of Deionized Water**:
  - $FeCl_3 \cdot 6H_2O$: 24.2 g (for 5,000 mg/L Fe^{3+})
  - $NdCl_3 \cdot 6H_2O$: 3.7 g (for 1,500 mg/L Nd^{3+})
  - $DyCl_3 \cdot 6H_2O$: 0.3 g (for 120 mg/L Dy^{3+})
  - $PrCl_3 \cdot 6H_2O$: 0.7 g (for 300 mg/L Pr^{3+})
  - H_3BO_3 (Boric Acid): 2.86 g (for 500 mg/L B)
  - $CoCl_2 \cdot 6H_2O$: 0.4 g (for 100 mg/L Co^{2+})
  - $CuCl_2 \cdot 2H_2O$: 0.13 g (for 50 mg/L Cu^{2+})
  - $NiCl_2 \cdot 6H_2O$: 0.2 g (for 50 mg/L Ni^{2+})
  - $AlCl_3 \cdot 6H_2O$: 0.37 g (for 50 mg/L Al^{3+})

2.3 Synthetic Recipe: Polishing Slurry Leachate

This recipe represents a more dilute waste stream, typical of final polishing steps, with lower metal content but potentially higher suspended solids.

- **Objective**: Simulate a lower-concentration, multi-metal stream.
- **Target pH**: 3.5 (Adjust with dilute HNO_3)
- **Salts per 1 Liter of Deionized Water**:
  - $FeCl_3 \cdot 6H_2O$: 2.4 g (for 500 mg/L Fe^{3+})
  - $NdCl_3 \cdot 6H_2O$: 0.37 g (for 150 mg/L Nd^{3+})
  - $DyCl_3 \cdot 6H_2O$: 0.03 g (for 12 mg/L Dy^{3+})
  - $PrCl_3 \cdot 6H_2O$: 0.07 g (for 30 mg/L Pr^{3+})
  - H_3BO_3: 0.29 g (for 50 mg/L B)
  - $CoCl_2 \cdot 6H_2O$: 0.04 g (for 10 mg/L Co^{2+})
  - **Note**: Add 1 g/L of fine Al_2O_3 powder to simulate abrasive particles.

2.4 Synthetic Recipe: Coolant Waste

This simulates contaminated metalworking fluid from grinding/cutting operations, characterized by a high organic load (COD) and emulsified oils.

- **Objective**: Test biosorption in the presence of high COD and coolant additives.
- **Target pH**: 8.5-9.0 (Inherent to coolant; adjust with NaOH if needed)
- **Preparation per 1 Liter**:
  - Add 50 mL (5% v/v) of a representative water-soluble mineral oil-based coolant (e.g., Mobilcut 230).
  - Add metal salts to achieve target contamination levels:
    - $FeSO_4 \cdot 7H_2O$: 0.5 g (for 100 mg/L Fe^{2+})
    - $NdCl_3 \cdot 6H_2O$: 0.07 g (for 30 mg/L Nd^{3+})
    - $CoCl_2 \cdot 6H_2O$: 0.02 g (for 5 mg/L Co^{2+})
  - **Expected COD**: >10,000 mg/L (from coolant base).

2.5 Synthetic Recipe: Cleaning Agent Waste

This recipe simulates alkaline degreasing or cleaning solutions containing chelating agents like EDTA, which compete with biomass for metal binding.

- **Objective**: Evaluate the impact of strong chelating agents on biosorption.
- **Target pH**: 10.0 (Adjust with NaOH)
- **Preparation per 1 Liter**:
  - `Disodium EDTA`: 0.37 g (for 1 mM EDTA)
  - Add metal salts to achieve target contamination levels:
    - $FeCl_3 \cdot 6H_2O$: 0.24 g (for 50 mg/L Fe^{3+})
    - $NdCl_3 \cdot 6H_2O$: 0.07 g (for 30 mg/L Nd^{3+})
    - $NiCl_2 \cdot 6H_2O$: 0.04 g (for 10 mg/L Ni^{2+})
  - **Note**: Metals will be strongly chelated by EDTA.

2.6 Master Recipe Comparison Table

This table summarizes the key characteristics of the four proposed synthetic leachate recipes.

Parameter	Grinding Sludge Leachate	Polishing Slurry Leachate	Coolant Waste	Cleaning Agent Waste
pH	2.0	3.5	8.5 - 9.0	10.0
Fe (mg/L)	5,000	500	100	50
Total REE (mg/L)	1,920	192	30	30
B (mg/L)	500	50	Trace (from coolant)	-
Co (mg/L)	100	10	5	-
Ni (mg/L)	50	-	-	10
Fe:REE Ratio	~2.6 : 1	~2.6 : 1	~3.3 : 1	~1.7 : 1
COD (mg/L)	Low	Low	High (>10,000)	Moderate
Organic Additives	None	None	5% (v/v) Water-Soluble Oil	1 mM EDTA

PART 3: Expanded Regulatory Discharge Limit Compilation

3.1 Complete Discharge Limit Table

The following table compiles wastewater discharge limits for key NdFeB contaminants across major regulatory frameworks. Values are daily maximums or ranges unless otherwise specified.

Contaminant	EU BAT-AEL (NFM BREF)	EU BAT-AEL (STM BREF)	EU BAT-AEL (Ferrous BREF)	US EPA 40 CFR 433	China GB 26451 / 25467
Iron (Fe)	Not specified	0.1 - 10 mg/L	1 - 5 mg/L	No specific limit	No specific limit
Boron (B)	Not specified	1 - 10 mg/L	Not specified	No specific limit	No specific limit
Cobalt (Co)	≤ 0.1 mg/L	Not specified	Not specified	No specific limit	No specific limit
Copper (Cu)	0.05 - 0.5 mg/L	0.2 - 2.0 mg/L	Not specified	3.38 mg/L (Daily Max)	No specific limit
Nickel (Ni)	≤ 0.5 mg/L	0.2 - 2.0 mg/L	0.01 - 0.2 mg/L	3.98 mg/L (Daily Max)	No specific limit
Aluminum (Al)	Not specified	1 - 10 mg/L	Not specified	No specific limit	No specific limit
Chromium (Cr)	≤ 0.2 mg/L	0.1 - 2.0 mg/L	0.01 - 0.1 mg/L	2.77 mg/L (Daily Max)	No specific limit
Total REE (Nd, Dy)	No specific limit	No specific limit	No specific limit	No specific limit	No specific limit
pH	Not specified	Not specified	Not specified	6.0 - 9.0	6.0 - 9.0
COD	Not specified	100 - 500 mg/L	30 - 90 mg/L	No specific limit	70 mg/L
TSS	Not specified	5 - 30 mg/L	5 - 30 mg/L	60 mg/L (Daily Max)	50 mg/L
Oil & Grease (HOI)	Not specified	1 - 5 mg/L	0.5 - 4 mg/L	52 mg/L (Daily Max)	No specific limit
Ammonia (as N)	Not specified	Not specified	Not specified	No specific limit	15 mg/L

Note: Limits are presented as ranges or daily maximums (mg/L). EU BAT-AELs are often expressed as ranges representing daily averages achievable with Best Available Techniques.

3.2 Updated Compliance Gap Matrix

This matrix assesses the feasibility of microalgae treatment by comparing the best-reported final concentrations from the evidence table against the strictest applicable regulatory limits identified above.

Contaminant	Strictest Limit (mg/L)	Best Microalgae Result (mg/L)	Compliance Gap	Feasibility Rating
Iron (Fe)	1.0 (EU Ferrous)	Not Assessed in high-Fe matrix	Large (Assumed)	**RED**
Boron (B)	1.0 (EU STM)	~55.9 (from 60 mg/L)	54.9 mg/L	**RED**
Cobalt (Co)	0.1 (EU NFM)	1.01 (from 63.6 mg/L)	0.91 mg/L	**YELLOW**
Neodymium (Nd)	No specific limit	Not Assessed vs. Limits	Not Applicable	**BLACK**
Dysprosium (Dy)	No specific limit	Not Assessed vs. Limits	Not Applicable	**BLACK**
Copper (Cu)	0.05 (EU NFM)	Not Assessed vs. Limits	Not Assessed	**BLACK**
Nickel (Ni)	0.01 (EU Ferrous)	Not Assessed vs. Limits	Not Assessed	**BLACK**
COD	30 (EU Ferrous)	Not Assessed in magnet leachate	Not Assessed	**BLACK**
TSS	5 (EU STM/Ferrous)	Not Assessed in magnet leachate	Not Assessed	**BLACK**

Feasibility Rating:

- **GREEN**: Compliance demonstrated in relevant matrix.
- **YELLOW**: Plausible; high removal shown but not to the strictest limit or in ideal matrix.
- **RED**: Evidence suggests removal is insufficient to meet limits.
- **BLACK**: No data available for this contaminant in a relevant matrix vs. limits.

3.3 Jurisdiction Comparison

A comparison of the regulatory frameworks reveals significant differences in stringency, which has major implications for facility siting and treatment technology selection.

- **Most Stringent**: The **European Union's BAT-AELs** are generally the strictest, particularly for common heavy metals. The Ferrous Metals Processing BREF sets a very low limit for Nickel (0.01-0.2 mg/L), and the Non-Ferrous Metals BREF imposes a tight limit for Cobalt (≤ 0.1 mg/L). The STM BREF sets a challenging limit for Boron (1-10 mg/L).
- **Less Stringent (for some parameters)**: The **US EPA 40 CFR 433** standard for metal finishing has higher daily maximum limits for some metals like Copper (3.38 mg/L) and Nickel (3.98 mg/L) compared to EU BAT-AELs, but it does not specify limits for key NdFeB contaminants like iron, boron, cobalt, or rare earth elements.
- **China's GB 26451** for the rare earth industry focuses on different parameters, setting limits for Ammonia (15 mg/L) and COD (70 mg/L) but lacks specific limits for many of the individual metals found in NdFeB waste in the available documentation.

Overall, compliance would be most difficult in the European Union due to its comprehensive and stringent metal-by-metal limits. The lack of specific REE discharge limits across all major frameworks is a significant regulatory gap. Boron and Nickel appear to be the most challenging metals to treat to the strictest EU standards based on existing evidence.

Data Analysis Pipeline: ICP-MS to Compliance Dashboard for NdFeB Magnet Leachate Microalgae Treatment

Summary

This deliverable specifies a reproducible, audit-friendly data workflow that starts with instrument exports from ICP-MS software (e.g., Agilent MassHunter and Thermo Qtegra ISDS) and ends with (i) statistically defensible screening/optimization models (DSD then RSM) and (ii) a jurisdiction-selectable compliance dashboard that compares final treated effluent concentrations against discharge limits. The architecture is grounded in sources describing MassHunter CSV export behavior, QC sequencing (blanks/standards/QC/sample ordering), CCV frequency expectations (every 10 samples), and the availability of DOE/RSM and desirability tooling in JMP and R's **rsm** ecosystem.

Pipeline overview

The end-to-end workflow is organized as a linear "data product" chain where each stage outputs a versioned table that the next stage consumes. This linearity is important because regulatory-style data validation distinguishes raw instrument outputs from validated deliverables that are recalculable and traceable.

Raw ICP-MS export (CSV/XML/PDF) → Python QC screening → validated, analysis-ready dataset → DSD screening analysis (JMP or R) → RSM modeling (JMP or R or Python) → multi-response desirability optimization → compliance dashboard (R Shiny or Python Dash alternative) → automated report outputs (interactive HTML / model summaries / exported CSV).

Stage 1

Stage 1 data ingestion and normalization

Agilent MassHunter can produce per-sample CSV exports, including "CPS Data to CSV" outputs that contain counts-per-second data (e.g., "5000 count/sec").

MassHunter export automation can be configured by selecting "Tabulate Chart after Sample Acquisition" and setting file format to .csv with CPS data format, which supports a consistent batch export pattern for downstream scripts.

For troubleshooting and reproducibility, downstream importers may require strict consistency in exported formatting (e.g., timestamp/format alignment), and import errors should be addressed by ensuring the export format matches tool expectations.

Thermo's Qtegra ISDS software is described as being used to control the ICP-MS instrument and to "generate, process, and report analytical data," and it includes "flags and limits" features to highlight values outside acceptable concentration ranges for easier review.

MassHunter performance-report exports are explicitly supported as PDF and XML (for technical support and troubleshooting), which provides an additional machine-readable artifact even when the primary quantitative export is a CSV table.

If vendor software does not support a desired batch export in a directly accessible format, community guidance notes that MSConvert from ProteoWizard can be used for conversion workflows.

Stage 1 tabular schema

To enable automated QC, model fitting, and compliance checking, the pipeline uses a "long" table schema (one analyte per row per sample) that can be reshaped to a wide matrix for modeling as needed. This mirrors the general goal in mass-spectrometry pipelines of turning a complicated table into a usable matrix object for downstream analysis.

A practical column set is consistent with the project's instrument-export requirement that CSV includes at least: Sample ID, Analyte, Isotope, Concentration, Units, Internal Standard Recovery, Dilution Factor, and Analysis Date/Time.

The ingestion layer should also capture batch/run metadata needed for validation such as batch ID and QC sample type, which can be inferred from file names in some tooling.

One automation-ready schema is therefore:

Column	Purpose
Sample_ID	Primary key for joining to lab notebook/run logs
Batch_ID	Links results to a specific acquisition/QC sequence
Analysis_DateTime	Enables drift visualization and run-order effects
Analyte	Metal/element name for compliance mapping
Isotope	Supports interference-aware QA and isotope selection
Concentration	Core quantitative result for modeling and compliance
Units	Prevents silent unit errors (µg/L vs mg/L)
Dilution_Factor	Enables dilution correction for high-Fe matrix handling
Internal_Std_Recovery_pct	Supports internal-standard QC rules
Sample_Type	Sample/QC/blank classification for QC stratification

Stage 1 parsing options

If MassHunter exports are available as CSV tables, they can be parsed into analysis-friendly data frames in R using dedicated parsers that explicitly target MassHunter CSV outputs.

For example, **parse_agilent** describes itself as providing R functions to parse MassHunter CSV outputs and return results as a data frame, with requirements that include R, tidyverse, and data.table.

Similarly, **MasshunteR** imports the MassHunter Quant CSV output table, can import a compound–ISTD mapping CSV, normalizes by internal standards, calculates concentrations, auto-classifies samples by file name, and produces QC charts.

For broader vendor-format parsing and export to standard formats, **chromConverter** provides a unified interface to parsers and can export chromatogram-derived data to csv/mzml/cdf when an export format is specified, returning data as matrix/data.frame objects.

Stage 2

Stage 2 QC screening and data validation

The QC layer is designed to support "Stage 3 validation" style requirements, where selected samples may need recalculation/re-quantification from raw outputs and QC review.

For metals-by-ICP-MS validation workflows, guidance emphasizes that instruments should be calibrated for all target analytes plus any required for interference correction, and that internal standards should be added to all calibration standards, blanks, instrument QC checks, samples, and batch QC.

A core QC scheduling expectation is that continuing calibration verification (CCV) standards must be analyzed after every 10 field samples and at the end of the analytical run, which enables drift monitoring and correction triggers.

Complementary guidance also describes CCV frequency (every 10 samples) and notes that acceptance criteria are commonly set (e.g., ±10% of the true value) for CCV checks, which supports implementing a deterministic rule engine in code.

Validation workflows include review of sequence logs to confirm that all QC samples were analyzed in the correct order within a batch, which is why Batch_ID and run order should be preserved in the data model.

Field blank detections require context-sensitive qualification rules that depend on relative blank vs sample concentrations, reinforcing that the QC layer must include blank association logic.

Finally, guidance on modern ICP-MS workflows explicitly emphasizes "automated screening and blank drift correction" to maintain data integrity, which supports implementing automated blank/drift checks prior to any DOE/RSM modeling.

Stage 2 blank subtraction and drift correction

Drift and blank effects can be monitored and corrected using periodic check standards and blank measurements embedded in the run. This is supported by descriptions of drift checks that compare results for a standard measured before and after sample analysis, and by sequences where a standard is repeated periodically throughout a mixed sample-type run.

A conservative, implementation-ready approach is to compute (i) blank-corrected concentrations, (ii) drift factors inferred from periodic CCV/QC standard recoveries, and (iii) corrected concentrations, then to qualify or reject results when QC checks fail beyond acceptance windows defined by the project's QAPP/method.

Stage 2 QC automation pseudocode

The following Python pseudocode illustrates how a rule-based validator can be implemented on exported CSV tables, using configurable thresholds and method/QAPP-defined acceptance windows for internal standards and calibrations.

```
import pandas as pd

# ---- Config (set these in a QAPP / method config file) ----
IS_LOW, IS_HIGH = 70, 130            # example window for Internal_Std_Recovery_pct
CCV_LOW, CCV_HIGH = 90, 110          # example window for CCV % recovery
RPD_MAX = 20                         # example duplicate precision threshold
BLANK_MULTIPLIER = 2                 # example: flag if blank > 2x MDL

# ---- Load ----
df = pd.read_csv('icpms_long.csv')

# Required columns per pipeline spec
```

```
required = {
  'Sample_ID','Batch_ID','Analysis_DateTime','Analyte','Isotope',
  'Concentration','Units','Dilution_Factor','Internal_Std_Recovery_pct','Sample_Type'
}
missing = required - set(df.columns)
if missing:
    raise ValueError(f"Missing columns: {missing}")

# Apply dilution correction
df['Conc_corr'] = df['Concentration'] * df['Dilution_Factor']

# Flag internal standard recovery out of window
df['FLAG_IS'] = (df['Internal_Std_Recovery_pct'] < IS_LOW) | (df['Internal_Std_Recovery_pct'] > IS

# Flag CCV / CCB failures (requires QC rows tagged as Sample_Type)
ccv = df[df['Sample_Type'].eq('CCV')].copy()
ccv['FLAG_CCV'] = (ccv['Conc_corr'] < CCV_LOW) | (ccv['Conc_corr'] > CCV_HIGH)

# Example duplicate RPD calculation (requires duplicates tagged with a common ID)
# df['RPD'] = ...
# df['FLAG_RPD'] = df['RPD'] > RPD_MAX

# Example blank rule (requires MDL column or analyte-specific MDL table)
# blank = df[df['Sample_Type'].eq('Blank')]
# df = df.merge(blank[['Batch_ID','Analyte','Conc_corr']].rename(columns={'Conc_corr':'BlankConc'}
#               on=['Batch_ID','Analyte'], how='left')
# df['FLAG_BLANK'] = df['BlankConc'] > (BLANK_MULTIPLIER * df['MDL'])

# Output flagged dataset
qc_cols = ['FLAG_IS']
out = df.assign(QC_FAIL=lambda x: x[qc_cols].any(axis=1))
out.to_csv('icpms_validated_long.csv', index=False)
```

This script reflects the documented need to (i) preserve batch sequencing (because QC samples occur in a defined order), (ii) incorporate internal standards into all samples/QC/blanks, and (iii) check CCV frequency (every 10 samples) and qualify data when QC acceptance windows are violated.

Stage 3

Stage 3 DSD screening analysis

Definitive screening designs (DSDs) are used to screen many factors efficiently when a full quadratic model would require too many runs, and they require careful analysis because they are not "automatic" in the way simpler designs can be.

DSDs can be interpreted in JMP Pro (e.g., a DSD created and interpreted using JMP Pro 15.0.0), which supports an integrated DoE-to-modeling workflow.

A DSD can also include additional runs to estimate quadratic effects and can be randomized in blocks to reduce bias, which is relevant when species or biomass-form is treated as a block factor in the biosorption program.

In R, DSD and subsequent model fitting can be implemented in complementary tooling; for example, DSDs have a known relationship between run count and number of variables in some descriptions (e.g., N = 2k + 1 or 2k + 3 for k factors).

Stage 3 JMP workflow

JMP is documented as providing response-surface design and modeling workflows, including importing/exporting CSV, executing Python/R via JSL, and automatically updating analyses/graphs, which are capabilities that support "re-run the entire workflow when new ICP-MS batches arrive."

Within JMP's response-surface modeling, a documented fitting approach is "Fit Model > Standard Least Squares > add quadratic and 2FI terms," which aligns with second-order RSM practice following screening.

JMP also supports dashboard creation and can be used to package QC and modeling outputs into interactive deliverables.

Stage 3 R workflow

In R, the **rsm** package provides functions for designing and analyzing sequential experiments, including Box-Behnken design generation (bbd), CCD generation (ccd), and fitted response-surface model objects with ANOVA including lack-of-fit tests.

The **rsm** ecosystem supports converting variables to coded units with coded.data for numerical stability and proper scaling, which is important when mixing variables like pH, biomass dosage, and contact time.

Stage 4

Stage 4 RSM modeling

RSM for biosorption optimization typically fits second-order (quadratic) models using structured designs such as Box-Behnken or central composite designs, and these designs are documented in JMP and R tooling.

Box-Behnken designs in JMP are described as classical RSM designs that avoid extreme points and test each continuous factor at low, middle, and high values to fit a second-order model with quadratic and interaction terms.

In R, **rsm** provides visualization tools (contour.lm, persp.lm, image.lm) to visualize fitted surfaces, which supports interpretation of factor interactions.

For Python-based modeling/optimization, SciPy and statsmodels can be used to fit quadratic regression and run numerical optimization (e.g., scipy.optimize.minimize), providing an alternative path when custom constraint handling is needed.

Stage 5

Stage 5 multi-response desirability optimization

Because the objective is wastewater discharge compliance across multiple regulated contaminants, the optimization target is inherently multi-response, and desirability approaches provide a way to combine multiple response goals into a single objective.

JMP explicitly supports integrating a desirability profiler into the workflow to optimize multiple responses simultaneously, and JMP's Prediction Profiler supports "Optimization and Desirability > Maximize Desirability."

A standard formulation for overall desirability combines individual desirabilities using the geometric mean, D=(d1 d2 … dm)^(1/m), which supports multi-response trade-offs (e.g., meeting Cu, Ni, Zn simultaneously).

The **rsm** package documentation also explicitly frames desirability as the mechanism for simultaneous optimization, including setting targets to discharge limits for each metal, while noting that multiple-response optimization is not built into **rsm** itself and may require a companion "desirability" package.

Stage 6

Stage 6 compliance dashboard

A compliance dashboard should accept validated, tabular outputs (CSV) and compute exceedance indicators by comparing each analyte concentration to jurisdiction-specific limits, which aligns with tools designed to visualize discharge limits and wastewater monitoring data.

EPA's TADA ecosystem is explicitly described as an R package and a set of R Shiny applications; it provides tools that help visualize wastewater monitoring data and discharge limits and can be run as a Shiny web app.

TADAShiny's "Module 1: Data Discovery and Cleaning" retrieves data and applies QC/wrangling/cleaning steps, including flagging suspect results using validation reference tables, harmonization of synonyms, unit conversions, censored data substitutions, filtering, and data visualizations, which maps well to an effluent-monitoring dashboard workflow.

For a light-weight project-specific dashboard, Shiny is described as a free and open source tool for building highly customizable interactive web-based applications in R, and it supports CSV/tabular inputs as a standard data type.

A practical dashboard requirement for this program is consistent with the project's specification of a Shiny app that uploads a CSV of treated effluent results and displays analyte-versus-limit comparisons with traffic-light indicators by jurisdiction.

Stage 6 dashboard minimum feature set

A minimum viable compliance dashboard includes (i) file upload, (ii) jurisdiction selection (EU/US/China), (iii) traffic-light compliance table, and (iv) time series plots with discharge limit lines, which are common in monitoring dashboards that emphasize visualization for compliance use.

To support automated data refresh, an ingestion approach can be modeled on systems that consume delimited (CSV) testing results and synchronize them into a database for dashboards, which demonstrates how scheduled uploads can enable near-real-time and historic visualization.

Stage 7

Stage 7 reporting and archiving

Regulatory-style metals workflows often require packaging deliverables that include raw data artifacts plus structured electronic deliverables, and the EPA SOP explicitly describes a completed data package containing an Executive Narrative, Sample Summary Report, and an Electronic Data Deliverable (EDD) template with defined column headers.

DoD/EDQW guidance further emphasizes that electronic data output should comply with the EDD data structure defined in the QAPP and that automated validation outputs should be documented (including identification of automated review software and its output report), which supports writing automated QC summary reports as artifacts alongside the validated CSV.

For model and visualization outputs, R-based workflows explicitly demonstrate exporting designs to CSV, reading analysis CSV files, and exporting model summaries and contour plots to text and image files, which supports automated report generation from scripts.

JMP supports saving reports as interactive HTML, which can serve as a reproducible and shareable reporting artifact for QC summaries, DSD screening results, and RSM/desirability outputs.

Software stack

The recommended stack is explicitly grounded in sources that describe (i) vendor export + QC requirements, (ii) JMP/RSM/desirability tooling, and (iii) dashboard tooling.

Component	Role in pipeline	Evidence
Agilent MassHunter	Export per-sample CSV and/or XML/PDF artifacts for downstream analysis	
Thermo Qtegra ISDS	Generate/process/report data; flag out-of-range results via flags/limits	
Python + pandas	Automated QC screening on tabular exports; scheduling/integration patterns exist in automated ICP workflows	
JMP	DoE/RSM modeling, desirability optimization, automation hooks (JSL Python/R), dashboards, interactive HTML export	
R rsm	Generate BBD/CCD; fit response-surface models; provide ANOVA and lack-of-fit; provide contour/3D visualization	
R Shiny	Interactive compliance dashboards with tabular inputs; supports highly customizable web apps	
EPA TADA	Existing R Shiny + R package pattern for QC/cleaning and discharge-limit visualization	

Implementation notes

The pipeline should be implemented so that QC and compliance checking occur before any DSD/RSM modeling, because screening/optimization conclusions are only defensible if the underlying concentrations are qualified and traceable to a validated data package.

When integrating compliance limits, the dashboard can embed limit tables for key jurisdictions from authoritative sources, such as 40 CFR Part 433 numeric limits for Cu, Ni, Zn, total Cr, TSS, oil & grease, and pH range, which enables automated exceedance checks on the validated dataset.

Similarly, EU BREF-derived values (where applicable) can be encoded as targets for desirability optimization and compliance visualization (e.g., STM BREF ranges for Fe, B, Cu, Ni, Al, Cr, Zn, COD, total hydrocarbons, and suspended solids), with the important caveat that BREF tables provide ranges and installation-specific contexts rather than a single universal permit number.

Technical Deliverables: Experimental Design, Budget Analysis, and Technology Selection

This report presents three key deliverables for the research program on microalgae-based treatment of NdFeB magnet manufacturing wastewater: 1) a detailed, importable experimental run plan for the initial screening phase; 2) a sensitivity analysis of the project budget based on critical cost uncertainties; and 3) a weighted decision matrix to guide technology selection based on different regulatory and economic priorities.

Part 1: JMP/Minitab-Importable DSD Run Order Table

Design Specification

The initial experimental screening phase will employ a Definitive Screening Design (DSD) to efficiently evaluate seven key factors. This design allows for the identification of significant main effects, two-factor interactions, and quadratic (curvature) effects with a minimal number of runs.

- **Factors**: 7 continuous/categorical variables
- **Base Runs**: 15 runs per DSD matrix
- **Blocking Factor**: 6 microalgae species
- **Replication**: 3 full replicates per species
- **Total Runs**: 15 runs/replicate × 3 replicates/species × 6 species = **270 total experimental runs**

Factor Definition Table

The seven factors have been defined with low (-1), center (0), and high (+1) levels appropriate for screening. For the binary factor 'Fe Pre-removal', the center point (0) is treated as the 'low' level (No).

Factor	Name	Type	Low (-1)	Center (0)	High (+1)	Units
A	pH	Continuous	3.0	5.0	7.0	pH units
B	Leachate Dilution	Continuous	10	55	100	% (v/v)
C	Fe Pre-removal	Categorical	No	No	Yes	Binary
D	Coolant Conc.	Continuous	0	2.5	5.0	% (v/v)
E	EDTA Conc.	Continuous	0	0.5	1.0	mM
F	Biomass Dosage	Continuous	0.5	2.75	5.0	g/L
G	Contact Time	Continuous	1	12.5	24	hours

Complete 270-Run Table

The following table provides the complete, randomized run order for the 270 experiments. The design is blocked by species, and the run order has been randomized within each of the 18 blocks (6 species × 3 replicates) to mitigate systematic error. This table is formatted for direct import into statistical software like JMP or Minitab as a CSV file.

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_EDTA_mM	F_Dosage_g/L	G_Time_h
1	C. vulgaris	1	9	3	100	Yes	5	0	5	1
2	C. vulgaris	1	2	3	100	No	0	1	5	24
3	C. vulgaris	1	13	5	55	No	2.5	0.5	2.75	12.5
4	C. vulgaris	1	5	7	10	Yes	0	1	0.5	24
5	C. vulgaris	1	11	5	55	No	2.5	0.5	2.75	12.5
6	C. vulgaris	1	4	7	100	Yes	0	0	0.5	1
7	C. vulgaris	1	7	7	100	No	5	1	0.5	1
8	C. vulgaris	1	1	7	10	No	5	0	5	24
9	C. vulgaris	1	15	3	10	No	0	0	0.5	1
10	C. vulgaris	1	10	7	10	No	0	1	5	1
11	C. vulgaris	1	6	3	100	No	5	0	0.5	24
12	C. vulgaris	1	3	3	10	Yes	5	1	0.5	1
13	C. vulgaris	1	12	5	55	No	2.5	0.5	2.75	12.5
14	C. vulgaris	1	8	3	10	Yes	0	0	5	1
15	C. vulgaris	1	14	7	100	Yes	5	1	5	24
16	C. vulgaris	2	3	3	10	Yes	5	1	0.5	1
17	C. vulgaris	2	11	5	55	No	2.5	0.5	2.75	12.5
18	C. vulgaris	2	1	7	10	No	5	0	5	24
19	C. vulgaris	2	15	3	10	No	0	0	0.5	1
20	C. vulgaris	2	6	3	100	No	5	0	0.5	24
21	C. vulgaris	2	8	3	10	Yes	0	0	5	1
22	C. vulgaris	2	13	5	55	No	2.5	0.5	2.75	12.5

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg/L	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
23	C. vulgaris	2	5	7	10	Yes	0	1	0.5	24	
24	C. vulgaris	2	9	3	100	Yes	5	0	5	1	
25	C. vulgaris	2	12	5	55	No	2.5	0.5	2.75	12.5	
26	C. vulgaris	2	14	7	100	Yes	5	1	5	24	
27	C. vulgaris	2	2	3	100	No	0	1	5	24	
28	C. vulgaris	2	7	7	100	No	5	1	0.5	1	
29	C. vulgaris	2	4	7	100	Yes	0	0	0.5	1	
30	C. vulgaris	2	10	7	10	No	0	1	5	1	
31	C. vulgaris	3	12	5	55	No	2.5	0.5	2.75	12.5	
32	C. vulgaris	3	6	3	100	No	5	0	0.5	24	
33	C. vulgaris	3	1	7	10	No	5	0	5	24	
34	C. vulgaris	3	9	3	100	Yes	5	0	5	1	
35	C. vulgaris	3	15	3	10	No	0	0	0.5	1	
36	C. vulgaris	3	4	7	100	Yes	0	0	0.5	1	
37	C. vulgaris	3	11	5	55	No	2.5	0.5	2.75	12.5	
38	C. vulgaris	3	2	3	100	No	0	1	5	24	
39	C. vulgaris	3	8	3	10	Yes	0	0	5	1	
40	C. vulgaris	3	13	5	55	No	2.5	0.5	2.75	12.5	
41	C. vulgaris	3	10	7	10	No	0	1	5	1	
42	C. vulgaris	3	7	7	100	No	5	1	0.5	1	
43	C. vulgaris	3	3	3	10	Yes	5	1	0.5	1	
44	C. vulgaris	3	14	7	100	Yes	5	1	5	24	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_pct	E_EDTA_mM	F_Dosage_g_L	G_Time_h
45	C. vulgaris	3	5	7	10	Yes	0	1	0.5	24
46	G. sulphuraria	1	14	7	100	Yes	5	1	5	24
47	G. sulphuraria	1	1	7	10	No	5	0	5	24
48	G. sulphuraria	1	7	7	100	No	5	1	0.5	1
49	G. sulphuraria	1	3	3	10	Yes	5	1	0.5	1
50	G. sulphuraria	1	11	5	55	No	2.5	0.5	2.75	12.5
51	G. sulphuraria	1	9	3	100	Yes	5	0	5	1
52	G. sulphuraria	1	2	3	100	No	0	1	5	24
53	G. sulphuraria	1	13	5	55	No	2.5	0.5	2.75	12.5
54	G. sulphuraria	1	5	7	10	Yes	0	1	0.5	24
55	G. sulphuraria	1	10	7	10	No	0	1	5	1
56	G. sulphuraria	1	4	7	100	Yes	0	0	0.5	1
57	G. sulphuraria	1	15	3	10	No	0	0	0.5	1
58	G. sulphuraria	1	6	3	100	No	5	0	0.5	24
59	G. sulphuraria	1	12	5	55	No	2.5	0.5	2.75	12.5

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_pct	E_EDTA_mM	F_Dosage_g/L	G_Time_h
60	G. sulphuraria	1	8	3	10	Yes	0	0	5	1
61	G. sulphuraria	2	12	5	55	No	2.5	0.5	2.75	12.5
62	G. sulphuraria	2	1	7	10	No	5	0	5	24
63	G. sulphuraria	2	4	7	100	Yes	0	0	0.5	1
64	G. sulphuraria	2	9	3	100	Yes	5	0	5	1
65	G. sulphuraria	2	15	3	10	No	0	0	0.5	1
66	G. sulphuraria	2	6	3	100	No	5	0	0.5	24
67	G. sulphuraria	2	11	5	55	No	2.5	0.5	2.75	12.5
68	G. sulphuraria	2	8	3	10	Yes	0	0	5	1
69	G. sulphuraria	2	3	3	10	Yes	5	1	0.5	1
70	G. sulphuraria	2	14	7	100	Yes	5	1	5	24
71	G. sulphuraria	2	5	7	10	Yes	0	1	0.5	24
72	G. sulphuraria	2	10	7	10	No	0	1	5	1
73	G. sulphuraria	2	2	3	100	No	0	1	5	24

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
74	G. sulphuraria	2	7	7	100	No	5	1	0.5	1	
75	G. sulphuraria	2	13	5	55	No	2.5	0.5	2.75	12.5	
76	G. sulphuraria	3	8	3	10	Yes	0	0	5	1	
77	G. sulphuraria	3	1	7	10	No	5	0	5	24	
78	G. sulphuraria	3	14	7	100	Yes	5	1	5	24	
79	G. sulphuraria	3	5	7	10	Yes	0	1	0.5	24	
80	G. sulphuraria	3	11	5	55	No	2.5	0.5	2.75	12.5	
81	G. sulphuraria	3	3	3	10	Yes	5	1	0.5	1	
82	G. sulphuraria	3	13	5	55	No	2.5	0.5	2.75	12.5	
83	G. sulphuraria	3	6	3	100	No	5	0	0.5	24	
84	G. sulphuraria	3	10	7	10	No	0	1	5	1	
85	G. sulphuraria	3	2	3	100	No	0	1	5	24	
86	G. sulphuraria	3	7	7	100	No	5	1	0.5	1	
87	G. sulphuraria	3	12	5	55	No	2.5	0.5	2.75	12.5	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg_L	E_pH	F_EDTA_mM	G_Dosage_g_L	H_Time_h
88	G. sulphuraria	3	15	3	10	No	0	0	0.5	1	
89	G. sulphuraria	3	9	3	100	Yes	5	0	5	1	
90	G. sulphuraria	3	4	7	100	Yes	0	0	0.5	1	
91	Desmodesmus sp.	1	7	7	100	No	5	1	0.5	1	
92	Desmodesmus sp.	1	1	7	10	No	5	0	5	24	
93	Desmodesmus sp.	1	12	5	55	No	2.5	0.5	2.75	12.5	
94	Desmodesmus sp.	1	14	7	100	Yes	5	1	5	24	
95	Desmodesmus sp.	1	3	3	10	Yes	5	1	0.5	1	
96	Desmodesmus sp.	1	10	7	10	No	0	1	5	1	
97	Desmodesmus sp.	1	5	7	10	Yes	0	1	0.5	24	
98	Desmodesmus sp.	1	13	5	55	No	2.5	0.5	2.75	12.5	
99	Desmodesmus sp.	1	8	3	10	Yes	0	0	5	1	
100	Desmodesmus sp.	1	11	5	55	No	2.5	0.5	2.75	12.5	
101	Desmodesmus sp.	1	4	7	100	Yes	0	0	0.5	1	
102	Desmodesmus sp.	1	9	3	100	Yes	5	0	5	1	
103	Desmodesmus sp.	1	6	3	100	No	5	0	0.5	24	
104	Desmodesmus sp.	1	2	3	100	No	0	1	5	24	
105	Desmodesmus sp.	1	15	3	10	No	0	0	0.5	1	
106	Desmodesmus sp.	2	1	7	10	No	5	0	5	24	
107	Desmodesmus sp.	2	14	7	100	Yes	5	1	5	24	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
108	Desmodesmus sp.	2	8	3	10	Yes	0	0	5	1	
109	Desmodesmus sp.	2	4	7	100	Yes	0	0	0.5	1	
110	Desmodesmus sp.	2	11	5	55	No	2.5	0.5	2.75	12.5	
111	Desmodesmus sp.	2	7	7	100	No	5	1	0.5	1	
112	Desmodesmus sp.	2	13	5	55	No	2.5	0.5	2.75	12.5	
113	Desmodesmus sp.	2	2	3	100	No	0	1	5	24	
114	Desmodesmus sp.	2	10	7	10	No	0	1	5	1	
115	Desmodesmus sp.	2	5	7	10	Yes	0	1	0.5	24	
116	Desmodesmus sp.	2	15	3	10	No	0	0	0.5	1	
117	Desmodesmus sp.	2	6	3	100	No	5	0	0.5	24	
118	Desmodesmus sp.	2	12	5	55	No	2.5	0.5	2.75	12.5	
119	Desmodesmus sp.	2	3	3	10	Yes	5	1	0.5	1	
120	Desmodesmus sp.	2	9	3	100	Yes	5	0	5	1	
121	Desmodesmus sp.	3	11	5	55	No	2.5	0.5	2.75	12.5	
122	Desmodesmus sp.	3	2	3	100	No	0	1	5	24	
123	Desmodesmus sp.	3	8	3	10	Yes	0	0	5	1	
124	Desmodesmus sp.	3	15	3	10	No	0	0	0.5	1	
125	Desmodesmus sp.	3	4	7	100	Yes	0	0	0.5	1	
126	Desmodesmus sp.	3	10	7	10	No	0	1	5	1	
127	Desmodesmus sp.	3	13	5	55	No	2.5	0.5	2.75	12.5	
128	Desmodesmus sp.	3	1	7	10	No	5	0	5	24	
129	Desmodesmus sp.	3	7	7	100	No	5	1	0.5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
130	Desmodesmus sp.	3	12	5	55	No	2.5	0.5	2.75	12.5	
131	Desmodesmus sp.	3	5	7	10	Yes	0	1	0.5	24	
132	Desmodesmus sp.	3	14	7	100	Yes	5	1	5	24	
133	Desmodesmus sp.	3	6	3	100	No	5	0	0.5	24	
134	Desmodesmus sp.	3	9	3	100	Yes	5	0	5	1	
135	Desmodesmus sp.	3	3	3	10	Yes	5	1	0.5	1	
136	A. platensis	1	11	5	55	No	2.5	0.5	2.75	12.5	
137	A. platensis	1	1	7	10	No	5	0	5	24	
138	A. platensis	1	6	3	100	No	5	0	0.5	24	
139	A. platensis	1	14	7	100	Yes	5	1	5	24	
140	A. platensis	1	3	3	10	Yes	5	1	0.5	1	
141	A. platensis	1	9	3	100	Yes	5	0	5	1	
142	A. platensis	1	12	5	55	No	2.5	0.5	2.75	12.5	
143	A. platensis	1	5	7	10	Yes	0	1	0.5	24	
144	A. platensis	1	2	3	100	No	0	1	5	24	
145	A. platensis	1	15	3	10	No	0	0	0.5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mM	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
146	A. platensis	1	7	7	100	No	5	1	0.5	1	
147	A. platensis	1	13	5	55	No	2.5	0.5	2.75	12.5	
148	A. platensis	1	10	7	10	No	0	1	5	1	
149	A. platensis	1	4	7	100	Yes	0	0	0.5	1	
150	A. platensis	1	8	3	10	Yes	0	0	5	1	
151	A. platensis	2	13	5	55	No	2.5	0.5	2.75	12.5	
152	A. platensis	2	7	7	100	No	5	1	0.5	1	
153	A. platensis	2	1	7	10	No	5	0	5	24	
154	A. platensis	2	10	7	10	No	0	1	5	1	
155	A. platensis	2	4	7	100	Yes	0	0	0.5	1	
156	A. platensis	2	12	5	55	No	2.5	0.5	2.75	12.5	
157	A. platensis	2	15	3	10	No	0	0	0.5	1	
158	A. platensis	2	3	3	10	Yes	5	1	0.5	1	
159	A. platensis	2	8	3	10	Yes	0	0	5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mM	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
160	A. platensis	2	5	7	10	Yes	0	1	0.5	24	
161	A. platensis	2	14	7	100	Yes	5	1	5	24	
162	A. platensis	2	2	3	100	No	0	1	5	24	
163	A. platensis	2	9	3	100	Yes	5	0	5	1	
164	A. platensis	2	6	3	100	No	5	0	0.5	24	
165	A. platensis	2	11	5	55	No	2.5	0.5	2.75	12.5	
166	A. platensis	3	14	7	100	Yes	5	1	5	24	
167	A. platensis	3	7	7	100	No	5	1	0.5	1	
168	A. platensis	3	1	7	10	No	5	0	5	24	
169	A. platensis	3	10	7	10	No	0	1	5	1	
170	A. platensis	3	4	7	100	Yes	0	0	0.5	1	
171	A. platensis	3	12	5	55	No	2.5	0.5	2.75	12.5	
172	A. platensis	3	15	3	10	No	0	0	0.5	1	
173	A. platensis	3	3	3	10	Yes	5	1	0.5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
174	A. platensis	3	8	3	10	Yes	0	0	5	1	
175	A. platensis	3	5	7	10	Yes	0	1	0.5	24	
176	A. platensis	3	11	5	55	No	2.5	0.5	2.75	12.5	
177	A. platensis	3	2	3	100	No	0	1	5	24	
178	A. platensis	3	9	3	100	Yes	5	0	5	1	
179	A. platensis	3	6	3	100	No	5	0	0.5	24	
180	A. platensis	3	13	5	55	No	2.5	0.5	2.75	12.5	
181	Nostoc sp.	1	14	7	100	Yes	5	1	5	24	
182	Nostoc sp.	1	7	7	100	No	5	1	0.5	1	
183	Nostoc sp.	1	2	3	100	No	0	1	5	24	
184	Nostoc sp.	1	9	3	100	Yes	5	0	5	1	
185	Nostoc sp.	1	4	7	100	Yes	0	0	0.5	1	
186	Nostoc sp.	1	11	5	55	No	2.5	0.5	2.75	12.5	
187	Nostoc sp.	1	6	3	100	No	5	0	0.5	24	
188	Nostoc sp.	1	12	5	55	No	2.5	0.5	2.75	12.5	
189	Nostoc sp.	1	1	7	10	No	5	0	5	24	
190	Nostoc sp.	1	15	3	10	No	0	0	0.5	1	
191	Nostoc sp.	1	10	7	10	No	0	1	5	1	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mg/L	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
192	Nostoc sp.	1	5	7	10	Yes	0	1	0.5	24	
193	Nostoc sp.	1	13	5	55	No	2.5	0.5	2.75	12.5	
194	Nostoc sp.	1	3	3	10	Yes	5	1	0.5	1	
195	Nostoc sp.	1	8	3	10	Yes	0	0	5	1	
196	Nostoc sp.	2	10	7	10	No	0	1	5	1	
197	Nostoc sp.	2	4	7	100	Yes	0	0	0.5	1	
198	Nostoc sp.	2	13	5	55	No	2.5	0.5	2.75	12.5	
199	Nostoc sp.	2	1	7	10	No	5	0	5	24	
200	Nostoc sp.	2	8	3	10	Yes	0	0	5	1	
201	Nostoc sp.	2	15	3	10	No	0	0	0.5	1	
202	Nostoc sp.	2	6	3	100	No	5	0	0.5	24	
203	Nostoc sp.	2	12	5	55	No	2.5	0.5	2.75	12.5	
204	Nostoc sp.	2	3	3	10	Yes	5	1	0.5	1	
205	Nostoc sp.	2	9	3	100	Yes	5	0	5	1	
206	Nostoc sp.	2	5	7	10	Yes	0	1	0.5	24	
207	Nostoc sp.	2	2	3	100	No	0	1	5	24	
208	Nostoc sp.	2	14	7	100	Yes	5	1	5	24	
209	Nostoc sp.	2	11	5	55	No	2.5	0.5	2.75	12.5	
210	Nostoc sp.	2	7	7	100	No	5	1	0.5	1	
211	Nostoc sp.	3	1	7	10	No	5	0	5	24	
212	Nostoc sp.	3	11	5	55	No	2.5	0.5	2.75	12.5	
213	Nostoc sp.	3	5	7	10	Yes	0	1	0.5	24	

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Pretreatment	D_Coagulant_mL	E_EDTA_mM	F_Dosage_g_L	G_Time_h
214	Nostoc sp.	3	13	5	55	No	2.5	0.5	2.75	12.5
215	Nostoc sp.	3	2	3	100	No	0	1	5	24
216	Nostoc sp.	3	9	3	100	Yes	5	0	5	1
217	Nostoc sp.	3	15	3	10	No	0	0	0.5	1
218	Nostoc sp.	3	6	3	100	No	5	0	0.5	24
219	Nostoc sp.	3	12	5	55	No	2.5	0.5	2.75	12.5
220	Nostoc sp.	3	4	7	100	Yes	0	0	0.5	1
221	Nostoc sp.	3	10	7	10	No	0	1	5	1
222	Nostoc sp.	3	14	7	100	Yes	5	1	5	24
223	Nostoc sp.	3	3	3	10	Yes	5	1	0.5	1
224	Nostoc sp.	3	8	3	10	Yes	0	0	5	1
225	Nostoc sp.	3	7	7	100	No	5	1	0.5	1
226	Coccomyxa sp.	3	15	3	10	No	0	0	0.5	1
227	Coccomyxa sp.	3	8	3	10	Yes	0	0	5	1
228	Coccomyxa sp.	3	3	3	10	Yes	5	1	0.5	1
229	Coccomyxa sp.	3	11	5	55	No	2.5	0.5	2.75	12.5
230	Coccomyxa sp.	3	6	3	100	No	5	0	0.5	24
231	Coccomyxa sp.	3	1	7	10	No	5	0	5	24
232	Coccomyxa sp.	3	13	5	55	No	2.5	0.5	2.75	12.5
233	Coccomyxa sp.	3	4	7	100	Yes	0	0	0.5	1
234	Coccomyxa sp.	3	9	3	100	Yes	5	0	5	1
235	Coccomyxa sp.	3	12	5	55	No	2.5	0.5	2.75	12.5

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_pct	C_Fe_Precipitant	D_Coagulant_pct	E_EDTA_mM	F_Dosage_g_L	G_Time_h
236	Coccomyxa sp.	1	2	3	100	No	0	1	5	24
237	Coccomyxa sp.	1	14	7	100	Yes	5	1	5	24
238	Coccomyxa sp.	1	7	7	100	No	5	1	0.5	1
239	Coccomyxa sp.	1	5	7	10	Yes	0	1	0.5	24
240	Coccomyxa sp.	1	10	7	10	No	0	1	5	1
241	Coccomyxa sp.	2	2	3	100	No	0	1	5	24
242	Coccomyxa sp.	2	13	5	55	No	2.5	0.5	2.75	12.5
243	Coccomyxa sp.	2	7	7	100	No	5	1	0.5	1
244	Coccomyxa sp.	2	4	7	100	Yes	0	0	0.5	1
245	Coccomyxa sp.	2	15	3	10	No	0	0	0.5	1
246	Coccomyxa sp.	2	9	3	100	Yes	5	0	5	1
247	Coccomyxa sp.	2	11	5	55	No	2.5	0.5	2.75	12.5
248	Coccomyxa sp.	2	1	7	10	No	5	0	5	24
249	Coccomyxa sp.	2	6	3	100	No	5	0	0.5	24
250	Coccomyxa sp.	2	14	7	100	Yes	5	1	5	24
251	Coccomyxa sp.	2	8	3	10	Yes	0	0	5	1
252	Coccomyxa sp.	2	3	3	10	Yes	5	1	0.5	1
253	Coccomyxa sp.	2	10	7	10	No	0	1	5	1
254	Coccomyxa sp.	2	5	7	10	Yes	0	1	0.5	24
255	Coccomyxa sp.	2	12	5	55	No	2.5	0.5	2.75	12.5
256	Coccomyxa sp.	3	7	7	100	No	5	1	0.5	1
257	Coccomyxa sp.	3	13	5	55	No	2.5	0.5	2.75	12.5

RunOrder	Block (Species)	Replicate	DSD_Run	A_pH	B_Dilution_%	C_Fe_Pretreatment	D_Coagulant_mM	E_pH	F_EDTA_mM	G_Dosage_g/L	H_Time_h
258	Coccomyxa sp.	3	2	3	100	No	0	1	5	24	
259	Coccomyxa sp.	3	10	7	10	No	0	1	5	1	
260	Coccomyxa sp.	3	5	7	10	Yes	0	1	0.5	24	
261	Coccomyxa sp.	3	12	5	55	No	2.5	0.5	2.75	12.5	
262	Coccomyxa sp.	3	3	3	10	Yes	5	1	0.5	1	
263	Coccomyxa sp.	3	15	3	10	No	0	0	0.5	1	
264	Coccomyxa sp.	3	8	3	10	Yes	0	0	5	1	
265	Coccomyxa sp.	3	1	7	10	No	5	0	5	24	
266	Coccomyxa sp.	3	6	3	100	No	5	0	0.5	24	
267	Coccomyxa sp.	3	14	7	100	Yes	5	1	5	24	
268	Coccomyxa sp.	3	9	3	100	Yes	5	0	5	1	
269	Coccomyxa sp.	3	4	7	100	Yes	0	0	0.5	1	
270	Coccomyxa sp.	3	11	5	55	No	2.5	0.5	2.75	12.5	

Response Variables

The primary response variables to be measured for each of the 270 runs include:

- **Residual Metal Concentrations (mg/L)**: Fe, B, Co, Cu, Ni, Al, Nd, Dy, Pr (measured by ICP-MS/OES).
- **Final Process Parameters**: Final pH, Final COD.
- **Biomass Health**: Cell viability or equivalent metric for living biomass experiments.

Import Instructions

The table above can be saved as a CSV file and imported directly into JMP (`File > Open > CSV`) or Minitab (`Stat > DOE > Define Custom Design`). Ensure columns are assigned the correct data type (continuous or nominal/categorical) after import.

Quality Control Runs

In addition to the 270 experimental runs, a set of 24 Quality Control (QC) runs should be interspersed throughout the experiment (4 per block):

- **2 Blanks**: Leachate matrix without biomass to assess abiotic precipitation.
- **1 Spike**: Leachate matrix with biomass, spiked with a known concentration of key metals to assess matrix interference and recovery.
- **1 Duplicate**: An exact repeat of a randomly chosen DSD run within the block to assess reproducibility. This brings the total number of analytical runs for Stage 1 to 294.

Part 2: Budget Sensitivity Analysis

This analysis examines the sensitivity of the $2.27M baseline project budget to two critical variables: the sourcing strategy for ICP-MS analysis and the final classification of the generated biomass sludge.

Baseline Budget Breakdown

The estimated baseline budget of $2.27M is allocated as follows (based on the previous report's analysis):

- **Personnel (2 FTE for 24 months)**: $480,000
- **Laboratory Consumables & Reagents**: $250,000
- **Capital Equipment (Pilot Reactor, etc.)**: $250,000
- **Analytical Costs (Baseline: Outsourced)**: $990,000 (3,000 samples @ $330/sample avg.)
- **Waste Disposal (Baseline: Hazardous)**: $40,000 (0.1 tons @ $400/ton + fees)
- **Overhead/Indirect Costs (40%)**: $804,000
- **Total: $2,814,000** (Note: The baseline is re-calculated here based on updated assumptions for clarity.)

ICP-MS Sourcing Scenarios

The total analytical cost is a major budget driver. We assume a total of 3,000 samples over the 24-month project.

- **Scenario A: Fully Outsourced**: Analysis is performed by an external commercial lab.
  - Cost per sample: $65 - $165 (average $115) per sample.
  - Total Cost: 3,000 samples * $115/sample = **$345,000**
- **Scenario B: In-House ICP-MS**: A mid-range instrument is purchased and operated internally.
  - Capital Cost: ~$250,000 - $350,000 (average $300,000).
  - Annual Operating Cost (Service, Gas, Consumables): 20-30% of capital, or ~$75,000/year. Total for 2 years = $150,000.
  - Per-Sample Cost: ~$15 - $30 (average $22.50).
  - Total Cost: $300,000 (CAPEX) + $150,000 (Annual) + (3,000 samples * $22.50/sample) = **$517,500**
- **Scenario C: Shared University Facility**: Analysis is performed using a university core facility.
  - Cost per sample: $81 - $150 per hour, or per-sample rates ranging from ~$60 to $200 for external academics. Assume an average of **$100/sample**.
  - Total Cost: 3,000 samples * $100/sample = **$300,000**

Break-Even Analysis: The primary trade-off is between the high capital cost of an in-house instrument and the higher per-sample cost of outsourcing. The break-even point where the total cost of Scenario B (In-House) equals Scenario C (Shared Facility) is approximately **2,833 samples**. Given the projected 3,000 samples, purchasing an instrument is slightly more expensive over the project lifetime but provides greater flexibility and faster turnaround times.

Sludge Classification Scenarios

Based on a pilot-scale run generating an estimated 0.1 tons of dry, metal-laden biomass.

- **Scenario X: Hazardous Waste**: The biomass fails TCLP tests and must be disposed of at a licensed facility.
  - Cost per ton: $300 - $800 (average $550).
  - Total Cost: 0.1 tons * $550/ton + fees = **~$1,500**
- **Scenario Y: Non-Hazardous Waste**: The biomass passes TCLP tests and can be disposed of in a standard industrial landfill.
  - Cost per ton: $50 - $150 (average $100).
  - Total Cost: 0.1 tons * $100/ton + fees = **~$1,000**

Combined Sensitivity Matrix

The following matrix shows the total estimated project budget under all six scenarios. The baseline is Scenario A-X (Outsourced Analytics, Hazardous Sludge).

Scenario	ICP-MS Sourcing	Sludge Classification	Total Analytical Cost	Total Disposal Cost	Total Project Budget	% Change from Baseline
A-X (Baseline)	Outsourced	Hazardous	$345,000	$1,500	**$2,284,100**	0.0%
A-Y	Outsourced	Non-Hazardous	$345,000	$1,000	**$2,283,400**	-0.03%
B-X	In-House	Hazardous	$517,500	$1,500	**$2,525,100**	+10.5%
B-Y	In-House	Non-Hazardous	$517,500	$1,000	**$2,524,400**	+10.5%
C-X	Shared Facility	Hazardous	$300,000	$1,500	**$2,221,100**	-2.8%
C-Y	Shared Facility	Non-Hazardous	$300,000	$1,000	**$2,220,400**	-2.8%

Tornado Diagram Data

The variables are ranked by their total impact on the budget (difference between highest and lowest cost scenarios for that variable).

1. **ICP-MS Sourcing**: $517,500 (In-House) - $300,000 (Shared) = **$217,500**
2. **Sludge Classification**: $1,500 (Hazardous) - $1,000 (Non-Hazardous) = **$500**

The analysis clearly shows that the choice of analytical sourcing has an impact on the budget that is orders of magnitude greater than the impact of sludge classification at this project scale. The most budget-favorable path is using a shared academic facility, while the most expensive is purchasing an instrument in-house for this specific project duration and sample volume.

Part 3: Weighted Technology Selection Decision Matrix

MCDA Methodology

A Weighted Sum Model (WSM) is used for its simplicity and transparency in evaluating the trade-offs between different wastewater treatment technologies. This method scores each technology against a set of weighted criteria, allowing for a quantitative ranking of the alternatives.

Criteria and Weights

The following criteria and weights were selected to represent a balanced view of performance, cost, and operational factors.

Criterion	Weight	Rationale
Regulatory Compliance	0.30	The primary goal is achieving discharge limits. This is the most critical factor.
Lifecycle Cost	0.25	Includes both CAPEX and OPEX, reflecting the total economic impact.
Operational Complexity	0.15	Ease of operation, automation potential, and skill level required.
Sludge Impact	0.15	Volume, hazardous classification, and disposal cost of waste solids.
Technology Readiness (TRL)	0.10	Maturity and reliability of the technology for industrial applications.
Physical Footprint	0.05	Land area required for the installation.

Scoring Rubric

Each technology is scored on a scale of 1 (worst) to 5 (best) for each criterion based on the evidence gathered.

Score	Compliance	Cost	Complexity	Sludge Impact	TRL	Footprint
5	Meets all stringent limits	Lowest lifecycle cost	Fully automated, simple	Minimal, non-hazardous	TRL 9, widely deployed	Very Compact
4	Meets most limits	Low lifecycle cost	Mostly automated	Low volume, non-haz	TRL 8, proven	Compact
3	Meets moderate limits	Moderate lifecycle cost	Standard automation	Moderate volume or haz	TRL 6-7, pilot proven	Moderate
2	Meets few limits	High lifecycle cost	Manual, specialist needed	High volume or haz	TRL 5, component proven	Large

Score	Compliance	Cost	Complexity	Sludge Impact	TRL	Footprint
1	Fails key limits	Highest lifecycle cost	Very complex, unreliable	Very high volume, hazardous	TRL < 4, lab concept	Very Large

Technology Scoring and Justification

The four technologies are scored based on the rubric, with justifications derived from the literature.

Technology	Compliance Score	Cost Score	Complexity Score	Sludge Score	TRL Score	Footprint Score
Microalgae Biosorption	**1** (Fails on B, unknown on others)	**2** (High biomass cost, uncertain CAPEX)	**1** (Biological process control is complex)	**3** (Moderate volume, likely hazardous)	**2** (TRL 3-5 for this application)	**2** (Large ponds or complex PBRs)
Chemical Precipitation	**3** (Meets most metal limits, fails B)	**3** (Low CAPEX, high sludge/reagent OPEX)	**4** (Robust, well-automated process)	**1** (Very high volume, hazardous sludge)	**5** (TRL 9)	**4** (Compact)
Ion Exchange (as Polisher)	**4** (Meets very strict limits, incl. B)	**2** (High OPEX due to regenera-tion/resin cost)	**3** (Automated but requires chemical handling)	**4** (Low solid waste, liquid regenerant)	**5** (TRL 9)	**5** (Very compact)
Activated Carbon (as Polisher)	**2** (Good for organics, poor for most ions)	**3** (Moderate CAPEX/OPEX)	**5** (Very simple passive system)	**4** (Low solid waste, requires regeneration)	**5** (TRL 9)	**4** (Compact)

Decision Matrix Table (Baseline Weights)

Technology	Compliance (x0.30)	Cost (x0.25)	Complexity (x0.15)	Sludge (x0.15)	TRL (x0.10)	Footprint (x0.05)	**Total Score**	**Rank**
Microalgae Biosorp-tion	0.30	0.50	0.15	0.45	0.20	0.10	**1.70**	4
Chemical Precipita-tion	0.90	0.75	0.60	0.15	0.50	0.20	**3.10**	**1**
Ion Exchange (Polisher)	1.20	0.50	0.45	0.60	0.50	0.25	**3.50**	N/A*

Technology	Compliance (x0.30)	Cost (x0.25)	Complexity (x0.15)	Sludge (x0.15)	TRL (x0.10)	Footprint (x0.05)	**Total Score**	**Rank**
Activated Carbon (Polisher)	0.60	0.75	0.75	0.60	0.50	0.20	**3.40**	N/A*

**Ion Exchange and Activated Carbon are polishing steps and not directly comparable as standalone primary treatments. However, their high scores indicate their value as part of a hybrid system.*

Jurisdiction-Specific Sensitivity

The criteria weights are adjusted to reflect different regional priorities.

	EU (Strict)	**US (Moderate)**	**China (Cost)**
Compliance	0.40	0.30	0.20
Cost	0.15	0.25	0.40
Complexity	0.15	0.15	0.15
Sludge	0.20	0.15	0.10
TRL	0.05	0.10	0.10
Footprint	0.05	0.05	0.05
Winning Technology (Primary)	**Chemical Precipitation** (Score: 3.10)	**Chemical Precipitation** (Score: 3.10)	**Chemical Precipitation** (Score: 3.25)

Final Recommendation

Across all tested weighting scenarios, **Chemical Precipitation** remains the highest-ranked primary treatment technology. Its maturity, reliability, and effectiveness at removing the bulk of regulated metals outweigh its significant drawbacks in sludge production and inability to remove boron.

The analysis strongly suggests that a **hybrid approach** is necessary for full compliance, especially in jurisdictions with strict boron limits like the EU. The optimal configuration would be:

1. **Primary Treatment**: Chemical precipitation to remove the bulk of Fe, REEs, and other precipitable metals.
2. **Secondary/Polishing Treatment**: An Ion Exchange system using boron-selective resins to meet the stringent boron limits and remove any residual trace metals.

Microalgae biosorption, in its current state (TRL 3-5), is not a viable standalone solution for achieving regulatory compliance for complex NdFeB leachates. Its low scores in compliance, complexity, and TRL make it a high-risk option. However, if the challenges of iron interference and co-contaminant toxicity can be overcome through the proposed research, it could potentially serve as a lower-cost, greener polishing step to replace or augment ion exchange in the future, particularly if metal recovery from the biomass becomes economically viable.

Technical Report: Experimental Design and Cost Analysis for Microalgae-Based Treatment of NdFeB Leachate

PART 1: Reduced Fractional Factorial Screening Design

1.1 Full Factorial Problem Statement

A full factorial experimental design to screen all potential variables for microalgae biosorption of NdFeB leachate is infeasible due to the excessive number of required experiments. The original proposed design includes seven key factors: Species (6 levels), Biomass Form (3 levels: living, dried, immobilized), Leachate Dilution (4 levels), pH (6 levels), Iron (Fe) Pre-removal (2 levels), Coolant concentration (3 levels), and EDTA concentration (3 levels). A full exploration of this parameter space would require $6 \times 3 \times 4 \times 6 \times 2 \times 3 \times 3 = 7{,}776$ unique experimental conditions. With a minimum of three replicates for statistical validity, this totals 23,328 individual runs, a number that is prohibitive in terms of time, cost, and resources for a laboratory screening program.

1.2 Recommended Design: Two-Stage Approach

To overcome the limitations of a full factorial design, a two-stage approach is recommended, leveraging modern Design of Experiments (DoE) methodologies for efficiency.

Stage 1: Definitive Screening Design (DSD) The initial screening phase should employ a Definitive Screening Design (DSD). DSDs are highly efficient three-level designs that can estimate main effects, two-factor interactions, and quadratic effects with the lowest possible number of experiments. For the seven continuous or discrete three-level factors identified, a DSD requires only $2k + 1 = 2(7) + 1 = 15$ experimental runs per species. By testing all six candidate species, this stage would consist of 6 species × 15 runs/species = 90 unique conditions. With triplicates, the total run count is a manageable 270 experiments. This represents a >98% reduction in experimental effort compared to the full factorial design while preserving the ability to identify significant linear effects and key second-order effects.

Stage 2: Response Surface Methodology (RSM) Optimization Following the DSD screening, the top 3-4 most influential factors identified will be further optimized using a Response Surface Methodology (RSM) such as a Central Composite Design (CCD) or Box-Behnken Design (BBD). These designs are specifically suited for fitting second-order polynomial equations to map the response surface and find optimal conditions. For example, optimizing four factors using a CCD would require approximately 30 runs per species. Focusing on the 2-3 best-performing species from Stage 1, this optimization phase would involve an additional 180-270 runs (including triplicates), allowing for a detailed understanding of process optima with high precision.

1.3 Detailed Run Matrix for Stage 1

The seven factors and their three levels (+1, 0, -1) for the DSD are defined as follows:

- **A (pH):** 2 (-1), 4 (0), 6 (+1)
- **B (Leachate Dilution):** 10% (-1), 50% (0), 100% (+1)
- **C (Fe Pre-removal):** No (-1), Partial (0), Full (+1)
- **D (Coolant Conc.):** 0% (-1), 1% (0), 5% (+1)
- **E (EDTA Conc.):** 0 mM (-1), 0.1 mM (0), 1.0 mM (+1)
- **F (Biomass Dosage):** 0.5 g/L (-1), 3 g/L (0), 10 g/L (+1)
- **G (Contact Time):** 30 min (-1), 4 h (0), 24 h (+1)

The 15-run DSD matrix for these seven factors is shown below. This matrix would be executed for each of the six selected microalgae species.

Run	A (pH)	B (Dilution)	C (Fe Pre-removal)	D (Coolant)	E (EDTA)	F (Dosage)	G (Time)
1	+1	+1	+1	+1	+1	+1	+1
2	-1	-1	-1	-1	-1	-1	-1
3	+1	+1	-1	-1	-1	+1	+1
4	-1	-1	+1	+1	+1	-1	-1
5	+1	-1	+1	+1	-1	-1	+1
6	-1	+1	-1	-1	+1	+1	-1
7	+1	-1	-1	+1	+1	+1	-1
8	-1	+1	+1	-1	-1	-1	+1
9	-1	+1	+1	+1	-1	+1	-1
10	+1	-1	-1	-1	+1	-1	+1
11	-1	-1	+1	-1	+1	+1	+1
12	+1	+1	-1	+1	-1	-1	-1
13	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0

1.4 Sample Count Budget

The reduced experimental design leads to a practical sample budget for the 24-month program. The total number of Inductively Coupled Plasma Mass Spectrometry (ICP-MS) samples is estimated to be around 1,200.

- **Phase 1 (Characterization):** 50 samples
- **Phase 2 (Screening):**
  - Stage 1 (DSD): 270 runs (90 conditions × 3 replicates) + 30 blanks = 300 samples
  - Stage 2 (RSM): ~270 runs (90 conditions × 3 replicates) + 30 blanks = ~300 samples
  - Confirmatory Runs: 50 samples
- **Phase 4 (Pilot Validation):** ~500 samples (e.g., daily sampling over 30 days with triplicates and process points)

Given external academic rates for ICP-MS analysis can be $40-$120 per sample, plus digestion fees of $25-$45, this budget emphasizes the need for an efficient experimental plan.

1.5 Statistical Power

The DSD approach provides excellent statistical power for its run size. It is specifically designed to resolve all main effects independently of any two-factor interactions (2FI) or quadratic effects, a property not shared by traditional fractional factorial designs. This ensures that the primary drivers of biosorption performance can be identified without confounding. For key interactions, such as those involving Fe, pH, and species, the DSD allows for the estimation of these effects if they are large. With triplicates and an alpha of 0.05, the statistical power to detect a factor with an effect size of 20% is expected to be greater than 0.80.

1.6 Species Strategy

Based on the evidence from the literature review, the following six species are recommended for the screening phase, justified by their documented performance:

1. ***Chlorella vulgaris***: This species has been tested on actual NdFeB magnet leachate and demonstrated a high maximum uptake capacity for Nd (qmax of 188.68 mg/g). It also shows high removal for other metals like Mn (99.4%) and Co (100% from real wastewater).
2. ***Galdieria sulphuraria***: As an extremophile, this species is highly tolerant of the low pH and high metal concentrations characteristic of NdFeB leachates. Its ability to grow in acidic conditions (pH 1.5-2.5) makes it a strong candidate for treating raw, unadjusted leachate.
3. ***Desmodesmus sp. / Scenedesmus sp.***: Species from this genus, such as *Scenedesmus almeriensis*, have been tested for boron and arsenic removal, showing tolerance and moderate uptake (up to 38.6% for B). They are robust and commonly studied in wastewater treatment.
4. ***Arthrospira platensis (Spirulina)***: This cyanobacterium has demonstrated rapid biosorption kinetics for rare earth elements like Dysprosium (38% removal in 5 minutes) and a maximum uptake capacity of 3.24 mg/g.
5. ***Nostoc sp.***: This cyanobacterium showed the highest adsorption capacity for a range of REEs (Ce, Nd, Tb, La) in screening studies, reaching 84.2–91.5 mg/g.
6. ***Coccomyxa sp.***: Strains of this genus isolated from extreme environments have shown high stress tolerance and the ability to remove >90% of REEs from water within a few hours, making them promising for harsh industrial effluents.

PART 2: Month-9 Go/No-Go Decision Tree

2.1 Decision Framework

A quantitative, stage-gate decision framework based on Technology Readiness Level (TRL) and Best Available Techniques (BAT) assessment principles will guide the project's progression. Go/No-Go decisions at each gate will be based on achieving predefined, numeric performance criteria directly linked to the final goal of regulatory compliance for NdFeB leachate.

2.2 Gate 1: Biological Viability (Month 4)

This initial gate assesses the fundamental ability of the selected microalgae to survive in the toxic leachate environment.

- **PASS CRITERION**: At least two of the six tested species demonstrate greater than 50% viability when exposed to 25% (v/v) real NdFeB leachate for 48 hours. Viability is measured by chlorophyll fluorescence (Fv/Fm > 0.3) or cell counts (>50% of control).
- **ACTION (PASS)**: Proceed to Gate 2 with the viable species.
- **FAIL CRITERION**: Fewer than two species meet the viability threshold.
- **CONTINGENCY (FAIL)**: The project pivots to focus exclusively on non-living (dried, pre-treated) biomass for all subsequent experiments, as non-living biomass has also shown significant potential as a biosorbent.

2.3 Gate 2: Single-Metal Removal (Month 6)

This gate evaluates the performance of the most promising configurations (species/biomass form) from Stage 1 screening against individual, key contaminants at representative concentrations. Thresholds are set to demonstrate a meaningful step towards regulatory compliance.

- **PASS CRITERION**: For a synthetic multi-metal solution simulating the leachate, at least one experimental condition (for one or more species) achieves removal for ≥4 of the 6 target contaminants below their specified thresholds:
  - **Fe**: < 5 mg/L (from an initial ~500 mg/L)
  - **B**: < 10 mg/L (from an initial ~50 mg/L)
  - **Co**: < 0.5 mg/L (from an initial ~10 mg/L)
  - **Cu**: < 0.5 mg/L (from an initial ~50 mg/L)
  - **Ni**: < 0.5 mg/L (from an initial ~50 mg/L)
  - **COD**: No net increase > 500 mg/L (to control for organic leaching from biomass)
- **ACTION (PASS)**: Proceed to Gate 3 with the best-performing conditions for full multi-metal compliance testing.
- **FAIL CRITERION**: No condition achieves the threshold for at least four contaminants.
- **CONTINGENCY (FAIL)**: Re-evaluate the experimental matrix. If failure is due to a single recalcitrant metal (e.g., Boron), the project scope may be narrowed to focus on a hybrid system where microalgae are used as a polishing step after conventional treatment removes the difficult contaminant.

2.4 Gate 3: Multi-Metal Simultaneous Compliance (Month 9)

This is the critical Go/No-Go decision point for the project, assessing the technology's potential to achieve simultaneous compliance for all key contaminants in a real, complex leachate matrix.

- **GREEN (GO)**: At least one optimized condition simultaneously reduces all 6 target contaminants (Fe, B, Co, Cu, Ni, COD) below their respective regulatory discharge limits (e.g., EU BAT-AEL or US EPA 40 CFR 433).
- **YELLOW (PROCEED WITH CAUTION)**: At least one condition reduces 4-5 of the 6 contaminants to below their limits, and the remaining contaminant(s) are within 2x of their limit. This triggers a focused effort on a hybrid system or polishing step for the non-compliant metal(s).
- **RED (NO-GO)**: Fewer than 4 contaminants are brought into compliance, or any single contaminant remains more than 10x above its discharge limit. The microalgae-only approach is deemed technically infeasible.
- **BLACK (ABORT)**: No measurable removal is observed for key contaminants, or total biomass death occurs in all tested conditions with real leachate. The project is terminated.

2.5 Decision Tree Flowchart

```
[Start Program]
    |
    |--> [Gate 1: Biological Viability (Month 4)]
    |    - Test: >=2 species >50% viability in 25% leachate?
    |    |
    |    |-- (PASS) --> [Proceed to Gate 2 with viable species]
    |    |
      |      '-- (FAIL) --> [PIVOT: Focus exclusively on non-living biomass] --
> [Proceed to Gate 2]
    |
    |--> [Gate 2: Single-Metal Removal (Month 6)]
    |    - Test: >=4 of 6 target contaminants below threshold in synthetic leachate?
    |    |
    |    |-- (PASS) --> [Proceed to Gate 3 with optimized conditions]
```

```
    |    |
    |    '-- (FAIL) --> [CONTINGENCY: Re-scope to Hybrid System (e.g., chemical pre-
treatment + microalgae polishing)] --> [Re-evaluate at Gate 3]
    |
    |--> [Gate 3: Multi-Metal Compliance (Month 9)]
    |    - Test: Simultaneous removal in real leachate vs. regulatory limits.
    |    |
    |    |-- (GREEN - GO) --> [Proceed to Phase 4 Pilot Validation]
    |    |
   |   |-- (YELLOW - CAUTION) --> [PROCEED: Focus on Hybrid System/Polishing Step for non-
compliant metals]
    |    |
   |   |-- (RED - NO-GO) --> [CONCLUSION: Microalgae-only treatment is infeasible. Evaluate hybrid opti
    |    |
    |    '-- (BLACK - ABORT) --> [TERMINATE PROJECT]
```

2.6 Contingency Actions

- **Gate 1 Fail**: Shift all resources from culturing living cells to procuring and preparing dried/immobilized biomass. This mitigates risks associated with leachate toxicity to living organisms.
- **Gate 2 Fail**: Immediately initiate a parallel study on conventional pre-treatment (e.g., chemical precipitation) for the most recalcitrant contaminant(s). The microalgae project timeline is adjusted to focus on its role as a polishing step for the remaining metals.
- **Gate 3 Fail (RED/BLACK)**: A final report is generated concluding that microalgae treatment is not a viable standalone or primary treatment technology for this waste stream. Resources are reallocated to evaluating alternative technologies like ion exchange or advanced oxidation, which have high efficiency but are often more expensive.

PART 3: Comparative Cost-Effectiveness Analysis

3.1 Scenario: 10 m^3/day Grinding Sludge Leachate

A comparative cost analysis is performed for a hypothetical scenario involving the treatment of 10 m^3/day (2,642 gal/day) of highly concentrated NdFeB grinding sludge leachate. The influent is characterized by:

- **Fe**: 5,000 mg/L
- **Total REE**: 1,920 mg/L (Nd, Dy, Pr)
- **B**: 500 mg/L
- **Co**: 100 mg/L
- **Cu**: 50 mg/L
- **Ni**: 50 mg/L
- **pH**: 2.0

This composition is representative of acidic leachates from magnet manufacturing waste, which are dominated by iron and present significant treatment challenges.

3.2 Technology 1: Chemical Precipitation + Clarification

This conventional approach involves raising the pH to 9-10 with lime ($Ca(OH)_2$) or caustic soda (NaOH) to precipitate metal hydroxides. It is a simple, low-cost, and robust method for bulk heavy metal removal.

- **Capital Cost**: Low to moderate. A basic system for this flow rate, including tanks, pumps, pH control, and a clarifier, would likely be in the range of $50,000 - $150,000.
- **Operating Cost**: Moderate. Dominated by chemical consumption (lime/NaOH) and sludge disposal. Estimated OpEx: $5 - $15 per m^3.
- **Performance**: Highly effective for Fe, Cu, Ni, and Co, achieving >95-99% removal to meet typical discharge limits.
- **Limitations**: Ineffective for Boron and does not selectively separate REEs, which co-precipitate with iron hydroxides. Generates large volumes of hazardous sludge, with disposal costs ranging from $160/drum to over $1,000/ton if classified as hazardous.

3.3 Technology 2: Precipitation + Ion Exchange

This two-stage process uses chemical precipitation to remove the bulk of Fe and other base metals, followed by a polishing step with selective ion exchange (IX) resins to remove residual metals and target specific contaminants like boron and REEs.

- **Capital Cost**: High. Includes the precipitation system plus the IX system (vessels, resin, regeneration equipment), likely totaling $200,000 - $400,000 for this scale. Specialty boron or REE-selective resins can cost $500 - $2,000 per cubic foot.
- **Operating Cost**: High. Includes chemicals for precipitation, IX resin regeneration (acids, bases), and disposal of both sludge and spent regenerant brine. Estimated OpEx: $15 - $40 per m^3.
- **Performance**: Very high removal efficiency (>99%) for most metals. Can achieve very low discharge limits, including for boron (to <2 mg/L) and can selectively capture REEs if the correct resin is used.
- **Limitations**: High operational complexity and cost. Resin fouling is a significant concern, and the regeneration process creates a concentrated, hazardous liquid waste stream that requires further management.

3.4 Technology 3: Microalgae Biosorption (Standalone)

This scenario evaluates using only microalgae biosorption to treat the raw leachate.

- **Capital Cost**: Moderate. Requires photobioreactors or ponds, harvesting equipment (centrifuge/filter press), and biomass handling systems. A pilot-scale (50-100L) continuous reactor alone can cost $50,000-$100,000.
- **Operating Cost**: Varies widely. Biomass production is a key cost driver, with estimates ranging from $43 to over $1,100 per ton. Energy for mixing/pumping and harvesting is also significant. Estimated OpEx: Potentially $5 - $25 per m^3 if low-cost biomass is available.
- **Performance**: Highly unlikely to be effective as a standalone process. The extremely high iron (5,000 mg/L) and low pH (2.0) would likely be toxic to most species and cause severe competitive inhibition, preventing the removal of other target metals. Boron removal by microalgae is generally low, especially in competitive environments.
- **Limitations**: Not robust for this high-strength, complex waste stream. Performance is highly sensitive to pH, metal toxicity, and competition. The resulting metal-laden biomass would be classified as hazardous waste, incurring high disposal costs.

3.5 Technology 4: Hybrid Precipitation + Microalgae Polishing

This is the most plausible scenario involving a biological component. Bulk iron (>99%) is first removed via chemical precipitation by raising the pH to ~3.5-4.0. The decanted, iron-depleted supernatant is then treated with microalgae biomass to remove (polish) remaining Co, Ni, Cu, and potentially some REEs.

- **Capital Cost**: Moderate to high. Combines a simplified precipitation system (targeting only iron) with a smaller-scale biosorption system. Estimated total CAPEX: $100,000 - $250,000.
- **Operating Cost**: Moderate. Chemical costs are lower than full precipitation, as pH adjustment is less extreme. Biosorption OpEx is for polishing a lower-volume, less toxic stream. Estimated combined OpEx: $8 - $20 per m^3.
- **Performance**: Potentially high. The pre-treatment step removes the main interferent (Fe), allowing the microalgae to effectively target the remaining metals like Co and Ni, for which high removal efficiencies have been demonstrated. Boron removal would remain a challenge.
- **Limitations**: Still generates two waste streams: a large volume of iron sludge and a smaller volume of metal-laden biomass. The process is more complex than precipitation alone. Boron compliance would likely require an additional dedicated unit (e.g., boron-selective IX resin).

3.6 Comparative Cost Table

Technology	Capital Cost (Est.)	OpEx (Est. $/m³)	Sludge Generation	Total Cost/m³ (Est.)	Key Metals Compliant?
1. Chemical Precipitation	Low-Moderate	$5 - $15	High (Hazardous Sludge)	$10 - $25	**Yes**: Fe, Co, Cu, Ni. **No**: B, REEs.
2. Precipitation + Ion Exchange	High	$15 - $40	High (Sludge + Brine)	$25 - $50	**Yes**: All target metals, including Boron.
3. Microalgae Biosorption (Standalone)	Moderate	$5 - $25+	Moderate (Hazardous Biomass)	$15 - $40+	**No**: Infeasible due to toxicity and competition.
4. Hybrid (Precipitation + Microalgae)	Moderate-High	$8 - $20	High (Sludge + Biomass)	$15 - $35	**Yes**: Fe, Co, Cu, Ni. **Partial**: REEs. **No**: B.

3.7 Cost-Effectiveness Ratio

To compare these technologies beyond total cost, a cost-effectiveness ratio can be calculated. This would involve dividing the total annualized cost of each system by the total mass of contaminants removed (e.g., $/kg of metal removed) or by the number of parameters brought into compliance. A lower ratio indicates higher efficiency. For this scenario, Technology 2 (Precip + IX) would likely have the highest total cost but also the best performance, while Technology 4 (Hybrid) may offer the most balanced cost-effectiveness if boron compliance is not strictly required or is handled separately.

3.8 Non-Cost Factors

Beyond direct costs, other factors influence technology selection:

- **Footprint**: Precipitation and biosorption systems (especially ponds) require significant space, whereas IX systems are more compact.
- **Operational Complexity**: The hybrid and IX systems require more skilled operators than a simple precipitation plant.
- **Sludge Toxicity**: Sulfide precipitates are generally less leachable and more stable than hydroxide precipitates, but generate H_2S gas risks. Metal-laden biomass would need to pass TCLP testing to determine its disposal classification.
- **Energy Consumption**: Pumping and aeration for biological systems and high-pressure pumps for RO are major energy consumers.
- **Robustness**: Chemical precipitation is highly robust and tolerant of fluctuations in influent quality, while biological systems can be more sensitive to shock loads and toxicity.

Target Funding Agencies and Programs

Primary and secondary funding opportunities relevant to this research program, which integrates environmental engineering, critical mineral sustainability, and innovative wastewater treatment, have been identified from U.S. federal agencies.

Primary Targets:

- **National Science Foundation (NSF) - Environmental Engineering (ENG/CBET)**: This core program supports fundamental research in wastewater treatment, biosorption, and resource recovery. Unsolicited proposals are accepted anytime under program code PD 20-1440, with awards typically lasting up to three years. The program emphasizes novelty and transformative potential, making it an excellent fit for the proposed research's innovative approach. Larger, interdisciplinary proposals may also align with the ECO-CBET program, which funds 4-year projects from $1,500,000 to $1,700,000.
- **Department of Energy (DOE) - FECM & EERE**: The DOE has multiple funding announcements directly targeting critical minerals and rare earth elements (REEs).
  - **DE-FOA-0002956 (Advancing Technology Development for Securing a Domestic Supply of Critical Minerals and Materials)**: Managed by the Office of Fossil Energy and Carbon Management (FECM), this $19.5M program offers awards from $500,000 to $2.5 million for projects recovering REEs from unconventional resources, including produced water and recycled materials. The program requires a 20% cost share, with a full application deadline of November 26, 2024.
  - **ARPA-E RECOVER Program**: This program is specifically designed to fund technologies that extract critical minerals from industrial wastewater, making it a perfect thematic fit. While selections for the initial $40 million round are pending, future rounds are anticipated.
- **Environmental Protection Agency (EPA) - Science to Achieve Results (STAR) Program**: The EPA's STAR program periodically issues Requests for Applications (RFAs) for innovative water treatment technologies. While current open opportunities are focused on nanosensors for PFAS, the program has a history of funding projects related to wastewater treatment and critical mineral recovery, making it a key agency to monitor for future relevant solicitations.

Project Abstract (250 words)

The domestic production of high-strength NdFeB permanent magnets, essential for defense and clean energy technologies, generates a complex and hazardous wastewater stream. This leachate contains not only valuable rare earth elements (REEs) but also high concentrations of iron, boron, cobalt, nickel, and residual process chemicals that are difficult to treat and must meet stringent regulatory discharge limits. Current conventional treatment, primarily chemical precipitation, is inefficient for co-contaminants like boron, generates large volumes of toxic sludge, and represents a missed opportunity for sustainable resource management. There is a critical gap in validated, cost-effective biological treatment methods for this specific, multi-contaminant industrial effluent. This proposal hypothesizes that an optimized microalgae-based biosorption process can achieve simultaneous regulatory compliance for all regulated metals and contaminants in NdFeB leachate. To test this, we will execute a 24-month, four-phase research program: (1) comprehensive chemical characterization of real industrial NdFeB waste streams; (2) high-throughput screening of six promising microalgae species and key process parameters using a Definitive Screening Design; (3) refinement of optimal conditions using Response Surface Methodology to maximize multi-contaminant compliance; and (4) validation of the optimized protocol in a 100-L continuous-flow pilot reactor using real industrial wastewater. This work is significant as it provides a green chemistry alternative that addresses a critical mineral supply chain vulnerability, reduces hazardous waste generation, and supports a circular economy. Expected outcomes include a compliance feasibility matrix, a validated pilot-scale treatment protocol, and a technology readiness assessment for

industrial adoption.

Specific Aims Page

Overall Goal

The overall goal of this project is to develop and validate a robust, microalgae-based biosorption process capable of treating complex NdFeB permanent magnet manufacturing wastewater to meet stringent U.S. and EU regulatory discharge limits.

Background and Rationale

The global production of NdFeB magnets, crucial for modern electronics, electric vehicles, and wind turbines, is projected to exceed 200,000 tonnes annually. The manufacturing process generates multi-component waste streams containing not only target REEs (Nd, Dy, Pr) but also high concentrations of iron (often >5,000 mg/L), boron (>500 mg/L), cobalt, nickel, and organic process chemicals. Discharging this effluent requires treatment to meet strict regulatory frameworks like the US EPA NPDES permits (40 CFR 433) and EU BAT-AELs, which set limits for metals such as nickel (<2.38 mg/L), copper (<2.07 mg/L), and chromium (<1.71 mg/L). Conventional chemical precipitation is widely used but is often ineffective for persistent contaminants like boron, generates large volumes of hazardous sludge, and represents a significant operational cost.

Microalgae-based biosorption offers a promising, cost-effective alternative. The literature demonstrates that various microalgae species, including *Chlorella vulgaris* and *Nostoc sp.*, can achieve high removal capacities for individual REEs, with some studies reporting uptake capacities exceeding 150 mg/g for neodymium. However, a critical knowledge gap exists: no systematic study has evaluated the performance of microalgae against the complete, complex chemical matrix of real NdFeB manufacturing leachate, which includes the combined inhibitory and competitive effects of extreme iron concentrations, high boron levels, and organic co-contaminants (coolants, chelators). This project directly addresses this gap by moving beyond single-metal synthetic solutions to test biosorption as a viable compliance technology for real-world industrial effluent. This work is directly aligned with national priorities to secure critical mineral supply chains, as outlined in initiatives by the DOE's Critical Minerals Institute (CMI).

Specific Aim 1: Characterize NdFeB Manufacturing Waste Streams and Establish Regulatory Compliance Targets

Hypothesis: A detailed chemical analysis of real NdFeB grinding, polishing, and cleaning waste streams will reveal a complex mixture of metals and organic compounds, with iron, boron, and COD being key parameters that exceed typical discharge limits. **Approach**: We will procure authentic wastewater samples from grinding, polishing, and coolant/cleaning operations from an industrial partner. Each stream will be comprehensively analyzed using ICP-MS/OES for a full suite of metals (Fe, B, Co, Cu, Ni, Al, Nd, Dy, Pr, Sm), ion chromatography, and standard methods for COD, TOC, oil and grease, and surfactants. We will compile a target compliance matrix based on the most stringent limits from EU BAT-AEL (NFM and STM BREFs) and US EPA 40 CFR 433 frameworks. **Expected Outcome**: A detailed report characterizing multiple real-world NdFeB waste streams and a definitive set of quantitative treatment targets (mg/L) for all regulated parameters. **Timeline**: Months 1-4.

Specific Aim 2: Screen Microalgae Species and Process Conditions Using Definitive Screening Design

Hypothesis: A statistically designed screening experiment will efficiently identify the most robust microalgae species and the most influential process factors (e.g., pH, iron pre-removal) for achieving multi-metal removal from real leachate. **Approach**: We will conduct a 270-run Definitive Screening Design (DSD) to investigate 7 factors: (1)

Algae Species (6 strains, including *Chlorella vulgaris*, *Galdieria sulphuraria*, *Desmodesmus sp.*); (2) Biomass Form (living, dried, immobilized); (3) Leachate Dilution (10-100%); (4) pH (3-7); (5) Iron Pre-removal (yes/no); (6) Coolant Presence (0-5%); and (7) EDTA Presence (0-1.0 mM). The primary response variable will be the residual concentration (mg/L) of each regulated metal, measured against the compliance targets established in Aim 1. **Expected Outcome**: Identification of the top 1-2 most effective microalgae species/biomass forms and the 3-4 most statistically significant process variables driving regulatory compliance. This aim includes a critical go/no-go gate at Month 9. **Timeline**: Months 4-9.

Specific Aim 3: Optimize Treatment Conditions via Response Surface Methodology for Multi-Metal Compliance

Hypothesis: The significant factors identified in Aim 2 can be optimized using Response Surface Methodology (RSM) to define an operational window that maximizes simultaneous compliance for the entire suite of regulated contaminants. **Approach**: Using the top species and factors from Aim 2, we will execute a Box-Behnken Design (90-162 runs). The RSM will model the quadratic effects and interactions of key variables (e.g., pH, biomass dosage, contact time) on the residual concentrations of all target metals. We will use a multi-response desirability function to numerically optimize the process for achieving the lowest collective deviation from all regulatory discharge limits simultaneously. **Expected Outcome**: A validated statistical model and a set of optimized operating parameters (e.g., pH 5.5, 2.5 g/L biomass, 60 min contact time) that predict the maximum achievable compliance for the complete leachate matrix. **Timeline**: Months 9-15.

Specific Aim 4: Validate at Pilot Scale with Real Wastewater and Assess Technology Readiness

Hypothesis: The optimized biosorption process from Aim 3 will demonstrate sustained, effective treatment of real NdFeB wastewater under continuous-flow conditions in a pilot-scale reactor, achieving compliance for a majority of regulated metals. **Approach**: We will construct and operate a 100-L continuous-flow packed-bed or stirred-tank bioreactor. The reactor will be fed with real industrial leachate and operated for 30 consecutive days at the optimized conditions determined in Aim 3. Daily influent and effluent samples will be analyzed to verify performance against the compliance target matrix. The generated metal-laden biomass (sludge) will be characterized for hazardous properties using the Toxicity Characteristic Leaching Procedure (TCLP). **Expected Outcome**: A comprehensive pilot-validation report with 30-day performance data, a final compliance matrix, sludge classification, and a Technology Readiness Level (TRL) assessment (Target TRL 4-5). **Timeline**: Months 15-24.

Innovation Statement

This project is innovative in several key aspects:

- It is the first systematic investigation of microalgae biosorption applied to the complete, chemically complex wastewater matrix from NdFeB magnet manufacturing, moving beyond simplified synthetic solutions.
- It employs a compliance-driven experimental design, using residual metal concentration (mg/L) versus regulatory limits as the primary response variable, rather than simple percent removal, which is more relevant for industrial application.
- It directly addresses the combined inhibitory effects of high iron, high boron, and organic process chemicals (coolants, chelators), which are critical but un-addressed challenges for biological treatment of this waste stream.
- The project culminates in a pilot-scale validation using real industrial wastewater, providing a crucial bridge between laboratory research and potential industrial deployment and generating a robust TRL assessment.

Broader Impacts

This research will have significant broader impacts. It will provide a foundation for a sustainable, green chemistry-based solution for a critical environmental challenge in the REE supply chain. The project will support the training of a graduate student and a postdoctoral researcher in advanced environmental biotechnology, statistical experimental design, and analytical chemistry. We will actively partner with our industrial collaborator to ensure the results are relevant and to facilitate a clear pathway for technology transfer. The findings will be disseminated through peer-reviewed publications and conference presentations, and will provide valuable data for informing future Best Available Techniques (BAT) designations for the metal finishing and electronics manufacturing sectors.

Budget Justification Summary

The requested budget of $2.27 million over 24 months is essential to accomplish the project's ambitious goals. Major cost categories include: **Personnel** (2.0 FTE: 1 postdoctoral researcher, 1 graduate student, plus PI/co-PI support); **Analytical Services**, which represents a significant portion of the budget due to the high cost and sample throughput required for ICP-MS analysis of complex multi-element matrices (commercial rates range from $65 to over $165 per sample); **Consumables**, including microalgae culture media, reagents for synthetic leachate, analytical standards, and materials for biomass immobilization; **Equipment**, for the construction and automation of the 100-L pilot-scale bioreactor; and **Travel/Dissemination** for site visits to our industrial partner and presenting findings at scientific conferences.

Elicit - Literature Review Evidence Table Microalgae Biosorption of NdFeB Leachate Contaminants

Metals Tested	Waste Type	Species
Nd (50–250 mg/L; stock standardized to 1000 mg/L Nd); Fe, B, Dy, Co analyzed but reported below detection limit in prepared solutions	Real NdFeB magnet leachate (aqueous solution derived from permanent Nd–Fe–B magnets; selective leaching used to remove Fe/B/Dy/Co interference)	**Chlorella vulgaris (SAG 211-12)**
Nd (initial 250 mg/L)	Mixed leachate solution derived from neodymium magnets (batch and continuous sorption systems)	**Chlorella vulgaris (dried biomass)**
Cu 5 ppm; La 5 ppm; Nd 5 ppm; Dy 5 ppm	Synthetic multi-metal solution (5 ppm each of Cu, La, Nd, Dy); tested at different pH	**Galdieria sulphuraria (SAG 21.92)**
La added to reach 20 μM and 100 μM (depending on condition/pH)	Synthetic La-spiked nutrient-replete growth medium (phosphate present)	**Galdieria sulphuraria (SAG 21.92)**
Eu^{3+} (0.111 mM and 0.191 mM), Sm^{3+} (0.139 mM), Nd^{3+} (0.103 mM)	Synthetic REE solutions: REE nitrate salts dissolved in modified Bold's Basal Medium (BBM)	**Anabaena sp. (SAG 12.82)**
Eu^{3+} (0.111 mM and 0.191 mM), Sm^{3+} (0.139 mM), Nd^{3+} (0.103 mM)	Synthetic REE solutions: REE nitrate salts dissolved in modified BBM	**Anabaena cylindrica (SAG 1403.2)**
REEs: Ce, Nd, Tb, La; competition tests: Ce in presence of Al, Pb, Ni, Zn (mixed-metal solutions)	Synthetic REE solutions (single-metal screening at 10 mM; plus equimolar mixed-metal competition tests)	**Nostoc sp. 20.02**
REEs: Ce, Nd, Tb, La; competition tests: Ce in presence of Al, Pb, Ni, Zn (mixed-metal solutions)	Synthetic REE solutions (single-metal screening at 10 mM; plus equimolar mixed-metal competition tests)	**Synechococcus elongatus (UTEX 2973)**
Single-metal solutions of Nd^{3+}, Eu^{3+} and others: Au^{3+}, Pb^{2+}, Pt^{4+}, Fe^{3+}, Cu^{2+}, Ni^{2+} (10 mM stocks; specific concentration series for Nd in isotherm tests)	Synthetic single-metal solutions (Nd^{3+}/Eu^{3+} and other metals as individual solutions; stock solutions 10 mM)	**Calothrix brevissima (SAG 34.79)**
Single-metal solutions of Nd^{3+}, Eu^{3+} and others: Au^{3+}, Pb^{2+}, Pt^{4+}, Fe^{3+}, Cu^{2+}, Ni^{2+} (10 mM stocks; Nd isotherm series also reported)	Synthetic single-metal solutions (Nd^{3+}/Eu^{3+} and other metals as individual solutions; stock solutions 10 mM)	**Chlorella kessleri (SAG 211-11g)**
Dy (10–100 mg/L)	Synthetic single-metal dysprosium solutions (Dy(NO3)2·6H2O; batch lab experiments)	**Spirulina platensis**
B 60 mg/L; As 12 mg/L; Cu 3 mg/L; Mn 3 mg/L; Zn 3 mg/L	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Chlamydomonas reinhardtii (strain 11-32b, SAG)**
B 60 mg/L; As 12 mg/L; Cu 3 mg/L; Mn 3 mg/L; Zn 3 mg/L	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Chlorella vulgaris**
B 60 mg/L; As 12 mg/L; Cu 3 mg/L; Mn 3 mg/L; Zn 3 mg/L	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Scenedesmus almeriensis**
As, B, Cu, Mn, Zn (mixture); initial concentrations not specified in this summary source	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Chlorophyceae spp. (indigenous strain from polluted environment)**
Industrial samples included: Co 0.316 mg/L (mine beneficiation); Tl 2.386 mg/L (smelter tailings); Cr 135.56 mg/L (smelter return water); simulated La 25–150 mg/L and Tl 150–800 mg/L also tested	Real industrial wastewaters (South Africa): mine beneficiation plant wastewater; smelter tailings/return waters; plus simulated La/Tl solutions	**Chlorella vulgaris (dried biomass)**
Simulated La 25–150 mg/L; simulated Tl 150–800 mg/L; industrial sample example: K 2036 mg/L (initial) reduced to 744.7 mg/L	Real industrial wastewaters (South Africa) and simulated La/Tl solutions	**Chlamydomonas reinhardtii**
B: 0.5, 10, 25, 50 mg/L	Synthetic boron-spiked water ("B stress" experiments at multiple B levels)	**Chlorella regularis**
Co^{2+} initial 63.6 mg/L (reported); also described as 60 mg/L added in methods	Synthetic single-metal cobalt solution added to photobioreactors (Co(NO3)2·6H2O; ~60–63.6 mg/L Co^{2+})	**Chlorella vulgaris**
Fe: 350 μg/L (0.35 mg/L)	Synthetic aqueous Fe test (study summarized in review table)	**Microcystis aeruginosa**
Ce (initial concentration not reported)	Synthetic Ce-containing aqueous solutions (reported in a review summary; primary study details not provided here)	**Nannochloropsis oculata**
Rare earth elements (specific REEs and initial concentrations not reported)	.Not mentioned in source	**Coccomyxa sp. (genus-level; strain not specified)**
La^{3+} (initial concentration not reported)	Synthetic La^{3+}-spiked water (study context not fully reported in excerpt)	**Euglena gracilis**

Compliance Status	Final Concentration Mg L	Removal Efficiency Pct
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Nd: % removal not tabulated; >90% of Nd biosorption occurred before equilibrium (equilibrium reached within ~30 min across temperatures)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	.Not mentioned in source (reported as uptake capacity, e.g., q = 157.21 mg/g at pH 5)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source (uptake reported as mg/g in some conditions)	Overall removal efficiency: <9% at pH 2.5; up to 86% at pH 5.0 (solution-wide metric; per-metal % not specified)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	No measurable La biosorption observed at initial pH 1.5 (under all growth conditions)
.Not assessed against discharge limits (not mentioned in source)	Eu residual reported as 76 ppb after >1 day; Sm residual 1.52×10^{-9} mM in early samples (Nd residual not numerically specified)	REE^{3+} biosorption reported as almost complete (>99%) within the first hour (with additional detail for Eu/Sm/Nd time courses)
.Not assessed against discharge limits (not mentioned in source)	Eu residual reported as 76 ppb after >1 day; Sm residual 1.52×10^{-9} mM in early samples (Nd residual not numerically specified)	REE^{3+} biosorption reported as almost complete (>99%) within the first hour (with additional detail for Eu/Sm/Nd time courses)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; reported as adsorption capacity (e.g., 84.2–91.5 $mg\cdot g^{-1}$, species-dependent)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; reported as adsorption capacity ($mg\cdot g^{-1}$) across tested conditions
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; Nd sorption capacity reported (Qmax Nd ≈ 0.47±0.01 $mmol\cdot g^{-1}$)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; Nd sorption capacity reported (Qmax Nd ≈ 0.37 $mmol\cdot g^{-1}$)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Max Dy removal 50% at pH 3.0 during 1 h; fast initial removal 38% within 5 min
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	B removal in multimetal solution decreased markedly (reported as low as 0.2% for Chlamydomonas reinhardtii in multimetallic solutions)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	B removal decreased from 21.4% (mono) to 6.6% (multi) for Chlorella vulgaris; metals like Mn can reach very high removals under some conditions (e.g., 99.4% for Mn at pH 7.0, 3 h)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Reported maximum removals include B up to 38.6% (at pH 5.5, 10 min) and As up to 40.7% (at pH 9.5, 3 h) for Scenedesmus almeriensis under selected conditions
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Reported maximum removals include Zn up to 91.9% (pH 5.5, 3 h) and Cu up to 88% (pH 7.0, 10 min) for Chlorophyceae spp. under selected conditions
.Not assessed against discharge limits (not mentioned in source)	Tl final 0.021 mg/L; Cr final 45.67 mg/L	Co: complete removal (100%) from 0.316 mg/L; Tl: 2.386→0.021 mg/L (~99%); Cr reduced but remained high (135.56→45.67 mg/L)
.Not assessed against discharge limits (not mentioned in source)	K final 744.7 mg/L (from 2036 mg/L)	K reduced from 2036 to 744.7 mg/L (~63%); La sorption capacity reported as qmax 143 mg/g (not % removal)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	.Not mentioned in source
Not assessed against discharge limits (synthetic solution; compliance not discussed)	Co^{2+}: 2.49 mg/L at 30 min; 1.01 mg/L at 10 h	at 30 min (63.6→2.49 mg/L); 98.78% at 10 h (to 1.01 mg/L) 96.44%
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Fe removal 54.14%
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Ce accumulation/removal up to 83%
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Reported generally as >90% metal removal within a few hours (metals not specified)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	La^{3+} removal efficiency 99.9%

Reactor Type	Biomass Form	Limiting Factor
Batch (Erlenmeyer flasks)	Oven-dried (non-living) biomass (24 h at 90°C)	pH control and (in general) Fe competition/precipitation: Fe can precipitate as pH rises, reducing competition; however Fe was removed by selective leaching so competitive effects were not present in tested leachate
Batch and continuous sorption systems	Dried biomass	Slow Nd sorption kinetics reduced capacity in continuous vs batch operation
Batch (conical flasks / capped bottles; 30 mL reaction volume)	Living biomass (grown, centrifuged, washed, resuspended to OD 10)	Orthophosphate limits REE solubility/availability and thus biosorption under growth-compatible conditions (phosphate + high acidity)
Batch culture flasks (100 mL cultures in conical flasks)	Living cultures (nutrient replete medium)	Orthophosphate limits REE solubility and availability for biosorption
Batch flask cultures (250 mL flasks with 150 mL medium)	Living biomass (biomass stayed alive during experiments; separated by centrifugation)	Nutrient/starvation state affects performance (biosorption more effective in culture media than starving culture); mechanism not fully resolved
Batch flask cultures (250 mL flasks with 150 mL medium)	Living biomass (biomass stayed alive during experiments; separated by centrifugation)	Nutrient/starvation state affects performance; mechanism not fully resolved
Batch tube incubation (2 mL solutions, shaking)	Lyophilized (dried) biomass	Mixed-metal competition reduces REE uptake; at higher concentrations other metals can replace Ce on binding sites
Batch tube incubation (2 mL solutions, shaking)	Lyophilized (dried) biomass	Mixed-metal competition reduces REE uptake; other metals can replace Ce at higher total concentrations
Batch sorption assays (biomass grown in shake flasks/bubble columns/15-L airlift reactor; sorption tests in lab vessels)	Lyophilized, ground biomass (stored at −20°C)	Nickel identified as least effectively bound ion (potential limiting/competing contaminant in mixed wastewaters)
Batch sorption assays (biomass grown in shake flasks/bubble columns/15-L airlift reactor; sorption tests in lab vessels)	Lyophilized, ground biomass (stored at −20°C)	Nickel identified as least effectively bound ion; common wastewater metals (Pb/Fe/Cu/Ni) noted as potentially problematic competitors requiring pre-removal in application contexts
Batch experiments (stirred, 50 mL solution)	Dried, homogenized biomass (105°C, 24 h; non-living)	pH strongly affects removal capacity; increased pH beyond optimum reduced removal capacity
Batch (50 mL in 120 mL flasks)	Living biomass used in uptake experiments (biomass obtained after centrifugation; initial biomass concentration given)	Boron was the lowest yield ratio (YR) after 3 h, suggesting strong interference by other metals in multimetal systems (except Chlorophyceae spp.)
Batch (50 mL in 120 mL flasks)	Living biomass used in uptake experiments (initial biomass concentration given)	Boron shows the lowest YR in multimetal systems (competition/interference)
.Not mentioned in source	.Not mentioned in source	Boron adsorption is strongly affected by co-metals; multimetal competition reduces B uptake across microalgae tested
.Not mentioned in source	.Not mentioned in source	B removal efficiencies decreased markedly in multimetal solutions for many microalgae (exception noted for Chlorophyceae spp.)
Batch (Erlenmeyer flasks; stirred)	Dried biomass (harvested, cleaned, oven-dried at 50°C overnight)	Potential desorption behavior for Tl (reduction then desorption back over time) and binding-site saturation limiting further reduction
Batch (Erlenmeyer flasks; stirred)	Dried biomass used in biosorption experiments (harvested, cleaned, oven-dried at 50°C)	Binding site saturation can limit further reduction; Tl behavior (desorption over time) noted as an issue in the overall study context
.Not mentioned in source	Living culture (biomass reached up to 4.5 g/L)	Tolerance reported up to 25 mg/L B with no growth decline; uptake reported as 3.35 mg/g (capacity may limit removal at higher B loads)
Glass tubular air-lift photobioreactor system (500 mL vessels)	Living culture (modified ‘endo’ medium; biomass concentrations in mg/L range)	Cobalt toxicity impacted viability (severe cell morphology deterioration); performance in real mixed wastewater matrices not tested
.Not mentioned in source	.Not mentioned in source	.Not mentioned in source
Batch systems mentioned in review context	.Not mentioned in source	.Not mentioned in source
.Not mentioned in source	Not clearly specified (living microalgae implied by stress tolerance statement)	Performance depends on REE identity/speciation, pH, and contact time (factors studied; no single limiting contaminant identified)
.Not mentioned in source	Living culture (implied by remediation/growth context; specific preparation not provided)	.Not mentioned in source

Kinetic Isotherm Fit	Biomass Dosage G L	Contact Time
Pseudo-second-order kinetics (R^5>0.99); Langmuir isotherm (qmax 188.68 mg/g; KL 0.029 L/mg)	g/L (dry biomass) 3–0.5	Up to 90 min (samples taken to 90 min); equilibrium within ~30 min
Not reported (q reported; kinetics described as slow)	g/L 0.5	Not reported (described qualitatively as slow kinetics)
.Not mentioned in source	.Not mentioned in source	Samples at 3 h and 20 h
.Not mentioned in source	.Not mentioned in source	At least 6 days (duration stated; time-resolved removal not reported)
Not reported (tracked by ICP-MS/LIBS; no model fit parameters given)	.Not mentioned in source	Minutes to ≤1 h for >99% uptake depending on REE (e.g., Nd within 30 min; Eu within ~1 h)
Not reported (tracked by ICP-MS/LIBS; no model fit parameters given)	.Not mentioned in source	Minutes to ≤1 h for >99% uptake depending on REE (e.g., Nd within 30 min; Eu within ~1 h)
Isotherms fitted (Langmuir/Freundlich); Langmuir reported as best correlation in this study	mg in 2 mL (≈5–10 g/L) 20–10	Standard incubation 3 h; kinetics sampled 2–60 min (rapid uptake reported)
Isotherms fitted (Langmuir/Freundlich); Langmuir reported as best correlation	mg in 2 mL (≈5–10 g/L) 20–10	Standard incubation 3 h; kinetics sampled 2–60 min
Isotherm fitting used (Langmuir/Freundlich); Langmuir-type behavior reported for some species in the study (kinetic models not reported)	g/L tested; commonly 5 g/L (10 mg in 2 mL) 20–5	h (Nd experiments) and 24 h (some Qmax experiments) depending on test 3
Nd isotherm for Chlorella kessleri better fitted with Freundlich isotherm (kinetic models not reported)	g/L tested; commonly 5 g/L (10 mg in 2 mL) 20–5	h (Nd experiments) and 24 h (some Qmax experiments) depending on test 3
Isotherm: Langmuir; kinetics: pseudo-second-order and Elovich (qmax Spirulina 3.24 mg/g)	g biosorbent in 50 mL (10 g/L) 0.5	min tested (max removal reported at 60 min) 120–5
No kinetic/isotherm model fits reported; rapid initial decrease within first 10 min noted	g/L initial biomass concentration 1.0	Samples collected at 10 min and 3 h (extended sampling also reported at 24 h and 72 h)
No kinetic/isotherm model fits reported; rapid initial decrease within first 10 min noted	g/L initial biomass concentration 1.0	Samples collected at 10 min and 3 h (also reported at 24 h and 72 h)
.Not mentioned in source	.Not mentioned in source	min and 3 h reported for maximum removals (metal/species dependent) 10
.Not mentioned in source	.Not mentioned in source	min and 3 h reported for maximum removals (metal dependent) 10
Langmuir performed better than Freundlich (linearized); kinetic modeling used non-linear Langmuir–Hinshelwood (AQUASIM) for La/Tl system	g per 100 mL (≈1.0 g/L) used for waste samples 1.0	Kinetic sampling 5–1440 min; industrial sample contact reported as 24 h
Langmuir fit preferred over Freundlich; La/Tl kinetics modeled using Langmuir–Hinshelwood equation in AQUASIM	g per 100 mL (≈1.0 g/L) used for waste samples (study also used 0.08 g per 150 mL in some tests; not standardized across all runs) 1.0	Kinetic sampling between 5–1440 min; some batch contacts reported up to 24 h
.Not mentioned in source	Biomass concentration reached up to 4.5 g/L (culture concentration; dosing strategy not specified)	.Not mentioned in source
.Not mentioned in source	Biomass concentrations used: 0.9–3.6 mg/L (0.0009–0.0036 g/L) and 4.2 mg/L stock reported	Sampling at 30 min and 2–10 h (key results at 30 min and 10 h)
.Not mentioned in source	.Not mentioned in source	days 4
.Not mentioned in source	.Not mentioned in source	.Not mentioned in source
.Not mentioned in source	.Not mentioned in source	Within a few hours" (exact time not reported)"
.Not mentioned in source	.Not mentioned in source	.Not mentioned in source

Elicit - Microalgae Biosorption of NdFeB Magnet Leachate Contaminants Study-Level Evidence Table (1)

Metals Tested	Waste Type	Species
Nd (50–250 mg/L; stock standardized to 1000 mg/L Nd); Fe, B, Dy, Co analyzed but reported below detection limit in prepared solutions	Real NdFeB magnet leachate (aqueous solution derived from permanent Nd–Fe–B magnets; selective leaching used to remove Fe/B/Dy/Co interference)	**Chlorella vulgaris (SAG 211-12)**
Nd (initial 250 mg/L)	Mixed leachate solution derived from neodymium magnets (batch and continuous sorption systems)	**Chlorella vulgaris (dried biomass)**
Cu 5 ppm; La 5 ppm; Nd 5 ppm; Dy 5 ppm	Synthetic multi-metal solution (5 ppm each of Cu, La, Nd, Dy); tested at different pH	**Galdieria sulphuraria (SAG 21.92)**
La added to reach 20 µM and 100 µM (depending on condition/pH)	Synthetic La-spiked nutrient-replete growth medium (phosphate present)	**Galdieria sulphuraria (SAG 21.92)**
Eu^{3+} (0.111 mM and 0.191 mM), Sm^{3+} (0.139 mM), Nd^{3+} (0.103 mM)	Synthetic REE solutions: REE nitrate salts dissolved in modified Bold's Basal Medium (BBM)	**Anabaena sp. (SAG 12.82)**
Eu^{3+} (0.111 mM and 0.191 mM), Sm^{3+} (0.139 mM), Nd^{3+} (0.103 mM)	Synthetic REE solutions: REE nitrate salts dissolved in modified BBM	**Anabaena cylindrica (SAG 1403.2)**
REEs: Ce, Nd, Tb, La; competition tests: Ce in presence of Al, Pb, Ni, Zn (mixed-metal solutions)	Synthetic REE solutions (single-metal screening at 10 mM; plus equimolar mixed-metal competition tests)	**Nostoc sp. 20.02**
REEs: Ce, Nd, Tb, La; competition tests: Ce in presence of Al, Pb, Ni, Zn (mixed-metal solutions)	Synthetic REE solutions (single-metal screening at 10 mM; plus equimolar mixed-metal competition tests)	**Synechococcus elongatus (UTEX 2973)**
Single-metal solutions of Nd^{3+}, Eu^{3+} and others: Au^{3+}, Pb^{2+}, Pt^{4+}, Fe^{3+}, Cu^{2+}, Ni^{2+} (10 mM stocks; specific concentration series for Nd in isotherm tests)	Synthetic single-metal solutions (Nd^{3+}/Eu^{3+} and other metals as individual solutions; stock solutions 10 mM)	**Calothrix brevissima (SAG 34.79)**
Single-metal solutions of Nd^{3+}, Eu^{3+} and others: Au^{3+}, Pb^{2+}, Pt^{4+}, Fe^{3+}, Cu^{2+}, Ni^{2+} (10 mM stocks; Nd isotherm series also reported)	Synthetic single-metal solutions (Nd^{3+}/Eu^{3+} and other metals as individual solutions; stock solutions 10 mM)	**Chlorella kessleri (SAG 211-11g)**
Dy (10–100 mg/L)	Synthetic single-metal dysprosium solutions ($Dy(NO_3)_2 \cdot 6H_2O$; batch lab experiments)	**Spirulina platensis**
B 60 mg/L; As 12 mg/L; Cu 3 mg/L; Mn 3 mg/L; Zn 3 mg/L	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Chlamydomonas reinhardtii (strain 11-32b, SAG)**
B 60 mg/L; As 12 mg/L; Cu 3 mg/L; Mn 3 mg/L; Zn 3 mg/L	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Chlorella vulgaris**
B 60 mg/L; As 12 mg/L; Cu 3 mg/L; Mn 3 mg/L; Zn 3 mg/L	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Scenedesmus almeriensis**
As, B, Cu, Mn, Zn (mixture); initial concentrations not specified in this summary source	Synthetic multi-contaminant water modeled on Loa River: B/As/Cu/Mn/Zn mixture (batch tests)	**Chlorophyceae spp. (indigenous strain from polluted environment)**
Industrial samples included: Co 0.316 mg/L (mine beneficiation); Tl 2.386 mg/L (smelter tailings); Cr 135.56 mg/L (smelter return water); simulated La 25–150 mg/L and Tl 150–800 mg/L also tested	Real industrial wastewaters (South Africa): mine beneficiation plant wastewater; smelter tailings/return waters; plus simulated La/Tl solutions	**Chlorella vulgaris (dried biomass)**
Simulated La 25–150 mg/L; simulated Tl 150–800 mg/L; industrial sample example: K 2036 mg/L (initial) reduced to 744.7 mg/L	Real industrial wastewaters (South Africa) and simulated La/Tl solutions	**Chlamydomonas reinhardtii**
B: 0.5, 10, 25, 50 mg/L	Synthetic boron-spiked water ("B stress" experiments at multiple B levels)	**Chlorella regularis**
Co^{2+} initial 63.6 mg/L (reported); also described as 60 mg/L added in methods	Synthetic single-metal cobalt solution added to photobioreactors ($Co(NO_3)_2 \cdot 6H_2O$; ~60–63.6 mg/L Co^{2+})	**Chlorella vulgaris**
Fe: 350 µg/L (0.35 mg/L)	Synthetic aqueous Fe test (study summarized in review table)	**Microcystis aeruginosa**
Ce (initial concentration not reported)	Synthetic Ce-containing aqueous solutions (reported in a review summary; primary study details not provided here)	**Nannochloropsis oculata**
Rare earth elements (specific REEs and initial concentrations not reported)	.Not mentioned in source	**Coccomyxa sp. (genus-level; strain not specified)**
La^{3+} (initial concentration not reported)	Synthetic La^{3+}-spiked water (study context not fully reported in excerpt)	**Euglena gracilis**

Compliance Status	Final Concentration Mg L	Removal Efficiency Pct
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Nd: % removal not tabulated; >90% of Nd biosorption occurred before equilibrium (equilibrium reached within ~30 min across temperatures)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	.Not mentioned in source (reported as uptake capacity, e.g., q = 157.21 mg/g at pH 5)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source (uptake reported as mg/g in some conditions)	Overall removal efficiency: <9% at pH 2.5; up to 86% at pH 5.0 (solution-wide metric; per-metal % not specified)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	No measurable La biosorption observed at initial pH 1.5 (under all growth conditions)
.Not assessed against discharge limits (not mentioned in source)	Eu residual reported as 76 ppb after >1 day; Sm residual 1.52×10^{-5} mM in early samples (Nd residual not numerically specified)	REE^{3+} biosorption reported as almost complete (>99%) within the first hour (with additional detail for Eu/Sm/Nd time courses)
.Not assessed against discharge limits (not mentioned in source)	Eu residual reported as 76 ppb after >1 day; Sm residual 1.52×10^{-5} mM in early samples (Nd residual not numerically specified)	REE^{3+} biosorption reported as almost complete (>99%) within the first hour (with additional detail for Eu/Sm/Nd time courses)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; reported as adsorption capacity (e.g., 84.2–91.5 $mg\cdot g^{-1}$, species-dependent)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; reported as adsorption capacity ($mg\cdot g^{-1}$) across tested conditions
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; Nd sorption capacity reported (Qmax Nd ≈ 0.47±0.01 $mmol\cdot g^{-1}$)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Not reported as % removal; Nd sorption capacity reported (Qmax Nd ≈ 0.37 $mmol\cdot g^{-1}$)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Max Dy removal 50% at pH 3.0 during 1 h; fast initial removal 38% within 5 min
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	B removal in multimetal solution decreased markedly (reported as low as 0.2% for Chlamydomonas reinhardtii in multimetallic solutions)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	B removal decreased from 21.4% (mono) to 6.6% (multi) for Chlorella vulgaris; metals like Mn can reach very high removals under some conditions (e.g., 99.4% for Mn at pH 7.0, 3 h)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Reported maximum removals include B up to 38.6% (at pH 5.5, 10 min) and As up to 40.7% (at pH 9.5, 3 h) for Scenedesmus almeriensis under selected conditions
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Reported maximum removals include Zn up to 91.9% (pH 5.5, 3 h) and Cu up to 88% (pH 7.0, 10 min) for Chlorophyceae spp. under selected conditions
.Not assessed against discharge limits (not mentioned in source)	Tl final 0.021 mg/L; Cr final 45.67 mg/L	Co: complete removal (100%) from 0.316 mg/L; Tl: 2.386→0.021 mg/L (~99%); Cr reduced but remained high (135.56→45.67 mg/L)
.Not assessed against discharge limits (not mentioned in source)	K final 744.7 mg/L (from 2036 mg/L)	K reduced from 2036 to 744.7 mg/L (~63%); La sorption capacity reported as qmax 143 mg/g (not % removal)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	.Not mentioned in source
Not assessed against discharge limits (synthetic solution; compliance not discussed)	Co^{2+}: 2.49 mg/L at 30 min; 1.01 mg/L at 10 h	at 30 min (63.6→2.49 mg/L); 98.78% at 10 h (to 1.01 mg/L) 96.44%
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Fe removal 54.14%
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Ce accumulation/removal up to 83%
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	Reported generally as >90% metal removal within a few hours (metals not specified)
.Not assessed against discharge limits (not mentioned in source)	.Not mentioned in source	La^{3+} removal efficiency 99.9%

Reactor Type	Biomass Form	Limiting Factor
Batch (Erlenmeyer flasks)	Oven-dried (non-living) biomass (24 h at 90°C)	pH control and (in general) Fe competition/precipitation: Fe can precipitate as pH rises, reducing competition; however Fe was removed by selective leaching so competitive effects were not present in tested leachate
Batch and continuous sorption systems	Dried biomass	Slow Nd sorption kinetics reduced capacity in continuous vs batch operation
Batch (conical flasks / capped bottles; 30 mL reaction volume)	Living biomass (grown, centrifuged, washed, resuspended to OD 10)	Orthophosphate limits REE solubility/availability and thus biosorption under growth-compatible conditions (phosphate + high acidity)
Batch culture flasks (100 mL cultures in conical flasks)	Living cultures (nutrient replete medium)	Orthophosphate limits REE solubility and availability for biosorption
Batch flask cultures (250 mL flasks with 150 mL medium)	Living biomass (biomass stayed alive during experiments; separated by centrifugation)	Nutrient/starvation state affects performance (biosorption more effective in culture media than starving culture); mechanism not fully resolved
Batch flask cultures (250 mL flasks with 150 mL medium)	Living biomass (biomass stayed alive during experiments; separated by centrifugation)	Nutrient/starvation state affects performance; mechanism not fully resolved
Batch tube incubation (2 mL solutions, shaking)	Lyophilized (dried) biomass	Mixed-metal competition reduces REE uptake; at higher concentrations other metals can replace Ce on binding sites
Batch tube incubation (2 mL solutions, shaking)	Lyophilized (dried) biomass	Mixed-metal competition reduces REE uptake; other metals can replace Ce at higher total concentrations
Batch sorption assays (biomass grown in shake flasks/bubble columns/15-L airlift reactor; sorption tests in lab vessels)	Lyophilized, ground biomass (stored at −20°C)	Nickel identified as least effectively bound ion (potential limiting/competing contaminant in mixed wastewaters)
Batch sorption assays (biomass grown in shake flasks/bubble columns/15-L airlift reactor; sorption tests in lab vessels)	Lyophilized, ground biomass (stored at −20°C)	Nickel identified as least effectively bound ion; common wastewater metals (Pb/Fe/Cu/Ni) noted as potentially problematic competitors requiring pre-removal in application contexts
Batch experiments (stirred, 50 mL solution)	Dried, homogenized biomass (105°C, 24 h; non-living)	pH strongly affects removal capacity; increased pH beyond optimum reduced removal capacity
Batch (50 mL in 120 mL flasks)	Living biomass used in uptake experiments (biomass obtained after centrifugation; initial biomass concentration given)	Boron was the lowest yield ratio (YR) after 3 h, suggesting strong interference by other metals in multimetal systems (except Chlorophyceae spp.)
Batch (50 mL in 120 mL flasks)	Living biomass used in uptake experiments (initial biomass concentration given)	Boron shows the lowest YR in multimetal systems (competition/interference)
.Not mentioned in source	.Not mentioned in source	Boron adsorption is strongly affected by co-metals; multimetal competition reduces B uptake across microalgae tested
.Not mentioned in source	.Not mentioned in source	B removal efficiencies decreased markedly in multimetal solutions for many microalgae (exception noted for Chlorophyceae spp.)
Batch (Erlenmeyer flasks; stirred)	Dried biomass (harvested, cleaned, oven-dried at 50°C overnight)	Potential desorption behavior for Tl (reduction then desorption back over time) and binding-site saturation limiting further reduction
Batch (Erlenmeyer flasks; stirred)	Dried biomass used in biosorption experiments (harvested, cleaned, oven-dried at 50°C)	Binding site saturation can limit further reduction; Tl behavior (desorption over time) noted as an issue in the overall study context
.Not mentioned in source	Living culture (biomass reached up to 4.5 g/L)	Tolerance reported up to 25 mg/L B with no growth decline; uptake reported as 3.35 mg/g (capacity may limit removal at higher B loads)
Glass tubular air-lift photobioreactor system (500 mL vessels)	Living culture (modified 'endo' medium; biomass concentrations in mg/L range)	Cobalt toxicity impacted viability (severe cell morphology deterioration); performance in real mixed wastewater matrices not tested
.Not mentioned in source	.Not mentioned in source	.Not mentioned in source
Batch systems mentioned in review context	.Not mentioned in source	.Not mentioned in source
.Not mentioned in source	Not clearly specified (living microalgae implied by stress tolerance statement)	Performance depends on REE identity/speciation, pH, and contact time (factors studied; no single limiting contaminant identified)
.Not mentioned in source	Living culture (implied by remediation/growth context; specific preparation not provided)	.Not mentioned in source

Kinetic Isotherm Fit	Biomass Dosage G L	Contact Time
Pseudo-second-order kinetics (R^2>0.99); Langmuir isotherm (qmax 188.68 mg/g; KL 0.029 L/mg)	g/L (dry biomass) 3–0.5	Up to 90 min (samples taken to 90 min); equilibrium within ~30 min
Not reported (q reported; kinetics described as slow)	g/L 0.5	Not reported (described qualitatively as slow kinetics)
.Not mentioned in source	.Not mentioned in source	Samples at 3 h and 20 h
.Not mentioned in source	.Not mentioned in source	At least 6 days (duration stated; time-resolved removal not reported)
Not reported (tracked by ICP-MS/LIBS; no model fit parameters given)	.Not mentioned in source	Minutes to ≤1 h for >99% uptake depending on REE (e.g., Nd within 30 min; Eu within ~1 h)
Not reported (tracked by ICP-MS/LIBS; no model fit parameters given)	.Not mentioned in source	Minutes to ≤1 h for >99% uptake depending on REE (e.g., Nd within 30 min; Eu within ~1 h)
Isotherms fitted (Langmuir/Freundlich); Langmuir reported as best correlation in this study	mg in 2 mL (≈5–10 g/L) 20–10	Standard incubation 3 h; kinetics sampled 2–60 min (rapid uptake reported)
Isotherms fitted (Langmuir/Freundlich); Langmuir reported as best correlation	mg in 2 mL (≈5–10 g/L) 20–10	Standard incubation 3 h; kinetics sampled 2–60 min
Isotherm fitting used (Langmuir/Freundlich); Langmuir-type behavior reported for some species in the study (kinetic models not reported)	g/L tested; commonly 5 g/L (10 mg in 2 mL) 20–5	h (Nd experiments) and 24 h (some Qmax experiments) depending on test 3
Nd isotherm for Chlorella kessleri better fitted with Freundlich isotherm (kinetic models not reported)	g/L tested; commonly 5 g/L (10 mg in 2 mL) 20–5	h (Nd experiments) and 24 h (some Qmax experiments) depending on test 3
Isotherm: Langmuir; kinetics: pseudo-second-order and Elovich (qmax Spirulina 3.24 mg/g)	g biosorbent in 50 mL (10 g/L) 0.5	min tested (max removal reported at 60 min) 120–5
No kinetic/isotherm model fits reported; rapid initial decrease within first 10 min noted	g/L initial biomass concentration 1.0	Samples collected at 10 min and 3 h (extended sampling also reported at 24 h and 72 h)
No kinetic/isotherm model fits reported; rapid initial decrease within first 10 min noted	g/L initial biomass concentration 1.0	Samples collected at 10 min and 3 h (also reported at 24 h and 72 h)
.Not mentioned in source	.Not mentioned in source	min and 3 h reported for maximum removals (metal/species dependent) 10
.Not mentioned in source	.Not mentioned in source	min and 3 h reported for maximum removals (metal dependent) 10
Langmuir performed better than Freundlich (linearized); kinetic modeling used non-linear Langmuir–Hinshelwood (AQUASIM) for La/Tl system	g per 100 mL (≈1.0 g/L) used for waste samples 1.0	Kinetic sampling 5–1440 min; industrial sample contact reported as 24 h
Langmuir fit preferred over Freundlich; La/Tl kinetics modeled using Langmuir–Hinshelwood equation in AQUASIM	g per 100 mL (≈1.0 g/L) used for waste samples (study also used 0.08 g per 150 mL in some tests; not standardized across all runs) 1.0	Kinetic sampling between 5–1440 min; some batch contacts reported up to 24 h
.Not mentioned in source	Biomass concentration reached up to 4.5 g/L (culture concentration; dosing strategy not specified)	.Not mentioned in source
.Not mentioned in source	Biomass concentrations used: 0.9–3.6 mg/L (0.0009–0.0036 g/L) and 4.2 mg/L stock reported	Sampling at 30 min and 2–10 h (key results at 30 min and 10 h)
.Not mentioned in source	.Not mentioned in source	days 4
.Not mentioned in source	.Not mentioned in source	.Not mentioned in source
.Not mentioned in source	.Not mentioned in source	Within a few hours" (exact time not reported)"
.Not mentioned in source	.Not mentioned in source	.Not mentioned in source

Microalgae Bioabsorption of NdFeB Magnet Leachate Contaminants for Wastewater Regulatory Compliance

1. State of evidence

The directly relevant evidence base for *magnet-derived* leachate treatment using microalgae biosorption is thin in the retrieved excerpts, with only a small subset using aqueous solutions explicitly derived from Nd–Fe–B magnets and tested with microalgal biomass. In those magnet-derived studies, the biosorbent was dried *Chlorella vulgaris* biomass and experiments were performed in batch flasks and also in continuous sorption configurations.

Most other included studies use synthetic single-metal or synthetic multi-metal solutions (typically REE nitrate/sulfate salts in defined media) and are therefore not faithful analogs of full-composition NdFeB manufacturing leachates that include Fe-dominant matrices and process chemicals. A smaller subset uses real industrial wastewater (non-magnet) matrices (e.g., mine beneficiation and smelter waters), showing that microalgal biosorption has been tested in complex real waters, but not specifically NdFeB manufacturing wastewaters in the provided excerpts.

The table below is an evidence map (not a performance ranking) that highlights which retrieved studies most closely resemble NdFeB-leachate compliance treatment.

Evidence category	Example species	Matrix type in excerpt	Why it matters for NdFeB wastewater compliance
Magnet-derived leachate biosorption	*Chlorella vulgaris*	Aqueous solution derived from real Nd–Fe–B magnets; selective leaching used such that Fe/B/Dy/Co were below detection in prepared solutions	Direct link to magnet source material, but not representative of Fe-rich leachates because Fe and other co-contaminants were removed in solution preparation
Synthetic multi-metal REE + base metals	*Galdieria sulphuraria*	5 ppm each of Cu, La, Nd, Dy at different pH	Demonstrates pH sensitivity and multi-metal effects for REE + Cu, but lacks Fe/B and industrial organics
Synthetic REE salts in growth media	*Anabaena* spp.	REE(NO3)3·6H2O salts in modified BBM (Nd, Sm, Eu concentrations reported)	Shows rapid REE uptake in culture media, but does not represent industrial leachate chemistry
Real industrial wastewater (non-magnet)	*Chlorella vulgaris*	"Simulated and industrial wastewater" from a mine and a smelter context (metals including Co, Tl, Cr reported)	Provides residual concentrations in real waters (useful for compliance framing), but not NdFeB composition

2. REE biosorption performance

For neodymium (Nd), dried *Chlorella vulgaris* showed high uptake capacities when tested on magnet-derived aqueous solutions, with Nd kinetics described as rapid (equilibrium within ~30 minutes) and with "more than 90%" of biosorption occurring before equilibrium. In the same Nd–*C. vulgaris* biosorption study, the Nd sorption kinetics fit a pseudo-second-order model (R^2>0.99) and the isotherm fit a Langmuir model with qmax 188.68 mg/g (KL 0.029 L/mg).

A separate magnet-derived "mixed leachate solution" study also used dried *Chlorella vulgaris* and reported a maximum Nd uptake of q = 157.21 mg/g at pH 5 with 0.5 g/L biosorbent dosage (initial Nd 250 mg/L). That work explicitly noted that slow Nd sorption kinetics limited performance in continuous assays relative to batch, underscoring that hydraulic/contact-time constraints are likely to matter for compliance operations.

Other taxa (including cyanobacteria) show very fast REE removal in defined media. In modified BBM with REE nitrate salts, *Anabaena* sp. (SAG 12.82) achieved almost complete (>99%) biosorption of trivalent REEs within the first hour, and specifically reported >99% Nd removal within 30 minutes after metal-salt addition (with initial Nd^{3+} 0.103 mM). For europium, the same study reported a residual concentration of 76 ppb after more than one day, indicating persistence of a small dissolved fraction after extended contact in that system.

Screening-style adsorption tests with lyophilized cyanobacterial biomass reported high REE capacities under acidic-to-near-neutral conditions. For example, *Nostoc* sp. 20.02 was reported to have the highest adsorption capacity (84.2–91.5 mg g^{-1}) in a screening study covering Ce, Nd, Tb, and La (single-element tests) and also included mixed-metal competition tests. Another study reported an Nd Qmax of 0.47 ± 0.01 mmol·g^{-1} for *Calothrix brevissima* (SAG 34.79) in single-element Nd solutions, with *Chlorella kessleri* also exhibiting high Nd capacity (0.37 ± 0.04 mmol·g^{-1}).

However, the evidence also shows that multi-ion competition can significantly reduce REE binding compared with single-element tests. In mixed-metal solutions, cerium adsorption capacity dropped significantly relative to single-element solutions, and at a metal concentration of 4 mM other metals (Al, Pb, Ni, Zn) "seemed to replace cerium" on binding sites as Ce binding capacity dropped across tested cyanobacterial biomasses.

3. Iron interference

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Modeling and parameterization is common, which is helpful for design but must be validated in realistic matrices. Examples include pseudo-second-order kinetics and Langmuir isotherms for Nd–*C. vulgaris* biosorption (qmax 188.68 mg/g), Langmuir isotherms for Dy biosorption (with pseudo-second-order and Elovich kinetics), and preference for Langmuir over Freundlich fits in industrial-wastewater biosorption datasets.

8. Critical evidence gaps

The excerpts collectively show strong proof-of-principle that microalgae can bind or accumulate REEs and some base metals rapidly under controlled conditions, but they also expose several gaps that prevent a confident conclusion that microalgae biosorption alone can bring complex NdFeB manufacturing leachates into discharge compliance.

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tested in that magnet-derived matrix. Closing this gap would require biosorption tests using full-composition NdFeB leachate (or a validated synthetic analog) that intentionally includes Fe and B at realistic concentrations and oxidation states, and that tracks dissolved vs precipitated fractions as pH is adjusted.

Second, phosphate chemistry can limit REE availability in living systems: *Galdieria sulphuraria* work concluded that REE biosorption was not efficient under conditions that allow sustained growth (high acidity and presence of phosphate), and confirmed that orthophosphate limits REE solubility and thus availability for biosorption. This implies that process-water nutrient content (or phosphate carryover) could reduce effective biosorption even when biomass is present, and should be explicitly controlled/measured in magnet-waste analogs.

Third, boron appears to be a likely rate-limiting compliance parameter in multi-contaminant mixtures, because its removal decreased dramatically in multimetal solutions and was identified as the lowest yield ratio after 3 hours due to interference by other metals. Even studies demonstrating boron tolerance and cellular uptake (e.g., no growth decline at 25 mg/L and uptake of 3.35 mg/g) do not yet provide the treated-water residual concentrations needed to validate discharge compliance for boron.

Fourth, time stability and rebound effects are insufficiently characterized. The reported Tl behavior—rapid initial reduction followed by desorption back into the system over time—shows that equilibrium and long-term stability can be problematic, and compliance-relevant protocols need to measure post-contact rebound and establish required residence times and/or polishing steps.

Finally, several studies note unresolved mechanisms (e.g., the "exact mechanism responsible for the REE entrapment is not clear"), which limits rational optimization for complex matrices where multiple adsorption and precipitation pathways may co-occur.

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Given that many of the strongest REE removals are demonstrated in defined media or selectively prepared solutions (including magnet-derived solutions without iron), a compliance-focused comparison would require side-by-side tests where the same real or fully representative leachate is treated by (i) optimized microalgae biosorption and (ii) best-practice precipitation/clarification, with both evaluated using residual concentrations (not only % removal) and with attention to rebound/desorption behavior over time.

Microalgae Bioabsorption of NdFeB Magnet Leachate Contaminants for Wastewater Regulatory Compliance

1. State of evidence

The directly relevant evidence base for *magnet-derived* leachate treatment using microalgae biosorption is thin in the retrieved excerpts, with only a small subset using aqueous solutions explicitly derived from Nd–Fe–B magnets and tested with microalgal biomass. In those magnet-derived studies, the biosorbent was dried *Chlorella vulgaris* biomass and experiments were performed in batch flasks and also in continuous sorption configurations.

Most other included studies use synthetic single-metal or synthetic multi-metal solutions (typically REE nitrate/sulfate salts in defined media) and are therefore not faithful analogs of full-composition NdFeB manufacturing leachates that include Fe-dominant matrices and process chemicals. A smaller subset uses real industrial wastewater (non-magnet) matrices (e.g., mine beneficiation and smelter waters), showing that microalgal biosorption has been tested in complex real waters, but not specifically NdFeB manufacturing wastewaters in the provided excerpts.

The table below is an evidence map (not a performance ranking) that highlights which retrieved studies most closely resemble NdFeB-leachate compliance treatment.

Evidence category	Example species	Matrix type in excerpt	Why it matters for NdFeB wastewater compliance
Magnet-derived leachate biosorption	*Chlorella vulgaris*	Aqueous solution derived from real Nd–Fe–B magnets; selective leaching used such that Fe/B/Dy/Co were below detection in prepared solutions	Direct link to magnet source material, but not representative of Fe-rich leachates because Fe and other co-contaminants were removed in solution preparation
Synthetic multi-metal REE + base metals	*Galdieria sulphuraria*	5 ppm each of Cu, La, Nd, Dy at different pH	Demonstrates pH sensitivity and multi-metal effects for REE + Cu, but lacks Fe/B and industrial organics
Synthetic REE salts in growth media	*Anabaena* spp.	REE(NO3)3·6H2O salts in modified BBM (Nd, Sm, Eu concentrations reported)	Shows rapid REE uptake in culture media, but does not represent industrial leachate chemistry
Real industrial wastewater (non-magnet)	*Chlorella vulgaris*	"Simulated and industrial wastewater" from a mine and a smelter context (metals including Co, Tl, Cr reported)	Provides residual concentrations in real waters (useful for compliance framing), but not NdFeB composition

2. REE biosorption performance

For neodymium (Nd), dried *Chlorella vulgaris* showed high uptake capacities when tested on magnet-derived aqueous solutions, with Nd kinetics described as rapid (equilibrium within ~30 minutes) and with "more than 90%" of biosorption occurring before equilibrium. In the same Nd–*C. vulgaris* biosorption study, the Nd sorption kinetics fit a pseudo-second-order model (R^2>0.99) and the isotherm fit a Langmuir model with qmax 188.68 mg/g (KL 0.029 L/mg).

A separate magnet-derived "mixed leachate solution" study also used dried *Chlorella vulgaris* and reported a maximum Nd uptake of q = 157.21 mg/g at pH 5 with 0.5 g/L biosorbent dosage (initial Nd 250 mg/L). That work explicitly noted that slow Nd sorption kinetics limited performance in continuous assays relative to batch, underscoring that hydraulic/contact-time constraints are likely to matter for compliance operations.

Other taxa (including cyanobacteria) show very fast REE removal in defined media. In modified BBM with REE nitrate salts, *Anabaena* sp. (SAG 12.82) achieved almost complete (>99%) biosorption of trivalent REEs within the first hour, and specifically reported >99% Nd removal within 30 minutes after metal-salt addition (with initial Nd^{3+} 0.103 mM). For europium, the same study reported a residual concentration of 76 ppb after more than one day, indicating persistence of a small dissolved fraction after extended contact in that system.

Screening-style adsorption tests with lyophilized cyanobacterial biomass reported high REE capacities under acidic-to-near-neutral conditions. For example, *Nostoc* sp. 20.02 was reported to have the highest adsorption capacity (84.2–91.5 mg g^{-1}) in a screening study covering Ce, Nd, Tb, and La (single-element tests) and also included mixed-metal competition tests. Another study reported an Nd Qmax of 0.47 ± 0.01 mmol·g^{-1} for *Calothrix brevissima* (SAG 34.79) in single-element Nd solutions, with *Chlorella kessleri* also exhibiting high Nd capacity (0.37 ± 0.04 mmol·g^{-1}).

However, the evidence also shows that multi-ion competition can significantly reduce REE binding compared with single-element tests. In mixed-metal solutions, cerium adsorption capacity dropped significantly relative to single-element solutions, and at a metal concentration of 4 mM other metals (Al, Pb, Ni, Zn) "seemed to replace cerium" on binding sites as Ce binding capacity dropped across tested cyanobacterial biomasses.

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1. REACTORS AND VESSELS

1.1 Continuous Stirred-Tank Bioreactor (CSTR)

- **Application**: The CSTR will serve as the primary biosorption reactor for continuous-flow treatment during the pilot-scale validation phase (Aim 4) of the research program.
- **Specifications**: A working volume of 50-100 liters is required. The reactor must be constructed from acid-resistant materials, such as High-Density Polyethylene (HDPE), Polypropylene (PP), or 316L stainless steel, to withstand the acidic nature of the leachate (pH 2-7). Key features should include a jacket for temperature control, internal baffles, a mechanical impeller (e.g., pitch-blade turbine), multiple sampling ports, and a bottom drain for complete emptying.
- **Vendor Options**:
  1. **TECNIC ePilot BR**: A stainless-steel pilot bioreactor platform available in volumes up to 50 L, featuring a hygienic 316L design, an integrated control tower, and built-in peristaltic pumps and sensors for pH and Dissolved Oxygen (DO).
  2. **Qubit Systems Z150 Photobioreactor**: Available in capacities up to 100 L, this glass and stainless steel system includes integrated controls for LED lighting, temperature, pH, and optional DO sensors, and supports up to eight peristaltic pumps.
  3. **Custom Fabricators (e.g., Wenzhou Ace Machinery, Qingdao Jingcheng Weiye)**: These vendors offer customizable CSTRs in stainless steel or other materials, allowing for tailored sizes and port configurations. Agitator options include turbine and anchor designs.
- **Estimated Price**: The cost varies significantly with features and material. A basic customizable stainless steel CSTR may start around $4,700, while fully instrumented, GMP-compliant systems from vendors like TECNIC or Sartorius can range from $50,000 to over $100,000. A budget of **$8,000 - $25,000** is appropriate for a well-instrumented, non-GMP research pilot reactor.

1.2 Packed-Bed Biosorption Columns

- **Application**: These columns are necessary for fixed-bed column studies to determine breakthrough curves (Aim 3) and for continuous-flow treatment with immobilized biomass.
- **Specifications**: Borosilicate glass columns are required for chemical resistance and visibility of the biomass bed. For bench-scale studies, columns with a 5 cm inner diameter and 50 cm length are suitable. For pilot-scale runs, larger columns of 10 cm inner diameter and 100 cm length are needed. Essential features include adjustable bed supports (e.g., porous polymer or sintered glass), PTFE endcaps, and sampling ports. The columns must be rated for low-pressure operation (<1 bar) or gravity flow.
- **Vendor Options**:
  1. **Bio-Rad Econo-Column®**: A standard choice for laboratory chromatography, available in a wide range of sizes, including 5.0 cm diameter columns with lengths from 10 cm to 50 cm. They are autoclavable and NaOH-compatible.
  2. **Ace Glass / Kimble/DWK**: These vendors supply a variety of glass chromatography columns that can be adapted for biosorption studies.
  3. **Custom Glassblowing**: For specific dimensions or port configurations for the 10 cm pilot column, a custom glass shop may be necessary.
- **Quantity**: 6 bench-scale columns (5 cm ID) and 2 pilot-scale columns (10 cm ID).
- **Estimated Price**: Bench-scale Bio-Rad columns (5.0 cm ID) range from approximately $214 (10 cm length) to $356 (50 cm length) each. Custom-fabricated 10 cm x 100 cm pilot columns are estimated at **$1,500 - $3,000 each**.

1.3 Feed and Effluent Storage Tanks

- **Application**: These tanks will serve as reservoirs for the raw leachate feed and for collecting treated effluent before analysis and disposal.
- **Specifications**: 100-200 L capacity tanks made of chemical-resistant HDPE or cross-linked polyethylene (XLPE) are required. They should be graduated for volume monitoring and feature a spigot or bottom drain for easy connection to pump tubing. Opaque walls are preferred to prevent algae growth in stored effluent.
- **Vendor Options**: Nalgene/Thermo Fisher, US Plastic Corp, Snyder Industries, Poly Processing, and Sigma-Aldrich (e.g., TANKPE100, a 100L HDPE tank).
- **Quantity**: 4 tanks (2 for feed, 2 for effluent).
- **Estimated Price**: **$100 - $300 each**.

2. PUMPS AND FLUID HANDLING

2.1 Peristaltic Pumps

- **Application**: Essential for precise, contamination-free delivery of leachate to reactors and columns, enabling accurate control of hydraulic retention time (HRT).
- **Specifications**: Pumps should offer variable flow rates in the range of 0.1 - 500 mL/min, be reversible, and support multi-channel heads (2-4 channels) for simultaneous experiments. Pump tubing must be chemically resistant to acidic leachate; options include Norprene®, C-Flex®, and Pharmed® BPT.
- **Vendor Options**:
  1. **Masterflex® L/S® Series (Cole-Parmer)**: A widely used laboratory standard, available in console and digital drive models with various pump heads (e.g., Easy-Load® II). Flow rates can range from 0.36 to 3400 mL/min depending on the drive and tubing combination.
  2. **Watson-Marlow 120, 300, and 500 Series**: Known for precision and reliability, these pumps offer flow rates suitable for lab and pilot applications. The 120 series offers low-flow precision (0.0001 to 170 mL/min), while the 300 and 500 series offer higher flows up to 2000 mL/min or more.
- **Quantity**: 4 pump drives with multi-channel heads.
- **Estimated Price**: Pump drives with heads range from **$1,500 - $4,000 each**. Tubing costs approximately $50 - $100 per 25 ft roll; a budget of 20 rolls is recommended.

2.2 pH Dosing Pumps

- **Application**: For automated addition of acid (e.g., HCl) or base (e.g., NaOH) to maintain the target pH in the CSTR.
- **Specifications**: Micro-dosing pumps with low flow rates (0.01 - 10 mL/min) and chemically resistant wetted parts are required. These are typically controlled by a signal from an external pH controller.
- **Vendor Options**: Hanna Instruments (BL series), LMI Milton Roy, ProMinent. The integrated peristaltic pumps on some bioreactor systems can also be configured for this purpose.
- **Quantity**: 2.
- **Estimated Price**: **$500 - $1,500 each**.

3. SENSORS AND ONLINE MONITORING

3.1 pH Probes and Controllers

- **Application**: Continuous online monitoring and control of pH in the bioreactors and columns.

- **Specifications**: Industrial-grade, combination pH electrodes with a BNC connector are needed. They must be durable and suitable for harsh chemical environments, covering a range of pH 0-14 with an accuracy of at least +/- 0.01. A separate controller with a 4-20mA output is required to interface with the dosing pumps.
- **Vendor Options**:
  1. **Hach**: IntelliCAL™ series probes and SC200/SC4500/SC1000 digital controllers. Digital pH sensors are priced from approximately $1,614.
  2. **Mettler Toledo**: InLab® and InPro® series probes are designed for process applications and bioprocesses.
  3. **YSI / Xylem**: Offers SensoLyt® probes and IQ SensorNet controllers. Probes are priced around $1,013, with controllers at approximately $2,826.
- **Quantity**: 4 probes and 2 controllers.
- **Estimated Price**: Probes: **$300 - $800 each**. Controllers: **$2,000 - $5,000 each**.

3.2 Dissolved Oxygen (DO) Probes

- **Application**: To monitor DO levels, which is critical for assessing the health and metabolic state of living microalgae systems (bioaccumulation mode).
- **Specifications**: Optical (luminescent) DO sensors are recommended for their stability and low maintenance. They should cover a range of 0-20 mg/L with an accuracy of +/- 0.1 mg/L.
- **Vendor Options**: Hach (IntelliCAL™ LDO101, from $1,218), YSI (ProODO, ProSolo), Hamilton (VisiFerm), Mettler Toledo (InPro® 6800, priced at $695), Sensorex (DO1200, $347; DO6400, from $845).
- **Quantity**: 2.
- **Estimated Price**: **$800 - $2,000 each**.

3.3 Temperature Probes

- **Application**: To monitor and control the temperature of the leachate and biomass slurry in reactors and columns.
- **Specifications**: Standard Pt100 RTD probes with a stainless steel sheath, covering a range of 0-100℃. These are often integrated into pH or DO probes.
- **Quantity**: 4.
- **Estimated Price**: **$50 - $200 each** for standalone probes.

3.4 Data Logger and Multichannel Controller

- **Application**: For centralized, continuous logging of sensor data (pH, DO, temperature) at user-defined intervals (e.g., every minute).
- **Vendor Options**:
  1. **Campbell Scientific**: CR-series dataloggers (e.g., CR6, CR1000) are industry standards for robust environmental monitoring, with prices starting around $1,500-$1,900.
  2. **LabJack**: T-series devices (e.g., T4, T7) are cost-effective multifunction DAQ units suitable for laboratory use, with prices ranging from $280 to $550.
  3. **Hach / YSI**: Sensor controllers like the SC200 or IQ SensorNet 2020 3G often have built-in logging capabilities.
- **Quantity**: 1.
- **Estimated Price**: **$2,000 - $6,000**.

4. ILLUMINATION (ALGAE CULTIVATION)

4.1 LED Grow Light Panels

- **Application**: To provide Photosynthetically Active Radiation (PAR) for maintaining living microalgae cultures and for conducting bioaccumulation experiments.
- **Specifications**: Lights must provide a PAR spectrum (400-700 nm) with an intensity of 100-300 μmol/m²/s at the reactor surface. Dimmable, full-spectrum or cool white LEDs are preferred for their energy efficiency and low heat emission. Panels should be IP65 waterproof rated for safety in a wet lab environment.
- **Vendor Options**: Valoya (research-grade, spectrum-tunable), Mars Hydro (commercial-grade), Fluence, Eco Industrial Supplies (Algae and Aquatic Bar Lights, A$207), Qubit Systems (integrated with Z150 PBRs, up to 3000 μmol/m²/s), and UTEX (RGB-LED platform for small bioreactors, $295).
- **Quantity**: 8-12 panels.
- **Estimated Price**: **$100 - $500 each** for commercial-grade panels; **$500 - $2,000 each** for research-grade, spectrum-tunable models.

5. BIOMASS HARVESTING AND SEPARATION

5.1 Laboratory Centrifuge

- **Application**: For harvesting microalgae biomass from batch experiments for analysis and weighing.
- **Specifications**: A refrigerated benchtop or floor-standing centrifuge with a swing-bucket rotor capable of holding 4 x 500 mL or 4 x 1000 mL bottles and reaching at least 5000 rpm is required.
- **Vendor Options**: Thermo Scientific (Sorvall Legend XTR, priced at $8,595), Eppendorf (5810R, 5910 Ri), Beckman Coulter (Allegra series, e.g., X-12R for $6,995).
- **Quantity**: 1.
- **Estimated Price**: **$5,000 - $15,000**.

5.2 Tangential Flow Filtration (TFF)

- **Application**: As a method for continuous harvesting or dewatering of biomass from the pilot-scale CSTR, which is more scalable than centrifugation.
- **Specifications**: A laboratory-scale TFF system with a 0.1 - 0.45 μm hollow fiber or flat sheet cassette is needed. The membrane material should be chemically resistant (e.g., PVDF or PES), with a feed flow rate capacity of 1-10 L/min.
- **Vendor Options**: Repligen (KrosFlo), Pall (Centramate), Sartorius (Vivaflow). Lgem also mentions microfiltration/ultrafiltration as a downstream processing option.
- **Quantity**: 1 system.
- **Estimated Price**: **$5,000 - $15,000** for the system, with cassettes costing **$500 - $2,000 each**.

6. ANALYTICAL SUPPORT EQUIPMENT

6.1 Benchtop pH Meter

- **Vendor Options**: Mettler Toledo (SevenCompact S220), Thermo Fisher Scientific (Orion), YSI (Pro10).
- **Quantity**: 2. **Estimated Price**: ~$1,500 - $3,000 each.

6.2 Analytical Balance

- **Specifications**: 0.1 mg readability.
- **Vendor Options**: Mettler Toledo (ME204), Sartorius (Quintix).
- **Quantity**: 1. **Estimated Price**: ~$2,000 - $4,000.

6.3 Orbital Shaker

- **Application**: For conducting batch biosorption isotherm and kinetic experiments.
- **Vendor Options**: IKA (KS 4000i), New Brunswick (Innova 44).
- **Quantity**: 2. **Estimated Price**: ~$3,000 - $6,000 each.

6.4 Vacuum Filtration Apparatus

- **Specifications**: Buchner funnel, filter flask, and 0.45 µm MCE or PVDF membranes.
- **Quantity**: 4 setups. **Estimated Price**: ~$200 - $500 each.

6.5 Drying Oven

- **Application**: For drying biomass to determine dry weight (60-105℃).
- **Vendor Options**: Thermo Scientific (Heratherm), VWR.
- **Quantity**: 1. **Estimated Price**: ~$2,000 - $4,000.

7. SAFETY EQUIPMENT

7.1 Chemical Fume Hood

- **Specifications**: 6-foot width with an acid-resistant liner, necessary for preparing acidic leachate solutions.
- **Estimated Price**: ~$5,000 - $15,000 installed.

7.2 Emergency Shower and Eyewash

- **Specifications**: Must be ANSI Z358.1 compliant.
- **Estimated Price**: ~$500 - $2,000 installed.

7.3 Spill Containment

- **Specifications**: Polyethylene secondary containment pallets/trays for 100 L acid and leachate tanks.
- **Estimated Price**: ~$200 - $500 each.

8. BUDGET SUMMARY TABLE

The following table summarizes the estimated capital costs for the major equipment categories required for the 24-month pilot program.

Category	Items	Qty	Unit Cost Range (USD)	Total Estimate (USD)
Reactors & Vessels	CSTR, Packed-Bed Columns, Storage Tanks	-	-	$20,000 - $60,000
Pumps & Fluid Handling	Peristaltic Pump Drives & Heads, Dosing Pumps, Tubing	-	-	$8,000 - $22,000
Sensors & Monitoring	pH/DO/Temp Probes, Controllers, Data Logger	-	-	$7,000 - $25,000
Illumination	LED Grow Light Panels	12	$100 - $2,000	$1,200 - $24,000
Biomass Harvesting	Laboratory Centrifuge, TFF System & Cassettes	-	-	$10,500 - $32,000
Analytical Support	Benchtop Meters, Balance, Shakers, Ovens	-	-	$10,000 - $25,000
Safety Equipment	Fume Hood, Eyewash/Shower, Spill Containment	-	-	$6,000 - $18,000
Grand Total Estimated Cost				**$62,700 - $206,000**

9. PROCUREMENT TIMELINE

Procurement should be phased according to the project Gantt chart.

- **Phase 0 (Months 1-2)**: Procure all bench-scale and analytical support equipment. This includes orbital shakers, benchtop meters, balances, filtration apparatus, drying ovens, and safety equipment. Bench-scale packed-bed columns (5 cm ID) should also be ordered.
- **Phase 3 / Early Phase 4 (Months 12-16)**: Procure all major pilot-scale equipment. This includes the 50-100 L CSTR, pilot-scale packed-bed columns (10 cm ID), peristaltic pumps, sensors, controllers, data logger, and the TFF system.
- **Long Lead-Time Items**: Custom-fabricated glass columns, specialized research-grade LED panels, and some high-capacity centrifuges or bioreactors may have lead times of several weeks to months and should be ordered early in their respective procurement phases. Regular communication with vendors is essential to align with the project timeline.

1.1 Scope

This document outlines a comprehensive procurement list of reagents, standards, and consumables required for a 24-month research program focused on microalgae biosorption of NdFeB magnet manufacturing wastewater. The list covers all experimental phases, including initial leachate characterization, Definitive Screening Design (DSD) experiments (approximately 294 runs including Quality Control), Response Surface Methodology (RSM) optimization (90–162 runs), continuous-flow column studies, a 30-day pilot-scale validation (50–100 L reactor), and final sludge characterization. The total volume of synthetic leachate required for the experimental phases is estimated to be 200–300 liters.

1.2 ICP-MS/OES Analytical Reagents

Accurate, trace-level quantification of rare earth and transition metals is critical. This requires high-purity reagents, certified reference materials (CRMs) for calibration, and appropriate internal standards to correct for matrix effects and instrument drift.

1.2.1 Calibration Standards

Calibration will be performed using multi-element certified reference materials. Standards for rare earth elements (REEs) will be procured separately from transition metals to avoid interferences during preparation. A second-source verification standard is required to validate the primary calibration curve.

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed
REE Standard	16-element mix (Sc, Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu) at 50 mg/L in 2% HNO_3.	Sigma-Aldrich (TraceCERT®)	67349	100 mL	~$300	2
Transition Metal Standard	30-element mix including Fe, Co, Ni, Cu, Al in dilute HNO_3.	Sigma-Aldrich (Certipur®)	1.11355	100 mL	~$250	2
Boron Standard	Single-element 1000 mg/L Boron in H_2O, traceable to NIST SRM.	Sigma-Aldrich (Certipur®)	1.70211	100 mL	~$100	1

Second-Source REE Standard	10 ppm Rare Earth mix in 5% HNO_3.	Inorganic Ventures	CMS-1-125ML	125 mL	$163	1

1.2.2 Internal Standards

An internal standard solution will be added online to all blanks, standards, and samples to correct for instrument drift and matrix suppression. The selected elements cover the full mass range of the target analytes.

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed
Internal Standard Mix	6-Li, Sc, Ge, Rh, In, Tb, Lu, Bi at 100 mg/L in 10% HNO_3.	Agilent Technologies	5188-6525	100 mL	$282	2
Single-Element Rh Standard	Rhodium 10 mg/L in 0.5 M HNO_3 (for custom mixes).	Sigma-Aldrich (Certipur®)	1.08525	100 mL	~$150	1
Single-Element Bi Standard	Bismuth 10 µg/mL in 2-5% HNO_3.	AccuStandard	IMS-111-100	100 mL	$30	1

1.2.3 Acids and Rinse Solutions

Trace-metal-grade acids are mandatory for sample digestion, preservation, and the preparation of blanks and standards to minimize elemental contamination. A complexing agent is required for the boron rinse solution to mitigate memory effects.

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed
Nitric Acid (HNO_3)	TraceMetal™ Grade, 67-70%. Evaluated for 65 metals at <1 ppb.	Fisher Chemical	A509P500	500 mL	$225	10
Hydrochloric Acid (HCl)	TraceMetal™ Grade, 32-35%.	Fisher Chemical	A508P500	500 mL	$198	4

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed
D-Mannitol	Reagent grade, for boron rinse solution preparation.	Sigma-Aldrich	M9546	1 kg	~$120	1

1.2.4 Digestion Consumables

Microwave digestion requires specialized PTFE vessels capable of withstanding high temperatures and pressures. Based on an estimated 600–800 total digestions for the project, several sets of vessels and liners will be required.

Item	Description	Vendor	Catalog No. (Example)	Est. Price (USD)	Qty Needed
Microwave Digestion Vessels	XP-1500 Plus Standard Vessel Assembly, 100 mL, for CEM MARS 6 system.	CEM Corporation	432020	~$500 / assembly	3 sets of 12
Vessel Liners	TFM Liner for XP-1500 Plus Vessels. Liners are consumable and require periodic replacement.	CEM Corporation	432175	~$150 / liner	24

1.3 Synthetic Leachate Salts

A high-fidelity synthetic leachate will be prepared for screening and optimization experiments. The target recipe is based on typical compositions of real NdFeB leachates: Fe 5000 mg/L, Nd 1500 mg/L, Dy 120 mg/L, Pr 300 mg/L, B 500 mg/L, Co 100 mg/L, Cu 50 mg/L, Ni 50 mg/L, and Al 50 mg/L. Reagent-grade salts are sufficient for this purpose.

Metal	Salt Form (Example)	Vendor	Catalog No. (Example)	Mass per Liter (g/L) for 300L Batch	Total Mass (kg)
Iron (Fe)	Iron(III) Chloride Hexahydrate ($FeCl_3 \cdot 6H_2O$)	Sigma-Aldrich	236489	24.2	7.26

Metal	Salt Form (Example)	Vendor	Catalog No. (Example)	Mass per Liter (g/L) for 300L Batch	Total Mass (kg)
Neodymium (Nd)	Neodymium(III) Chloride Hexahydrate ($NdCl_3 \cdot 6H_2O$)	Sigma-Aldrich	228968	5.2	1.56
Praseodymium (Pr)	Praseodymium(III) Chloride Hexahydrate ($PrCl_3 \cdot 6H_2O$)	Sigma-Aldrich	203360	0.8	0.24
Dysprosium (Dy)	Dysprosium(III) Chloride Hexahydrate ($DyCl_3 \cdot 6H_2O$)	Sigma-Aldrich	439607	0.3	0.09
Boron (B)	Boric Acid (H_3BO_3)	Fisher Chemical	A73-500	2.9	0.87
Cobalt (Co)	Cobalt(II) Chloride Hexahydrate ($CoCl_2 \cdot 6H_2O$)	Sigma-Aldrich	255599	0.4	0.12
Copper (Cu)	Copper(II) Chloride Dihydrate ($CuCl_2 \cdot 2H_2O$)	Sigma-Aldrich	307483	0.1	0.03
Nickel (Ni)	Nickel(II) Chloride Hexahydrate ($NiCl_2 \cdot 6H_2O$)	Sigma-Aldrich	654579	0.2	0.06
Aluminum (Al)	Aluminum Chloride Hexahydrate ($AlCl_3 \cdot 6H_2O$)	Sigma-Aldrich	237078	0.4	0.12

1.4 Microalgae Culture Reagents

BG-11 medium is a standard formulation for freshwater cyanobacteria and many green algae. Reagents can be purchased individually or as pre-made kits. Strains can be acquired from public culture collections such as UTEX (USA), SAG (Germany), or CCAP (UK).

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed
BG-11 Media Kit	Pre-measured components to prepare 100 Liters of medium.	UTEX	N/A	100 L Kit	$1,500	3
Sodium Nitrate ($NaNO_3$)	ACS Reagent, ≥99.0%	Fisher Chemical	BP360-500	500 g	~$75	1
Dipotassium Phosphate (K_2HPO_4)	BioXtra, ≥99.0%	Sigma-Aldrich	P3786	1 kg	~$150	1
Magnesium Sulfate ($MgSO_4 \cdot 7H_2O$)	ACS Reagent, ≥98.0%	Sigma-Aldrich	230391	500 g	~$80	1

1.5 Immobilization Materials

For experiments involving immobilized biomass, sodium alginate is used to form calcium alginate beads. A low-to-medium viscosity sodium alginate is suitable for this application.

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed
Sodium Alginate	BioReagent, from brown algae, suitable for immobilization of micro-organisms.	Sigma-Aldrich	71238	100 g	~$140	3
Calcium Chloride ($CaCl_2$)	Anhydrous, granular, ≥93.0%	Sigma-Aldrich	C1016	500 g	~$60	1

1.6 Filtration and Sampling

All aqueous samples for dissolved metals analysis must be filtered prior to acidification and analysis. PVDF syringe filters are recommended for their low protein binding and broad chemical compatibility.

Item	Description	Vendor	Catalog No. (Example)	Pack Size	Est. Price (USD)	Qty Needed (Est.)
Syringe Filters	Whatman GD/X 25 mm Sterile Syringe Filter, 0.45 µm PVDF membrane.	Whatman / Cytiva	6900-2504	50/pack	$277	60 packs (3000 total)
HDPE Bottles	High-Density Polyethylene bottles, wide mouth, acid-washed.	Nalgene/Fisher	Multiple	125 mL	~$2 / bottle	500
Polypropylene Tubes	Conical centrifuge tubes for boron sample storage to avoid glass contamination.	Falcon/VWR	Multiple	50 mL	~$0.50 / tube	1000

1.7 COD/BOD/TOC Reagents

Analysis of organic load will be conducted using commercially available spectrophotometric test kits.

Item	Description	Vendor (Example)	Notes
COD Vials	Dichromate method colorimetric vials (e.g., Low Range, High Range).	HACH	Requires spectrophotometer and digestion reactor.
BOD Reagents	Pre-measured pillows for standard 5-day BOD test.	HACH	Requires incubator and DO meter.
TOC Standards	Total Organic Carbon standards for instrument calibration.	Multiple	Instrument-specific.

1.8 Pilot-Scale Consumables

The 30-day pilot validation will require continuous operation of a 50-100 L reactor system. Key consumables include pump tubing, probes, and lighting.

Item	Description	Vendor (Example)	Notes
Peristaltic Pump Tubing	PharMed® BPT or equivalent for long life and chemical resistance.	Cole-Parmer	Size dependent on pump head and flow rate requirements.
pH / DO Probes	Combination pH electrodes and Dissolved Oxygen probes compatible with controller.	Mettler Toledo / Hanna	Requires periodic calibration and replacement.
LED Grow Lights	Full-spectrum LED panels suitable for promoting microalgal growth.	Multiple	Wattage and spectrum depend on reactor design.

1.9 Summary Budget Table

The following table provides a high-level summary of the estimated consumables budget for the 24-month program, based on the quantities and prices listed above.

Category	Estimated Total Cost (USD)	% of Consumables Budget
ICP-MS Analytical Reagents (Standards, Acids)	$7,500	17.0%
Synthetic Leachate Salts	$4,200	9.5%
Microalgae Culture Reagents (Kits & Bulk)	$5,500	12.5%
Immobilization & General Lab Chemicals	$1,000	2.3%
Filtration & Sampling (Filters, Bottles, Tubes)	$17,820	40.5%
Digestion Consumables (Vessels, Liners)	$7,000	15.9%
Grand Total (Estimated)	**$44,020**	**100%**

This estimated consumables budget of approximately $44,000 represents a small fraction of a typical multi-year research program budget, underscoring that analytical throughput and personnel time are the primary cost drivers.

Research Strategy

1. SIGNIFICANCE

1.1 The Critical Minerals Imperative

The global transition to a clean energy economy is critically dependent on a stable supply of high-performance materials, particularly neodymium-iron-boron (NdFeB) permanent magnets. These magnets are indispensable components in electric vehicle (EV) motors, wind turbine generators, and advanced defense systems. The global market for permanent magnets is projected to grow from USD 35.0 billion in 2022 to USD 78.9 billion by 2032, with NdFeB magnets expected to comprise 45% of this market by 2030. This rapid expansion is creating unprecedented demand for rare earth elements (REEs) such as neodymium (Nd), praseodymium (Pr), and dysprosium (Dy). However, the supply chain is highly concentrated, with the U.S. importing nearly 70% of its permanent magnets from China, which also controls over 85% of global REE processing capacity, creating significant supply chain vulnerabilities. Policies such as the U.S. Inflation Reduction Act (IRA), the DOE Critical Minerals Initiative (CMI), and the EU Critical Raw Materials Act mandate the development of secure domestic and allied supply chains to mitigate this risk. An expansion in domestic manufacturing will lead to a proportional increase in industrial wastewater, presenting a new set of environmental challenges that must be addressed for the supply chain to be truly sustainable.

1.2 The Wastewater Compliance Challenge

NdFeB magnet manufacturing is a multi-stage process involving casting, grinding, cutting, and surface treatment, each generating a unique and chemically complex waste stream. When leached, these waste streams—particularly grinding sludge—produce an effluent characterized by extremely high concentrations of iron (Fe), often exceeding 30,000–60,000 mg/L, alongside significant levels of Nd (15,000–28,000 mg/L), boron (B, 500–1,200 mg/L), cobalt (Co, up to 1,700 mg/L), and other metals. These leachates must be treated to meet stringent discharge limits under regulatory frameworks like the US EPA 40 CFR 433 (Metal Finishing) and the EU Best Available Techniques (BAT) Associated Emission Levels (AELs). For example, typical EU BAT-AELs require Cu and Ni to be below 2.0 mg/L, while US EPA limits are similar (Cu <2.07 mg/L, Ni <2.38 mg/L).

Conventional treatment, dominated by chemical precipitation (e.g., lime or caustic soda), is widely used due to its simplicity but has major drawbacks for this specific waste stream. It generates vast quantities of hazardous sludge that is difficult to dewater and expensive to dispose of. Furthermore, hydroxide precipitation is often ineffective for removing boron, which has very strict discharge limits in many jurisdictions (e.g., 1.5 mg/L in EU drinking water directives), necessitating additional, costly polishing steps like ion exchange or reverse osmosis.

1.3 The Knowledge Gap

While microalgae-based biosorption has shown significant promise for removing heavy metals and REEs from aqueous solutions, a critical knowledge gap prevents its application to NdFeB manufacturing wastewater. A comprehensive review of the literature confirms that no peer-reviewed study has systematically tested the performance of microalgae against the full, complex chemical matrix of real NdFeB manufacturing leachate. The vast majority of existing biosorption research has been conducted using simplified, single-metal, or low-concentration synthetic solutions.

The combined and potentially synergistic toxic effects of the true leachate composition—extreme iron-to-REE ratios (>20:1), high boron concentrations, the presence of organic process chemicals (cooling agents, surfactants, chelators like EDTA), and multi-metal competition—on microalgae viability and biosorption capacity are entirely unknown. Boron, in particular, represents a rate-limiting challenge, as studies show its removal efficiency by microalgae in

multi-metal solutions can be extremely low, dropping to as little as 0.2% in some cases. This project will close this critical gap by providing the first comprehensive assessment of microalgae biosorption as a viable compliance technology for this challenging industrial effluent.

1.4 Why This Matters Now

As the United States and its allies begin onshoring NdFeB magnet manufacturing, driven by incentives in the IRA and a strategic need to secure critical mineral supply chains, new production facilities are being planned and constructed. These facilities require Best Available Techniques for wastewater treatment that are not only compliant but also sustainable and cost-effective. A validated biological treatment option would reduce reliance on chemical-intensive processes, minimize hazardous sludge production, lower operational costs, and align with national green chemistry and circular economy objectives. By defining the operational boundaries of microalgae-based treatment for this specific application, this research will provide invaluable data to engineers, regulators, and industry stakeholders. Even a negative result—a definitive finding that microalgae cannot achieve compliance—is a valuable outcome, as it would prevent misapplication of the technology and inform the necessary design of hybrid chemical-biological systems.

2. INNOVATION

This project is innovative in five key areas that distinguish it from prior research in the field of biosorption and wastewater treatment.

2.1 First Systematic Study of Microalgae vs. Full NdFeB Leachate

This research represents the first systematic, multi-species evaluation of microalgae biosorption against the complete chemical matrix of real-world NdFeB magnet manufacturing wastewater. By moving beyond simplified synthetic solutions, we will address the combined and currently unknown effects of an iron-dominant multi-metal system, high boron concentrations, and the presence of organic process chemicals like coolants and chelating agents.

2.2 Compliance-Framed Experimental Design

Our experimental approach is fundamentally reframed around industrial and regulatory reality. The primary response variable will not be the conventional metric of "percent removal," but rather the "final residual concentration (mg/L)" of each contaminant measured directly against regulatory discharge limits. This compliance-centric framework ensures that the results are immediately translatable to real-world engineering applications and directly answer the question of whether the technology can meet legal requirements. Optimization will utilize multi-response desirability functions that are programmed to minimize the deviation from these legal discharge limits simultaneously.

2.3 Statistically Rigorous Two-Stage DOE

The project replaces the inefficient one-factor-at-a-time (OFAT) approach, common in biosorption literature, with a powerful two-stage statistical Design of Experiments (DOE). In Stage 1, a Definitive Screening Design (DSD) will be used to efficiently screen seven different factors across six microalgae species in 270 runs, identifying the most critical process variables. In Stage 2, a Box-Behnken or Central Composite Design (BBD/CCD) will be employed to develop a detailed Response Surface Methodology (RSM) model, mapping the "compliance window" and identifying optimal conditions with high statistical confidence. This rigorous, statistically-driven approach is a significant methodological innovation in this research area.

2.4 Integrated Boron Co-Treatment Strategy

Recognizing boron as a uniquely challenging contaminant for biological systems, this project is the first to explicitly test whether microalgae can contribute to achieving stringent boron compliance in a complex, multi-metal industrial waste stream. Furthermore, the project innovatively plans for a hybrid system by integrating biosorption with a targeted polishing step (boron-selective ion exchange resin) to determine the most cost-effective pathway to full compliance.

2.5 Pilot Validation with Real Industrial Waste

While the majority of biosorption studies are confined to bench-scale experiments with synthetic solutions, this project culminates in a pilot-scale validation. The optimized process will be tested in a 50-100 L continuous-flow reactor using real industrial wastewater for 30 consecutive days. This crucial step will provide robust data on long-term performance, process stability, and sludge characteristics, advancing the technology from a laboratory concept to a TRL 4-5 validation.

3. APPROACH

3.1 Overview and Rationale

This project will be executed over a 24-month period through a four-phase program designed to systematically de-risk the technology and generate a definitive assessment of its industrial viability. The research plan integrates five go/no-go decision gates at months 4, 6, 9, 13, and 22, ensuring a milestone-driven approach and responsible use of funds. The overall approach follows the logical progression from fundamental characterization to pilot-scale validation, guided by the Specific Aims outlined in the proposal.

3.2 Aim 1: Waste Stream Characterization (Months 1-4)

We will procure authentic waste samples from an industrial partner, including grinding sludge, polishing slurry, spent coolant, and cleaning agent waste. Each stream will be leached under controlled conditions and subjected to comprehensive chemical analysis. A full suite of 15+ analytes (including Fe, B, Co, Cu, Ni, Al, and all relevant REEs) will be quantified using ICP-MS/OES according to EPA methods 200.8/6010D. Standard methods will be used for COD, BOD, TOC, oil and grease, and surfactant assays. Based on this analysis, we will formulate a synthetic leachate recipe that accurately mimics the composition of the real waste. This synthetic analog will be validated by comparing the biosorption response of a reference alga (*C. vulgaris*) in both real and synthetic leachates. The primary deliverable will be a comprehensive composition report, the validated synthetic leachate recipe, and a compliance target matrix detailing the most stringent discharge limits from EU and US regulations. *Risk & Mitigation:* Delays in obtaining industrial samples are possible. To mitigate this, we will begin initial screening experiments using the synthetic recipe based on extensive data from the literature while awaiting real samples.

3.3 Aim 2: DSD Screening (Months 4-9)

A 7-factor Definitive Screening Design will be executed to efficiently identify the most critical variables influencing compliance. The design will consist of 270 runs (+24 QC runs), evaluating seven factors across six species.

- **Factors**: (1) pH (3-7); (2) Leachate Dilution (10-100%); (3) Fe Pre-removal (Yes/No via pH 3.5 precipitation); (4) Coolant Concentration (0-5% v/v); (5) EDTA Concentration (0-1.0 mM); (6) Biomass Dosage (0.5-5.0 g/L); (7) Contact Time (15-240 min).

- **Species (as a blocking factor)**: Six species will be tested based on demonstrated robustness and REE biosorption capacity in the literature: *Chlorella vulgaris*, the extremophile *Galdieria sulphuraria*, *Desmodesmus sp.*, *Arthrospira platensis*, *Nostoc sp.*, and *Coccomyxa sp.*. The total analytical workload will be substantial, requiring approximately 2,646 ICP-MS determinations. Data will be analyzed via forward stepwise model selection (AICc) and half-normal plots to identify the top 3-4 significant factors and the top 1-2 species for optimization. *Go/No-Go Gate 3 (Month 9):* The project will be terminated if no tested species achieves a residual concentration less than 50% above the discharge limit for at least one regulated metal under any condition.

3.4 Aim 3: RSM Optimization (Months 9-15)

Using the top species and factors from Aim 2, we will conduct a Response Surface Methodology study using a Box-Behnken Design (90-162 runs). This will generate a quadratic model of the compliance response surface. The primary response variables will remain the residual concentrations (mg/L) of each regulated metal. We will employ a Derringer-Suich desirability function to numerically optimize the process for simultaneous compliance across all contaminants. In parallel, continuous-flow column studies using immobilized biomass will be performed to generate breakthrough curves and assess performance under dynamic conditions. Finally, we will test the integration of a polishing step using a boron-selective ion exchange resin (e.g., Amberlite IRA-743) downstream of the biosorption step to evaluate the feasibility of a hybrid system for complete boron removal.

3.5 Aim 4: Pilot Validation (Months 15-24)

The fully optimized process will be validated at a 50-100 L pilot scale using a continuous-flow reactor (either packed-bed or CSTR, depending on Aim 3 results). The reactor will be fed with real NdFeB manufacturing wastewater from our industrial partner and operated continuously for 30 days. Influent and effluent will be sampled daily and analyzed via ICP-MS to confirm sustained performance against the regulatory compliance matrix. The metal-laden biomass sludge produced will be systematically collected and subjected to Toxicity Characteristic Leaching Procedure (TCLP, EPA Method 1311) and EN 12457 testing to determine its classification under US RCRA and EU Waste Framework Directive regulations. A full mass balance will be conducted, and the process will be assessed to determine a final Technology Readiness Level (TRL), with a target of TRL 4-5.

3.6 Potential Problems and Alternative Strategies

- **Problem 1: High Leachate Toxicity**. If all tested species show no viability at >10% leachate concentration, we will pivot to testing non-living (dried or chemically pre-treated) biomass, which is not susceptible to toxicity.
- **Problem 2: Insufficient Boron Removal**. If microalgae alone cannot achieve boron compliance, we will focus on optimizing the hybrid system developed in Aim 3, conducting a cost-effectiveness analysis against an ion-exchange-only treatment train.
- **Problem 3: Iron Interference Overwhelms Biosorption**. If high iron concentrations saturate all biomass binding sites, we will mandate iron pre-removal via pH 3.5 precipitation as a required pre-treatment step for all further experiments.
- **Problem 4: Organic Co-Contaminants Prove Acutely Toxic**. If process chemicals like EDTA or coolants are found to be acutely toxic, we will investigate an activated carbon pre-treatment step to remove organic compounds prior to the biosorption stage.

3.7 Timeline and Milestones

The project will adhere to the 24-month Gantt timeline detailed in previous project documents, which includes five critical go/no-go gates. The critical path runs through the sequence of Aims 1 through 4, with zero float, ensuring a

focused and timely execution of the research plan.

3.8 Evaluation Plan

The success of this project will be evaluated based on the final compliance matrix generated in Aim 4.

- **Full Success**: The optimized microalgae-based process (alone or in a cost-effective hybrid configuration) achieves compliance for ≥6 of the 9 regulated metals at the pilot scale.
- **Partial Success**: The process achieves compliance for 3-5 of the regulated metals, providing a strong proof-of-concept and a clear pathway for developing a more comprehensive hybrid system.
- **Valuable Failure**: The process achieves compliance for <3 metals, definitively establishing the lower-bound capabilities of biological treatment for this waste stream. This negative result would be highly valuable to the field and would be published to guide future research.

Technical Report: Optimization Design, Analytical Protocol, and Project Timeline

This report details three critical components for the proposed 24-month research program: a Stage 2 Response Surface Methodology (RSM) for process optimization, a comprehensive Standard Operating Procedure (SOP) for ICP-MS analysis of all target contaminants, and a detailed Gantt timeline outlining all project phases, milestones, dependencies, and risks.

Stage 2 Response Surface Methodology (RSM) Design

Context: Transition from DSD Screening to RSM Optimization

The initial screening phase of this project utilizes a Definitive Screening Design (DSD) to efficiently identify the most influential factors from a list of seven variables. The DSD is a powerful tool for separating main effects from two-factor interactions and quadratic effects with a minimal number of experimental runs. Once the DSD analysis (Phase 2C of the project plan) identifies the 3-4 most significant factors impacting metal removal, the project will transition to Stage 2: Optimization. This stage employs Response Surface Methodology (RSM) to refine the operating window, characterize non-linear responses, and determine the precise factor settings required to achieve maximum regulatory compliance.

Design Selection: Box-Behnken vs CCD

For optimizing 3-4 factors, two primary RSM designs are suitable: the Box-Behnken Design (BBD) and the Central Composite Design (CCD).

- **Box-Behnken Design (BBD)**: A spherical, three-level design that is highly efficient. For 3 factors, a BBD requires only 15 runs, and for 4 factors, it requires 27 runs. A key advantage of BBD is that it avoids extreme corner points, which is beneficial for biological systems where extreme combinations of factors (e.g., highest temperature and highest leachate concentration) could be lethal to the microalgae.
- **Central Composite Design (CCD)**: A five-level design that is rotatable and provides excellent prediction quality. For 3 factors, a CCD typically requires 20 runs, and for 4 factors, around 31 runs. While slightly less efficient in terms of run count, CCDs provide robust data for fitting a full second-order (quadratic) model.

Recommendation: For this biological application, the **Box-Behnken Design (BBD)** is recommended. Its efficiency and avoidance of extreme factor combinations make it a more conservative and practical choice for optimizing a live biological process.

Hypothetical Factor Scenarios

Based on the likely outcomes of the Stage 1 DSD screening, three potential scenarios for Stage 2 RSM are outlined below:

- **Scenario A (3 significant factors)**: The DSD identifies pH, biomass dosage, and contact time as the most critical variables. This would be addressed with a 3-factor BBD requiring 15 unique experimental runs. Total runs for this phase: 15 runs/replicate \u00d7 3 replicates \u00d7 2 top-performing species = **90 runs**.
- **Scenario B (4 significant factors)**: The DSD identifies pH, biomass dosage, contact time, and the degree of Fe pre-removal as significant. This would require a 4-factor BBD with 27 unique runs. Total runs for this phase: 27 runs/replicate \u00d7 3 replicates \u00d7 2 species = **162 runs**.

- **Scenario C (3 factors + 1 blocked factor)**: The DSD identifies three main factors as significant, but also indicates that the presence/absence of the coolant has a large, but not interactive, effect. A split-plot BBD could be used to efficiently test the three main factors while "blocking" the runs by coolant presence.

Complete BBD Run Matrix for Scenario A (3 factors)

The standard 15-run BBD matrix for three factors is presented below. The factors are shown in coded form (-1 for low, 0 for center, +1 for high). Before execution, these coded values will be replaced with actual numeric values based on the optimal ranges identified in the DSD. For example, if the DSD found pH 5 to be optimal, the new range for the BBD might be 4.0 (-1), 5.0 (0), and 6.0 (+1).

Run	A: pH	B: Biomass Dosage (g/L)	C: Contact Time (h)
1	-1	-1	0
2	1	-1	0
3	-1	1	0
4	1	1	0
5	-1	0	-1
6	1	0	-1
7	-1	0	1
8	1	0	1
9	0	-1	-1
10	0	1	-1
11	0	-1	1
12	0	1	1
13	0	0	0
14	0	0	0
15	0	0	0

Complete BBD Run Matrix for Scenario B (4 factors)

The standard 27-run BBD matrix for four factors is presented below in coded form.

Run	A: pH	B: Biomass Dosage	C: Contact Time	D: Fe Pre-removal
1	-1	-1	0	0
2	1	-1	0	0
3	-1	1	0	0
4	1	1	0	0
...	...	...	...	...
25	0	0	0	0
26	0	0	0	0
27	0	0	0	0

(Note: Full 27-run matrix omitted for brevity; it follows a standard BBD structure with combinations of \u00b11 and 0 values, including center points.)

Power Analysis

A power analysis will be conducted to ensure the experimental design can detect practically significant effects. With an alpha level (\u03b1) of 0.05 and a target power (1-\u03b2) of 0.80, the design will have an 80% probability of detecting a true effect of a specified size. For the 3-factor BBD (Scenario A), each species will have 45 total runs (15 unique runs \u00d7 3 replicates). This provides sufficient degrees of freedom for the model terms (linear, quadratic, 2-factor interactions) and the residual error to confidently detect even moderate effect sizes, such as a 20-30% change in residual metal concentration relative to the process standard deviation.

Response Variables and Desirability Functions

The primary goal is simultaneous compliance for multiple metals. To optimize multiple responses concurrently, a Derringer-Suich desirability function will be used. Each residual metal concentration (Fe, B, Co, Cu, Ni, Nd, etc.) will be assigned a desirability score from 0 (unacceptable; above the discharge limit) to 1 (highly desirable; at or below the target). The overall desirability is the geometric mean of individual scores, and the RSM will seek to find the factor settings that maximize this composite score, thereby finding the best compromise for treating all contaminants simultaneously.

Model Validation Protocol

After fitting the quadratic model and identifying the predicted optimal conditions, a set of 5 confirmation runs will be conducted at these specific settings. The experimental results will be compared to the model's predicted values and their 95% confidence intervals. The model will be considered validated if the experimental mean falls within the predicted 95% CI for all critical metal responses.

ICP-MS Analytical Standard Operating Procedure

Scope and Applicable Methods

This SOP outlines the procedures for the quantitative determination of target analytes in aqueous samples derived from NdFeB magnet leachate.

- **Primary Method**: U.S. EPA Method 200.8, "Determination of Trace Elements in Waters and Wastes by Inductively Coupled Plasma-Mass Spectrometry".
- **Secondary Method**: U.S. EPA SW-846 Method 6020B, "Inductively Coupled Plasma-Mass Spectrometry".
- **Supplementary Method**: U.S. EPA SW-846 Method 6010D, "Inductively Coupled Plasma-Optical Emission Spectrometry" (for high-concentration elements like Fe and B).
- **Target Analytes**: Fe, B, Co, Cu, Ni, Al, Nd, Dy, Pr, Sm, La, Ce, Y, Cr, Zn.

Sample Collection and Preservation

- **Container**: Acid-washed HDPE or polypropylene bottles. Borosilicate glass must be avoided for samples intended for Boron analysis.
- **Preservation**: For dissolved metals, samples must be filtered through a 0.45 \u00b5m filter. All aqueous samples must be preserved by acidifying to pH < 2 with trace-metal-grade nitric acid (HNO3), typically 2 mL of (1+1) HNO3 per liter of sample.
- **Holding Time**: Properly preserved samples can be held for up to 6 months before analysis.

Sample Digestion Protocol

For total recoverable metals in samples containing particulates, acid digestion is mandatory.

- **Option A: Microwave-Assisted Digestion (per EPA 3015A)**
  - **Reagents:** 9 mL concentrated trace-metal-grade HNO3 + 3 mL concentrated trace-metal-grade HCl.
  - **Procedure:** To a 45 mL sample in a PTFE digestion vessel, add the acids. Ramp to 175\u00b0C over 5.5 minutes and hold for 4.5 minutes. After cooling, dilute the digestate to a final volume of 50 mL with ultrapure water.
- **Option B: Hotplate Digestion (per EPA 200.7/3010A)**
  - **Reagents:** Concentrated trace-metal-grade HNO3 and HCl.
  - **Procedure:** Add 2 mL of (1+1) HNO3 and 1 mL of (1+1) HCl to a 100 mL sample aliquot. Gently heat at 85\u00b0C to reduce the volume to ~20 mL, then reflux for 30 minutes. Do not boil. Cool and dilute to the final volume.
- **Special Consideration for Boron:** Borosilicate glassware is a major source of boron contamination and must be avoided for all sample handling, digestion, and storage. Use quartz or PTFE digestion vessels and polypropylene labware. A rinse solution containing 2% mannitol should be used between boron-containing samples to mitigate memory effects.

Calibration Standards

- **Multi-element Calibration:** Prepare from certified 1000 mg/L single or multi-element stock solutions. Standards must be matrix-matched with the same acid concentration as the final samples (e.g., 2% HNO3 + 0.5% HCl).
- **Calibration Levels:** A minimum of a blank and three standards are required. Example ranges:
  - **REEs & Trace Metals (\u00b5g/L):** 0, 0.1, 0.5, 1, 5, 10, 50, 100, 500.
  - **Boron (\u00b5g/L):** 0, 1, 5, 10, 50, 100, 500.
  - **High-Conc Metals (mg/L, by ICP-OES):** 0, 0.1, 0.5, 1, 5, 10, 50, 100.
- **Internal Standards:** An internal standard solution must be added to all blanks, standards, and samples. Recommended elements include Sc-45, Ge, Y-89, Rh-103, In-115, Tb-159, and Bi-209. Final concentration should be 10-50 \u00b5g/L.
- **Acceptance:** The correlation coefficient (R\u00b2) for each calibration curve must be \u2265 0.995.

Method Detection Limits (MDLs)

MDLs must be determined for each analyte according to EPA procedures. The following table lists typical target MDLs achievable with modern ICP-MS instrumentation.

Analyte	Isotope	Target MDL (\u00b5g/L)	Target Reporting Limit (\u00b5g/L)	Typical Discharge Limit (mg/L)
Fe	57	0.5	2.0	1.0 - 5.0
B	11	1.0	5.0	1.0 - 3.0
Co	59	0.005	0.02	0.1 - 0.5
Cu	63	0.02	0.1	0.2 - 1.0
Ni	60	0.01	0.05	0.2 - 0.5
Al	27	0.2	1.0	1.0 - 10.0
Nd	146	0.005	0.02	0.5 (Total REE)

Analyte	Isotope	Target MDL (\u00b5g/L)	Target Reporting Limit (\u00b5g/L)	Typical Discharge Limit (mg/L)
Dy	163	0.002	0.01	0.5 (Total REE)
Pr	141	0.001	0.005	0.5 (Total REE)

(MDLs derived from EPA 200.8 and typical instrument performance 405,406,602,783>)

Interference Management

- **Polyatomic Interferences**: Argon-based interferences (e.g., \u2074\u2070Ar\u00b9\u2076O\u207A on \u2075\u2076Fe\u207A) must be mitigated using Kinetic Energy Discrimination (KED) with helium collision gas or by selecting an alternative isotope (e.g., \u2075\u2077Fe). Oxide ratios (e.g., CeO/Ce) must be tuned to <2%.
- **Isobaric Interferences**: Overlapping isotopes (e.g., \u00b9\u2074\u00b2Nd on \u00b9\u2074\u00b2Ce) must be corrected by selecting non-interfered isotopes (e.g., \u00b9\u2074\u2076Nd) or by applying mathematical correction equations defined in the EPA method.
- **Boron Memory Effect**: Use a dedicated rinse solution of 2% HNO3 + 2% mannitol with an extended washout time (60-90 seconds) between samples.

QC Protocol

All QC checks must be performed with the specified frequency and meet the acceptance criteria.

QC Element	Frequency	Acceptance Criteria
Method Blank (MB)	1 per batch (\u226420 samples)	< MDL for all analytes
Lab Fortified Blank (LFB)	1 per batch	85-115% recovery
Lab Fortified Matrix (LFM)	1 per 10 samples	70-130% recovery
LFM Duplicate (LFMD)	1 per 10 samples	RPD < 20%
Continuing Calib. Verification (CCV)	Every 10 samples & end of run	90-110% of true value
Continuing Calib. Blank (CCB)	Every 10 samples & end of run	< 2x MDL
Internal Standard Recovery	Every sample	70-130% of initial calibration

Special Protocols for This Project

- **High-Fe Matrix Handling**: For samples with Fe > 100 mg/L, Fe must be analyzed by ICP-OES (Method 6010D). The sample must then be diluted 100-1000x for the analysis of trace REEs by ICP-MS to avoid matrix suppression and detector saturation.
- **Biomass-laden Samples**: For dissolved metals, samples must be filtered (0.45 \u00b5m) before acidification and analysis. For total recoverable metals, the unfiltered, homogenized sample must be digested per Section 2.3.
- **COD Measurement**: COD will be measured using the dichromate reactor digestion method (e.g., HACH method 8000 / ISO 15705). Chloride interference will be mitigated by the addition of mercuric sulfate (HgSO4) as specified by the method.

24-Month Gantt Timeline with Milestone Dependencies

Program Phases and Durations

The 24-month research program is structured into five primary phases, with four critical Go/No-Go decision gates.

Phase	Activity	Months	Duration	Dependencies
0	Project Setup, Procurement, Ethics/Safety	M1-M2	2 mo	None
1	Leachate Collection and Characterization	M2-M4	3 mo	Phase 0 complete
2A	Algae Culture Establishment & Viability	M2-M4	3 mo	Phase 0 complete (Parallel with Ph 1)
GATE 1	**Viability Assessment**	**M4**	-	**End of Phase 2A**
2B	DSD Screening Experiments (294 runs total)	M4-M8	4 mo	Phase 1 & 2A complete; Gate 1 PASS
GATE 2	**Single-Metal Threshold Check**	**M6**	-	**Mid-Phase 2B**
2C	DSD Data Analysis & Factor Selection	M8-M9	1 mo	Phase 2B complete
GATE 3	**Multi-Metal Compliance Go/No-Go**	**M9**	-	**Phase 2C complete**
3A	RSM Optimization (90-162 runs)	M9-M13	4 mo	Gate 3 = GREEN or YELLOW
3B	Continuous-Flow Column Studies	M12-M15	3 mo	Preliminary results from Phase 3A
3C	Polishing Step Integration Tests	M13-M15	2 mo	Results from Phase 3A available
GATE 4	**Optimization Viability Check**	**M13**	-	**Phase 3A midpoint**
4A	Pilot Reactor Design and Build (50-100 L)	M14-M16	3 mo	Phase 3A near-complete
4B	Pilot Operation (30-day continuous)	M16-M20	4 mo	Phase 4A + real waste supply secured
4C	Sludge Characterization (TCLP/EN 12457)	M18-M20	2 mo	Sufficient sludge from Phase 4B
5	Data Analysis, Modeling, Reporting	M20-M24	4 mo	Phase 4B & 4C complete

Phase	Activity	Months	Duration	Dependencies
GATE 5	**Final Technology Assessment**	**M22**	-	**Pilot data fully analyzed**

Visual Gantt Chart (Text-Based)

```
Month:    1  2  3  4  5  6  7  8  9  10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Phase 0:  ==
Phase 1:     ===
Phase 2A:    === G1
Phase 2B:       ====== G2 ======
Phase 2C:                      == G3
Phase 3A:                         ===== G4 ==
Phase 3B:                                  ====
Phase 3C:                                   ===
Phase 4A:                                      ===
Phase 4B:                                         ========
Phase 4C:                                                ===
Phase 5:                                                         ======== G5 ====
```

Critical Path

The critical path determines the minimum project duration and consists of sequentially dependent tasks with zero float. Delays on this path will delay the project completion date. **Critical Path**: Project Setup > Leachate Characterization > DSD Screening > DSD Analysis > **GATE 3** > RSM Optimization > Pilot Reactor Build > Pilot Operation > Final Data Analysis & Reporting. Non-critical path activities like Algae Culture Establishment can occur in parallel and have some float.

Resource Loading

- **Personnel**: 2.0 Full-Time Equivalents (FTE) are required.
  - 1.0 FTE: Postdoctoral researcher/lead technician responsible for experimental execution, data collection, and analysis.
  - 0.5 FTE: Principal Investigator (PI) for project oversight, planning, and reporting.
  - 0.5 FTE: Laboratory technician for media preparation, culture maintenance, and analytical support.
- **Key Equipment**:
  - Access to a fully equipped microalgae culture facility.
  - Bench-scale reactors (flasks, photobioreactors).
  - Analytical instrumentation: ICP-MS/OES, spectrophotometer, pH meters.
  - Pilot-scale (50-100 L) reactor for Phase 4.
- **Peak Demand**: The highest resource demand will be during the DSD screening phase (Months 4-8), requiring the analysis of ~294 samples (~70 ICP-MS samples per week), representing a significant analytical and operational workload.

Milestone Deliverables

Month	Milestone	Deliverable	Go/No-Go Criteria
M4	Leachate Characterized	Full analytical report of leachate composition.	N/A
M4	**Gate 1**	Viability report for 6 species.	>50% viability in 10% leachate for at least 3 species.
M9	**Gate 3**	DSD experimental report & factor analysis.	\u22651 species/condition achieves <150% of discharge limit for \u22653 metals.
M13	**Gate 4**	RSM optimization interim report.	Predicted optimum meets compliance for \u22654 of 8 regulated metals.
M20	Pilot Complete	30-day continuous operation dataset.	N/A
M22	**Gate 5**	Final Technology Assessment Report.	Pilot data demonstrates stable compliance or defines polishing needs.
M24	Project End	Final peer-reviewed publications and reports.	N/A

Risk Register and Contingencies

Risk	Probability	Impact	Mitigation
Leachate Supply Interruption	Low	High	Secure long-term agreement with magnet manufacturer; collect and store sufficient volume for key phases upfront.
Algal Culture Contamination	Medium	Medium	Maintain strict aseptic techniques; use axenic starter cultures; have backup cultures cryopreserved or on agar slants.
ICP-MS Instrument Downtime	Medium	High	Secure a service contract with the instrument vendor; establish a backup analysis agreement with a commercial or university core lab.

Risk	Probability	Impact	Mitigation
GATE 3 No-Go (RED)	Medium	High	**Contingency**: Pivot project to focus on pre-treated (dead/immobilized) biomass, which may be more robust. **Contingency**: Investigate a hybrid process (e.g., chemical precipitation followed by biosorption polishing). **Final Action**: Terminate project and publish negative results, which are valuable for the field.
Pilot Scale-up Issues	Medium	High	Conduct thorough bench-scale column studies (Phase 3B) to de-risk hydrodynamics and biomass stability before committing to the full 100 L pilot build.

```
- This comprehensive report provides a detailed plan for the next stages of the research program. It inc
  - **Part 1:** A statistically robust Response Surface Methodology (RSM) to efficiently optimize the k
  - **Part 2:** A detailed Standard Operating Procedure (SOP) for ICP-MS analysis, covering everything
quality, defensible data.
  - **Part 3:** A 24-month project plan presented as a Gantt chart with clearly defined phases, mileston
making.

This three-part deliverable provides the necessary framework to move the project from screening to opti
scale validation.
```

- I have created the three-part report as requested.
- Part 1 covers the RSM design, comparing BBD and CCD, providing hypothetical scenarios and example run matrices, and discussing power analysis and desirability functions.
- Part 2 is a detailed ICP-MS SOP, structured with all the requested subsections, from sample collection to special protocols for this project. It is heavily cited from the provided `sop_protocols` data.
- Part 3 is a complete 24-month project plan, including a text-based Gantt chart, critical path analysis, resource loading, milestone table, and risk register, synthesized from the `timeline_data`.
- All claims are cited from the provided quotes.
- The structure follows the user's instructions: one H1 title, H2 for sections, H3 for subsections.
- The style is professional, narrative prose, with tables used to structure complex information.
- I have fulfilled the prompt. I will now generate the output.

Part 1: Stage 2 Response Surface Methodology (RSM) Design

Context: Transition from DSD Screening to RSM Optimization

The initial screening phase of this project utilizes a Definitive Screening Design (DSD) to efficiently identify the most influential factors from a list of seven variables. The DSD is a powerful tool for separating main effects from two-factor interactions and quadratic effects with a minimal number of experimental runs. Once the DSD analysis (Phase 2C of the project plan) identifies the 3-4 most significant factors impacting metal removal, the project will transition to Stage 2: Optimization. This stage employs Response Surface Methodology (RSM) to refine the operating window, characterize non-linear responses, and determine the precise factor settings required to achieve maximum regulatory compliance.

Design Selection: Box-Behnken vs CCD

For optimizing 3-4 factors, two primary RSM designs are suitable: the Box-Behnken Design (BBD) and the Central Composite Design (CCD).

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- **Central Composite Design (CCD)**: A five-level design that is rotatable and provides excellent prediction quality. For 3 factors, a CCD typically requires 20 runs, and for 4 factors, around 31 runs. While slightly less efficient in terms of run count, CCDs provide robust data for fitting a full second-order (quadratic) model.

Recommendation: For this biological application, the **Box-Behnken Design (BBD)** is recommended. Its efficiency and avoidance of extreme factor combinations make it a more conservative and practical choice for optimizing a live biological process.

Hypothetical Factor Scenarios

Based on the likely outcomes of the Stage 1 DSD screening, three potential scenarios for Stage 2 RSM are outlined below:

- **Scenario A (3 significant factors)**: The DSD identifies pH, biomass dosage, and contact time as the most critical variables. This would be addressed with a 3-factor BBD requiring 15 unique experimental runs. Total runs for this phase: 15 runs/replicate \u00d7 3 replicates \u00d7 2 top-performing species = **90 runs**.
- **Scenario B (4 significant factors)**: The DSD identifies pH, biomass dosage, contact time, and the degree of Fe pre-removal as significant. This would require a 4-factor BBD with 27 unique runs. Total runs for this phase: 27 runs/replicate \u00d7 3 replicates \u00d7 2 species = **162 runs**.
- **Scenario C (3 factors + 1 blocked factor)**: The DSD identifies three main factors as significant, but also indicates that the presence/absence of the coolant has a large, but not interactive, effect. A split-plot BBD could be used to efficiently test the three main factors while "blocking" the runs by coolant presence.

Complete BBD Run Matrix for Scenario A (3 factors)

The standard 15-run BBD matrix for three factors is presented below. The factors are shown in coded form (-1 for low, 0 for center, +1 for high). Before execution, these coded values will be replaced with actual numeric values based on the optimal ranges identified in the DSD. For example, if the DSD found pH 5 to be optimal, the new range for the BBD might be 4.0 (-1), 5.0 (0), and 6.0 (+1).

Run	A: pH	B: Biomass Dosage (g/L)	C: Contact Time (h)
1	-1	-1	0
2	1	-1	0
3	-1	1	0
4	1	1	0
5	-1	0	-1
6	1	0	-1
7	-1	0	1
8	1	0	1
9	0	-1	-1
10	0	1	-1
11	0	-1	1
12	0	1	1
13	0	0	0
14	0	0	0
15	0	0	0

Complete BBD Run Matrix for Scenario B (4 factors)

The standard 27-run BBD matrix for four factors is presented below in coded form.

Run	A: pH	B: Biomass Dosage	C: Contact Time	D: Fe Pre-removal
1	-1	-1	0	0
2	1	-1	0	0
3	-1	1	0	0
4	1	1	0	0
…	…	…	…	…
25	0	0	0	0
26	0	0	0	0
27	0	0	0	0

(Note: Full 27-run matrix omitted for brevity; it follows a standard BBD structure with combinations of \u00b11 and 0 values, including center points.)

Power Analysis

A power analysis will be conducted to ensure the experimental design can detect practically significant effects. With an alpha level (\u03b1) of 0.05 and a target power (1-\u03b2) of 0.80, the design will have an 80% probability of detecting a true effect of a specified size. For the 3-factor BBD (Scenario A), each species will have 45 total runs (15 unique runs \u00d7 3 replicates). This provides sufficient degrees of freedom for the model terms (linear, quadratic, 2-factor interactions) and the residual error to confidently detect even moderate effect sizes, such as a 20-30% change in residual metal concentration relative to the process standard deviation.

Response Variables and Desirability Functions

The primary goal is simultaneous compliance for multiple metals. To optimize multiple responses concurrently, a Derringer-Suich desirability function will be used. Each residual metal concentration (Fe, B, Co, Cu, Ni, Nd, etc.) will be assigned a desirability score from 0 (unacceptable; above the discharge limit) to 1 (highly desirable; at or below the target). The overall desirability is the geometric mean of individual scores, and the RSM will seek to find the factor settings that maximize this composite score, thereby finding the best compromise for treating all contaminants simultaneously.

Model Validation Protocol

After fitting the quadratic model and identifying the predicted optimal conditions, a set of 5 confirmation runs will be conducted at these specific settings. The experimental results will be compared to the model's predicted values and their 95% confidence intervals. The model will be considered validated if the experimental mean falls within the predicted 95% CI for all critical metal responses.

Part 2: ICP-MS Analytical Standard Operating Procedure

Scope and Applicable Methods

This SOP outlines the procedures for the quantitative determination of target analytes in aqueous samples derived from NdFeB magnet leachate.

- **Primary Method**: U.S. EPA Method 200.8, "Determination of Trace Elements in Waters and Wastes by Inductively Coupled Plasma-Mass Spectrometry".
- **Secondary Method**: U.S. EPA SW-846 Method 6020B, "Inductively Coupled Plasma-Mass Spectrometry".
- **Supplementary Method**: U.S. EPA SW-846 Method 6010D, "Inductively Coupled Plasma-Optical Emission Spectrometry" (for high-concentration elements like Fe and B).
- **Target Analytes**: Fe, B, Co, Cu, Ni, Al, Nd, Dy, Pr, Sm, La, Ce, Y, Cr, Zn.

Sample Collection and Preservation

- **Container**: Acid-washed HDPE or polypropylene bottles. Borosilicate glass must be avoided for samples intended for Boron analysis.
- **Preservation**: For dissolved metals, samples must be filtered through a 0.45 \u00b5m filter. All aqueous samples must be preserved by acidifying to pH < 2 with trace-metal-grade nitric acid (HNO3), typically 2 mL of (1+1) HNO3 per liter of sample.
- **Holding Time**: Properly preserved samples can be held for up to 6 months before analysis.

Sample Digestion Protocol

For total recoverable metals in samples containing particulates, acid digestion is mandatory.

- **Option A: Microwave-Assisted Digestion (per EPA 3015A)**
  - **Reagents**: 9 mL concentrated trace-metal-grade HNO3 + 3 mL concentrated trace-metal-grade HCl.
  - **Procedure**: To a 45 mL sample in a PTFE digestion vessel, add the acids. Ramp to 175\u00b0C over 5.5 minutes and hold for 4.5 minutes. After cooling, dilute the digestate to a final volume of 50 mL with ultrapure water.
- **Option B: Hotplate Digestion (per EPA 200.7/3010A)**

  - **Reagents**: Concentrated trace-metal-grade HNO3 and HCl.
  - **Procedure**: Add 2 mL of (1+1) HNO3 and 1 mL of (1+1) HCl to a 100 mL sample aliquot. Gently heat at 85\u00b0C to reduce the volume to ~20 mL, then reflux for 30 minutes. Do not boil. Cool and dilute to the final volume.
- **Special Consideration for Boron**: Borosilicate glassware is a major source of boron contamination and must be avoided for all sample handling, digestion, and storage. Use quartz or PTFE digestion vessels and polypropylene labware. A rinse solution containing 2% mannitol should be used between boron-containing samples to mitigate memory effects.

Calibration Standards

- **Multi-element Calibration**: Prepare from certified 1000 mg/L single or multi-element stock solutions. Standards must be matrix-matched with the same acid concentration as the final samples (e.g., 2% HNO3 + 0.5% HCl).
- **Calibration Levels**: A minimum of a blank and three standards are required. Example ranges:
  - **REEs & Trace Metals (\u00b5g/L)**: 0, 0.1, 0.5, 1, 5, 10, 50, 100, 500.
  - **Boron (\u00b5g/L)**: 0, 1, 5, 10, 50, 100, 500.
  - **High-Conc Metals (mg/L, by ICP-OES)**: 0, 0.1, 0.5, 1, 5, 10, 50, 100.
- **Internal Standards**: An internal standard solution must be added to all blanks, standards, and samples. Recommended elements include Sc-45, Ge, Y-89, Rh-103, In-115, Tb-159, and Bi-209. Final concentration should be 10-50 \u00b5g/L.
- **Acceptance**: The correlation coefficient (R\u00b2) for each calibration curve must be \u2265 0.995.

Method Detection Limits (MDLs)

MDLs must be determined for each analyte according to EPA procedures. The following table lists typical target MDLs achievable with modern ICP-MS instrumentation.

Analyte	Isotope	Target MDL (\u00b5g/L)	Target Reporting Limit (\u00b5g/L)	Typical Discharge Limit (mg/L)
Fe	57	0.5	2.0	1.0 - 5.0
B	11	1.0	5.0	1.0 - 3.0
Co	59	0.005	0.02	0.1 - 0.5
Cu	63	0.02	0.1	0.2 - 1.0
Ni	60	0.01	0.05	0.2 - 0.5
Al	27	0.2	1.0	1.0 - 10.0
Nd	146	0.005	0.02	0.5 (Total REE)
Dy	163	0.002	0.01	0.5 (Total REE)
Pr	141	0.001	0.005	0.5 (Total REE)

(MDLs derived from EPA 200.8 and typical instrument performance 405,406,602,783>)

Interference Management

- **Polyatomic Interferences**: Argon-based interferences (e.g., \u2074\u2070Ar\u00b9\u2076O\u207A on \u2075\u2076Fe\u207A) must be mitigated using Kinetic Energy Discrimination (KED) with helium collision

gas or by selecting an alternative isotope (e.g., \u2075\u2077Fe). Oxide ratios (e.g., CeO/Ce) must be tuned to <2%.

- **Isobaric Interferences**: Overlapping isotopes (e.g., \u00b9\u2074\u00b2Nd on \u00b9\u2074\u00b2Ce) must be corrected by selecting non-interfered isotopes (e.g., \u00b9\u2074\u2076Nd) or by applying mathematical correction equations defined in the EPA method.
- **Boron Memory Effect**: Use a dedicated rinse solution of 2% HNO3 + 2% mannitol with an extended washout time (60-90 seconds) between samples.

QC Protocol

All QC checks must be performed with the specified frequency and meet the acceptance criteria.

QC Element	Frequency	Acceptance Criteria
Method Blank (MB)	1 per batch (\u226420 samples)	< MDL for all analytes
Lab Fortified Blank (LFB)	1 per batch	85-115% recovery
Lab Fortified Matrix (LFM)	1 per 10 samples	70-130% recovery
LFM Duplicate (LFMD)	1 per 10 samples	RPD < 20%
Continuing Calib. Verification (CCV)	Every 10 samples & end of run	90-110% of true value
Continuing Calib. Blank (CCB)	Every 10 samples & end of run	< 2x MDL
Internal Standard Recovery	Every sample	70-130% of initial calibration

Special Protocols for This Project

- **High-Fe Matrix Handling**: For samples with Fe > 100 mg/L, Fe must be analyzed by ICP-OES (Method 6010D). The sample must then be diluted 100-1000x for the analysis of trace REEs by ICP-MS to avoid matrix suppression and detector saturation.
- **Biomass-laden Samples**: For dissolved metals, samples must be filtered (0.45 \u00b5m) before acidification and analysis. For total recoverable metals, the unfiltered, homogenized sample must be digested per Section 2.3.
- **COD Measurement**: COD will be measured using the dichromate reactor digestion method (e.g., HACH method 8000 / ISO 15705). Chloride interference will be mitigated by the addition of mercuric sulfate (HgSO4) as specified by the method.

Part 3: 24-Month Gantt Timeline with Milestone Dependencies

Program Phases and Durations

The 24-month research program is structured into five primary phases, with four critical Go/No-Go decision gates.

Phase	Activity	Months	Duration	Dependencies
0	Project Setup, Procurement, Ethics/Safety	M1-M2	2 mo	None

Phase	Activity	Months	Duration	Dependencies
1	Leachate Collection and Characterization	M2-M4	3 mo	Phase 0 complete
2A	Algae Culture Establishment & Viability	M2-M4	3 mo	Phase 0 complete (Parallel with Ph 1)
GATE 1	**Viability Assessment**	**M4**	-	**End of Phase 2A**
2B	DSD Screening Experiments (294 runs total)	M4-M8	4 mo	Phase 1 & 2A complete; Gate 1 PASS
GATE 2	**Single-Metal Threshold Check**	**M6**	-	**Mid-Phase 2B**
2C	DSD Data Analysis & Factor Selection	M8-M9	1 mo	Phase 2B complete
GATE 3	**Multi-Metal Compliance Go/No-Go**	**M9**	-	**Phase 2C complete**
3A	RSM Optimization (90-162 runs)	M9-M13	4 mo	Gate 3 = GREEN or YELLOW
3B	Continuous-Flow Column Studies	M12-M15	3 mo	Preliminary results from Phase 3A
3C	Polishing Step Integration Tests	M13-M15	2 mo	Results from Phase 3A available
GATE 4	**Optimization Viability Check**	**M13**	-	**Phase 3A midpoint**
4A	Pilot Reactor Design and Build (50-100 L)	M14-M16	3 mo	Phase 3A near-complete
4B	Pilot Operation (30-day continuous)	M16-M20	4 mo	Phase 4A + real waste supply secured
4C	Sludge Characterization (TCLP/EN 12457)	M18-M20	2 mo	Sufficient sludge from Phase 4B
5	Data Analysis, Modeling, Reporting	M20-M24	4 mo	Phase 4B & 4C complete
GATE 5	**Final Technology Assessment**	**M22**	-	**Pilot data fully analyzed**

Visual Gantt Chart (Text-Based)

```
Month:    1  2  3  4  5  6  7  8  9  10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Phase 0:  ==
Phase 1:     ===
```

```
Phase 2A:     === G1
Phase 2B:        ====== G2 ======
Phase 2C:                        == G3
Phase 3A:                           ===== G4 ==
Phase 3B:                                     ====
Phase 3C:                                      ===
Phase 4A:                                         ===
Phase 4B:                                            ========
Phase 4C:                                                   ===
Phase 5:                                                            ======== G5 ====
```

Critical Path

The critical path determines the minimum project duration and consists of sequentially dependent tasks with zero float. Delays on this path will delay the project completion date. **Critical Path**: Project Setup > Leachate Characterization > DSD Screening > DSD Analysis > **GATE 3** > RSM Optimization > Pilot Reactor Build > Pilot Operation > Final Data Analysis & Reporting. Non-critical path activities like Algae Culture Establishment can occur in parallel and have some float.

Resource Loading

- **Personnel**: 2.0 Full-Time Equivalents (FTE) are required.
  - 1.0 FTE: Postdoctoral researcher/lead technician responsible for experimental execution, data collection, and analysis.
  - 0.5 FTE: Principal Investigator (PI) for project oversight, planning, and reporting.
  - 0.5 FTE: Laboratory technician for media preparation, culture maintenance, and analytical support.
- **Key Equipment**:
  - Access to a fully equipped microalgae culture facility.
  - Bench-scale reactors (flasks, photobioreactors).
  - Analytical instrumentation: ICP-MS/OES, spectrophotometer, pH meters.
  - Pilot-scale (50-100 L) reactor for Phase 4.
- **Peak Demand**: The highest resource demand will be during the DSD screening phase (Months 4-8), requiring the analysis of ~294 samples (~70 ICP-MS samples per week), representing a significant analytical and operational workload.

Milestone Deliverables

Month	Milestone	Deliverable	Go/No-Go Criteria
M4	Leachate Characterized	Full analytical report of leachate composition.	N/A
M4	**Gate 1**	Viability report for 6 species.	>50% viability in 10% leachate for at least 3 species.

Month	Milestone	Deliverable	Go/No-Go Criteria
M9	**Gate 3**	DSD experimental report & factor analysis.	\u22651 species/condition achieves <150% of discharge limit for \u22653 metals.
M13	**Gate 4**	RSM optimization interim report.	Predicted optimum meets compliance for \u22654 of 8 regulated metals.
M20	Pilot Complete	30-day continuous operation dataset.	N/A
M22	**Gate 5**	Final Technology Assessment Report.	Pilot data demonstrates stable compliance or defines polishing needs.
M24	Project End	Final peer-reviewed publications and reports.	N/A

Risk Register and Contingencies

Risk	Probability	Impact	Mitigation
Leachate Supply Interruption	Low	High	Secure long-term agreement with magnet manufacturer; collect and store sufficient volume for key phases upfront.
Algal Culture Contamination	Medium	Medium	Maintain strict aseptic techniques; use axenic starter cultures; have backup cultures cryopreserved or on agar slants.
ICP-MS Instrument Downtime	Medium	High	Secure a service contract with the instrument vendor; establish a backup analysis agreement with a commercial or university core lab.

Risk	Probability	Impact	Mitigation
GATE 3 No-Go (RED)	Medium	High	**Contingency**: Pivot project to focus on pre-treated (dead/immobilized) biomass, which may be more robust. **Contingency**: Investigate a hybrid process (e.g., chemical precipitation followed by biosorption polishing). **Final Action**: Terminate project and publish negative results, which are valuable for the field.
Pilot Scale-up Issues	Medium	High	Conduct thorough bench-scale column studies (Phase 3B) to de-risk hydrodynamics and biomass stability before committing to the full 100 L pilot build.

SOP: Synthetic NdFeB Magnet Manufacturing Leachate Preparation

1. Purpose and Scope

This Standard Operating Procedure (SOP) provides step-by-step instructions for preparing a consistent and reproducible synthetic wastewater that chemically replicates the leachate from NdFeB permanent magnet manufacturing waste, specifically grinding sludge. This synthetic leachate is intended for use in laboratory-scale experiments, including Definitive Screening Designs (DSD), Response Surface Methodology (RSM) optimization, column studies, and as a control solution. The total volume required for the planned 24-month research program is estimated to be 200–300 liters.

2. Target Composition

2.1 Base Recipe (100% Strength Grinding Sludge Leachate Analog)

The base recipe is formulated to match the high-iron, multi-REE, and multi-metal composition reported in literature for acidic leachates of NdFeB magnet production waste.

Parameter	Target (mg/L)	Source Salt	Salt Conc. (g/L)	CAS No.	MW (g/mol)
Fe(III)	5000	Ferric Chloride Hexahydrate ($FeCl_3 \cdot 6H_2O$)	24.17	10025-77-1	270.30
Nd(III)	1500	Neodymium(III) Chloride Hexahydrate ($NdCl_3 \cdot 6H_2O$)	5.50	13544-93-7	358.69
Dy(III)	120	Dysprosium(III) Chloride Hexahydrate ($DyCl_3 \cdot 6H_2O$)	0.42	15059-52-6	371.95
Pr(III)	300	Praseodymium(III) Chloride Hexahydrate ($PrCl_3 \cdot 6H_2O$)	1.14	19423-77-9	350.36
B	500	Boric Acid (H_3BO_3)	2.86	10043-35-3	61.83
Co(II)	100	Cobalt(II) Chloride Hexahydrate ($CoCl_2 \cdot 6H_2O$)	0.40	7791-13-1	237.93
Cu(II)	50	Copper(II) Chloride Dihydrate ($CuCl_2 \cdot 2H_2O$)	0.13	10125-13-0	170.48

Parameter	Target (mg/L)	Source Salt	Salt Conc. (g/L)	CAS No.	MW (g/mol)
Ni(II)	50	Nickel(II) Chloride Hexahydrate ($NiCl_2 \cdot 6H_2O$)	0.20	7791-20-0	237.69
Al(III)	50	Aluminum Chloride Hexahydrate ($AlCl_3 \cdot 6H_2O$)	0.41	7784-13-6	241.43
pH	1.5-2.0	Hydrochloric Acid (HCl), conc.	As needed	7647-01-0	36.46

2.2 Optional Organic Additive Package

To simulate process chemical carryover, the following can be added to the base recipe:

- **EDTA**: 0-1.0 mM (from a stock of Disodium EDTA dihydrate, $Na_2EDTA \cdot 2H_2O$)
- **Metalworking Fluid**: 0-5% v/v (e.g., a water-soluble or semi-synthetic commercial coolant)
- **Surfactant**: 0-50 mg/L (e.g., Sodium Dodecyl Sulfate, SDS)

2.3 Dilution Series

For experimental screening, prepare dilutions of the 100% strength stock solution at 10%, 25%, and 50% using ultrapure water acidified to pH <2 with HCl.

3. Materials and Equipment

- **Reagents**: All salts should be ACS reagent grade or better. Acids must be trace-metal grade. Use ultrapure deionized water (resistivity >18 MΩ·cm).
- **Equipment**: Analytical balance (4-decimal place), magnetic stirrer and stir bars, overhead stirrer (for >5 L), volumetric flasks (Class A), graduated cylinders (polypropylene or PTFE), HDPE or PP carboys/flasks for mixing and storage, calibrated pH meter with combination electrode, fume hood, hot plate (optional for boric acid dissolution).
- **Labware**: Use polypropylene (PP), polytetrafluoroethylene (PTFE), or perfluoroalkoxy (PFA) labware for all operations involving boron to prevent contamination from borosilicate glass.
- **Personal Protective Equipment (PPE)**: Safety glasses with side shields or splash goggles, acid-resistant nitrile gloves (double-gloving recommended), lab coat. All operations involving concentrated acids or weighing powders must be performed in a chemical fume hood.

4. Preparation Procedure

4.1 Critical Principle: Dissolution Order and pH Control

Maintaining a highly acidic environment (pH < 2) during preparation is critical to prevent metal hydroxide precipitation. Ferric iron (Fe(III)) begins to precipitate as $Fe(OH)_3$ at a pH around 3.5, while REE hydroxides precipitate

at higher pH (>7). Furthermore, EDTA forms an extremely stable complex with Fe(III) (log K = 25.1), which is significantly stronger than its complexes with REEs like Nd(III) (log K ≈ 16.6). Therefore, all metal salts must be fully dissolved in a strongly acidic solution before any optional EDTA is added.

4.2 Step-by-Step for 1 Liter of 100% Leachate

1. Add approximately 800 mL of ultrapure DI water to a 1 L HDPE or PP beaker/flask with a magnetic stir bar.
2. **Acidification**: In a fume hood, carefully add ~5 mL of concentrated HCl to the water to lower the pH to ~1.0. **Always add acid to water**.
3. **Iron Dissolution**: While stirring, slowly add 24.17 g of $FeCl_3 \cdot 6H_2O$. The dissolution is exothermic, and the solution will warm and turn a dark orange/brown color. Stir for 5-10 minutes until fully dissolved.
4. **Boric Acid Dissolution**: Add 2.86 g of H_3BO_3. This may dissolve slowly. If necessary, gently warm the solution to 40-50 ℃ to aid dissolution, but do not boil.
5. **REE Salt Dissolution**: Sequentially add the REE salts: 5.50 g $NdCl_3 \cdot 6H_2O$, 1.14 g $PrCl_3 \cdot 6H_2O$, and 0.42 g $DyCl_3 \cdot 6H_2O$. Allow each to dissolve completely.
6. **Transition Metal Salt Dissolution**: Sequentially add the remaining metal salts: 0.40 g $CoCl_2 \cdot 6H_2O$, 0.13 g $CuCl_2 \cdot 2H_2O$, and 0.20 g $NiCl_2 \cdot 6H_2O$.
7. **Aluminum Salt Dissolution**: In the fume hood, add 0.41 g of $AlCl_3 \cdot 6H_2O$.
8. **pH Verification**: After all salts are dissolved, check the pH to ensure it remains < 2. Adjust with a few drops of concentrated HCl if necessary.
9. **Final Volume**: Quantitatively transfer the solution to a 1 L volumetric flask (polypropylene or PFA recommended) and bring to volume with ultrapure DI water. Mix thoroughly.
10. **Record**: Label the container with the preparation date, batch ID, and final measured pH.

4.3 Adding Optional Organics

- **EDTA**: If required, add EDTA from a prepared stock solution (e.g., 100 mM) after all metal salts are fully dissolved.
- **Metalworking Fluid**: Add the required volume of coolant concentrate (e.g., 10 mL for 1% v/v in 1 L) to the final solution and stir for 10-15 minutes to ensure emulsification.

4.4 Scaling

For larger volumes (e.g., 10 L or 20 L), use an appropriate-sized HDPE carboy and an overhead mechanical stirrer. Increase dissolution times to 30-60 minutes to ensure complete mixing.

5. Stability Testing

Prepared synthetic leachate should be tested for stability before use in experiments.

- **Visual Inspection**: Check for any signs of cloudiness, turbidity, or precipitation at t=0, 1h, 4h, 24h, and 7 days.
- **ICP-MS Verification**: Take aliquots at the same time points, filter (0.45 μm), and analyze for key metals (Fe, Nd, B) to confirm concentrations remain stable.
- **Acceptance Criteria**: Concentrations should remain within 90-110% of the initial (t=0) values.
- **Shelf Life**: Due to the high concentration and complexity, the solution is prone to aging effects, particularly Fe(III) hydrolysis if the pH rises above 2.5. It is recommended to use the stock solution within 7 days of preparation when stored at 4 ℃ in the dark. Diluted solutions should be prepared fresh daily.

6. pH Adjustment for Experiments

For experiments requiring higher pH values, adjust the leachate pH slowly and immediately before use.

- Use dropwise addition of 1 M or 5 M NaOH while stirring vigorously and monitoring with a calibrated pH meter.
- **Fe Pre-Removal Protocol**: To test the effect of iron removal, adjust the leachate pH to 3.5 using NaOH. A reddish-brown precipitate of $Fe(OH)_3$ will form. Allow the solution to settle for 30 minutes, then filter through a 0.45 µm membrane. Measure the residual Fe in the filtrate by ICP-MS before re-adjusting the pH for the experiment.

7. Quality Control

- Verify the concentration of each prepared batch by ICP-MS/OES analysis, ensuring key analytes are within ±10% of the target concentration.
- If possible, matrix-match with a real, fully characterized leachate sample by comparing biosorption performance to ensure the synthetic analog behaves similarly.

8. Safety

- **Chemical Hazards**: Concentrated HCl and $FeCl_3$ are corrosive. $CoCl_2$ and $NiCl_2$ are suspected carcinogens. H_3BO_3 is a reproductive toxin. Handle all solid powders and concentrated acids in a chemical fume hood.
- **Waste Disposal**: All prepared solutions and resulting experimental waste containing these metals are classified as hazardous waste and must be disposed of according to institutional and local environmental regulations.

9. Record Keeping

A batch record must be maintained for each preparation, including:

- Date of preparation and preparer's initials.
- Lot numbers and masses of all salts and reagents used.
- Final volume.
- Initial and final pH and temperature readings.
- Appearance (color, clarity).
- ICP-MS QC verification results.
- Unique Batch ID.

10. References and Stability Constants

This SOP is based on typical compositions found in academic literature and guidance from standard analytical methods. For detailed calculations involving chelating agents, consult critically evaluated stability constant databases such as those from NIST. Always refer to the Safety Data Sheet (SDS) for each chemical before use.